# ***Computer Networks LAB***

# ***(CIC-355)***

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**Semester :** $5^{th}$ Semester

**Maharaja Agrasen Institute of Technology, PSP Area,**

**Sector – 22, Rohini, New Delhi - 110086**

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कार्याणि न मनोरथैः

# MAHARAJA AGRASEN INSTITUTE OF TECHNOLOGY

## Computer Science & Engineering Department

## VISION

"To be centre of excellence in education, research and technology transfer in the field of computer engineering and promote entrepreneurship and ethical values."

## MISSION

To foster an open, multidisciplinary and highly collaborative research environment for producing world-class engineers capable of providing innovative solutions to real-life problems and fulfil societal needs.

# Department of Computer Science & Engineering

## VISION

"To attain global excellence through education, innovation, research, and work ethics in the field of Computer Science and engineering with the commitment to serve humanity."

## MISSION

**M1** To lead in the advancement of computer science and engineering through internationally recognized research and education.

**M2** To prepare students for full and ethical participation in a diverse society and encourage lifelong learning.

**M3** To foster development of problem solving and communication skills as an integral component of the profession.

**M4** To impart knowledge, skills and cultivate an environment supporting incubation, product development, technology transfer, capacity building and entrepreneurship in the field of computer science and engineering.

**M5** To encourage faculty, student's networking with alumni, industry, institutions, and other stakeholders for collective engagement

MAHARAJA AGRASEN INSTITUTE OF TECHNOLOGY

MAIT

# Department of Computer Science and Engineering

## Rubrics for Lab Assessment

| Rubrics | | 0 Missing | 1 Inadequate | 2 Needs Improvement | 3 Adequate |
|---|---|---|---|---|---|
| R1 | Is able to identify the problem to be solved and define the objectives of the experiment. | No mention is made of the problem to be solved. | An attempt is made to identify the problem to be solved but it is described in a confusing manner, objectives are not relevant, objectives contain technical/ conceptual errors or objectives are not measurable. | The problem to be solved is described but there are minor omissions or vague details. Objectives are conceptually correct and measurable but may be incomplete in scope or have linguistic errors. | The problem to be solved is clearly stated. Objectives are complete, specific, concise, and measurable. They are written using correct technical terminology and are free from linguistic errors. |
| R2 | Is able to design a reliable experiment that solves the problem. | The experiment does not solve the problem. | The experiment attempts to solve the problem but due to the nature of the design the data will not lead to a reliable solution. | The experiment attempts to solve the problem but due to the nature of the design there is a moderate chance the data will not lead to a reliable solution. | The experiment solves the problem and has a high likelihood of producing data that will lead to a reliable solution. |
| R3 | Is able to communicate the details of an experimental procedure clearly and completely. | Diagrams are missing and/or experimental procedure is missing or extremely vague. | Diagrams are present but unclear and/or experimental procedure is present but important details are missing. | Diagrams and/or experimental procedure are present but with minor omissions or vague details. | Diagrams and/or experimental procedure are clear and complete. |
| R4 | Is able to record and represent data in a meaningful way. | Data are either absent or incomprehensible. | Some important data are absent or incomprehensible. | All important data are present, but recorded in a way that requires some effort to comprehend. | All important data are present, organized and recorded clearly. |
| R5 | Is able to make a judgment about the results of the experiment. | No discussion is presented about the results of the experiment . | A judgment is made about the results, but it is not reasonable or coherent. | An acceptable judgment is made about the result, but the reasoning is flawed or incomplete. | An acceptable judgment is made about the result, with clear reasoning. The effects of assumptions and experimental uncertainties are considered. |

# Lab Assessment Sheet

| S.No | Experiment | Marks | | | | | Total Marks | Date | Signature |
|---|---|---|---|---|---|---|---|---|---|
| | | R1 | R2 | R2 | R4 | R5 | | | |
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| S.No | Experiment | Marks | | | | | Total Marks | Date | Signature |
|---|---|---|---|---|---|---|---|---|---|
| | | R1 | R2 | R2 | R4 | R5 | | | |
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# Experiment – 1. *(14/08/2024)*

## Introduction Of Cisco Packet Tracer

Cisco Packet Tracer is a powerful network simulation tool developed by Cisco Systems, primarily used for teaching and learning networking concepts. It allows users to design, configure, and troubleshoot virtual networks with a wide array of networking devices like routers, switches, and end devices. Packet Tracer provides a hands-on experience without needing physical hardware, making it ideal for students, educators, and professionals. It supports learning for Cisco certifications such as CCNA and CCNP by providing realistic scenarios for practicing network configurations, protocols & techniques in a risk-free environment.

Cisco Packet Tracer is a network simulation tool developed by Cisco Systems. It allows users to create network topologies and simulate network configurations and troubleshooting without needing physical hardware. It's widely used in educational settings, especially in Cisco Networking Academy courses.

Cisco Packet Tracer is an essential tool for anyone involved in networking, from beginners to seasoned professionals. It offers a user-friendly interface that simulates real-world network environments, enabling users to experiment with complex network topologies. Beyond basic network configuration, Packet Tracer supports the simulation of advanced concepts like security protocols, Internet of Things (IoT) devices, and wireless networks. It also includes assessment features for educators, allowing them to create activities and tests that provide immediate feedback. The software fosters collaborative learning through its multiuser functionality, where users can work together on network simulations in real-time, making it a versatile and valuable resource in the field of networking.

## Steps to download and configure

## Downloading Cisco Packet Tracer

1. Visit Cisco Networking Academy:
   - Go to **Cisco Networking Academy.**
   - Log in or create a free account.
2. Find the Download:
   - Search for “Packet Tracer” or go to the “Resources” section.
3. Download the Installer:
   - Choose the version for your operating system (Windows, macOS, Linux) and download the file.

## Installing Cisco Packet Tracer

Windows:

- Run the `.exe` installer file.
- Follow the setup wizard and click "Install."

## Configuring Cisco Packet Tracer

1. Launch the Application:
   - Open Packet Tracer from your desktop or applications folder.
2. Create a New Project:
   - Click "File" > "New" to start a new network design.
3. Add and Connect Devices:
   - Drag devices from the panel to the workspace and use the "Connections" tool to link them.
4. Configure Devices:
   - Click on devices to open their configuration windows and set up IP addresses, routing, etc.
5. Save Your Work:
   - Click "File" > "Save" to keep your project.
6. Simulate and Test:
   - Use simulation mode to observe network behavior and troubleshoot.

## Applications of Cisco Packet Tracer:

1. Network Design and Simulation:
   - Users can design and build network topologies with virtual routers, switches, and other network devices.
   - Simulates network operations and protocols to understand their behavior and interaction.
2. Troubleshooting Practice:
   - Provides a platform to practice troubleshooting network issues.
   - Users can create network problems and test their problem-solving skills in a controlled environment.
3. Configuration Practice:
   - Allows users to practice configuring network devices with various Cisco IOS commands.
   - Users can simulate real-world configurations and scenarios.
4. Learning and Teaching:
   - Commonly used in Cisco Networking Academy courses to teach networking concepts.
   - Offers an interactive way to understand network fundamentals and advanced topics.
5. Testing and Experimentation:
   - Enables users to test new configurations and network setups without the risk of impacting live networks.
   - Users can experiment with different network designs and scenarios.

# Wireshark

Wireshark is a widely used network protocol analyzer that allows users to capture and examine data packets traveling through a network in real-time. It is an essential tool for network administrators, cybersecurity professionals, and developers, helping them troubleshoot network issues, analyze traffic, and detect security vulnerabilities. Wireshark supports deep inspection of hundreds of protocols and runs on various platforms, including Windows, macOS, and Linux. With its user-friendly interface and powerful filtering capabilities, Wireshark provides detailed insights into network operations, making it invaluable for diagnosing network performance issues and investigating suspicious activity in a network environment. Wireshark is a network protocol analyzer that captures and analyzes network packets. It's an open-source tool used for network troubleshooting, analysis, and development. Wireshark's open-source nature and continuous updates ensure it stays current with emerging protocols and threats, making it indispensable for modern network analysis.

Wireshark's versatility extends to its ability to decode and display a vast array of network protocols, making it a critical tool for both educational and professional environments. It allows users to reconstruct a sequence of events in network communication, which is particularly useful for debugging and forensic analysis. Wireshark also supports live data capture and offline analysis, with options to export captured data for further review. Its robust community and extensive documentation make it accessible to users of all skill levels, ensuring its continued relevance in the ever-evolving field of network analysis.

## Downloading Wireshark

1. Visit the Wireshark Website:
   - Go to **Wireshark's official website**.
2. Download the Installer:
   - Click on the "Download" button.
   - Choose the installer for your operating system (Windows, macOS, Linux).

## Installing Wireshark

Windows:

- Run the `.exe` installer file.
- Follow the setup wizard and click "Next" through the prompts.
- Install WinPcap or Npcap when prompted (needed for packet capturing).

## Configuring Wireshark

1. Launch Wireshark:
   - Open Wireshark from your applications or start menu.
2. Select an Interface to Capture:
   - Go to the "Capture" menu and select "Options" or click on the interface list on the main screen.
   - Choose the network interface you want to monitor (e.g., Wi-Fi, Ethernet).
3. Start Capturing Packets:
   - Click the "Start" button or double-click on the selected interface to begin capturing network traffic.
4. Apply Filters:
   - Use display filters to focus on specific traffic. For example, type `ip.addr == 192.168.1.1` to filter packets from/to a specific IP address.
5. Analyze Packets:
   - Click on packets in the list to view detailed information.
   - Use the packet details and byte view to analyze the packet contents and protocols.
6. Save and Export Data:
   - Click "File" > "Save As" to save the captured data & Export Data .

## Applications of Wireshark:

1. Network Troubleshooting:
   - Captures and inspects network packets to identify and diagnose issues like latency, dropped packets, or misconfigurations.
2. Protocol Analysis:
   - Allows detailed examination of various network protocols to understand how they work and interact.
   - Useful for protocol development and debugging.
3. Network Security:
   - Helps in identifying security threats and vulnerabilities by analyzing network traffic for suspicious activities or anomalies.
4. Performance Monitoring:
   - Monitors network performance by analyzing packet flow, bandwidth usage, and network congestion.
5. Educational Purposes:
   - Used to teach network protocol behavior and analysis techniques.
   - Provides practical insight into how data is transmitted across networks.

## Comparison between Wireshark and Cisco Packet Tracer

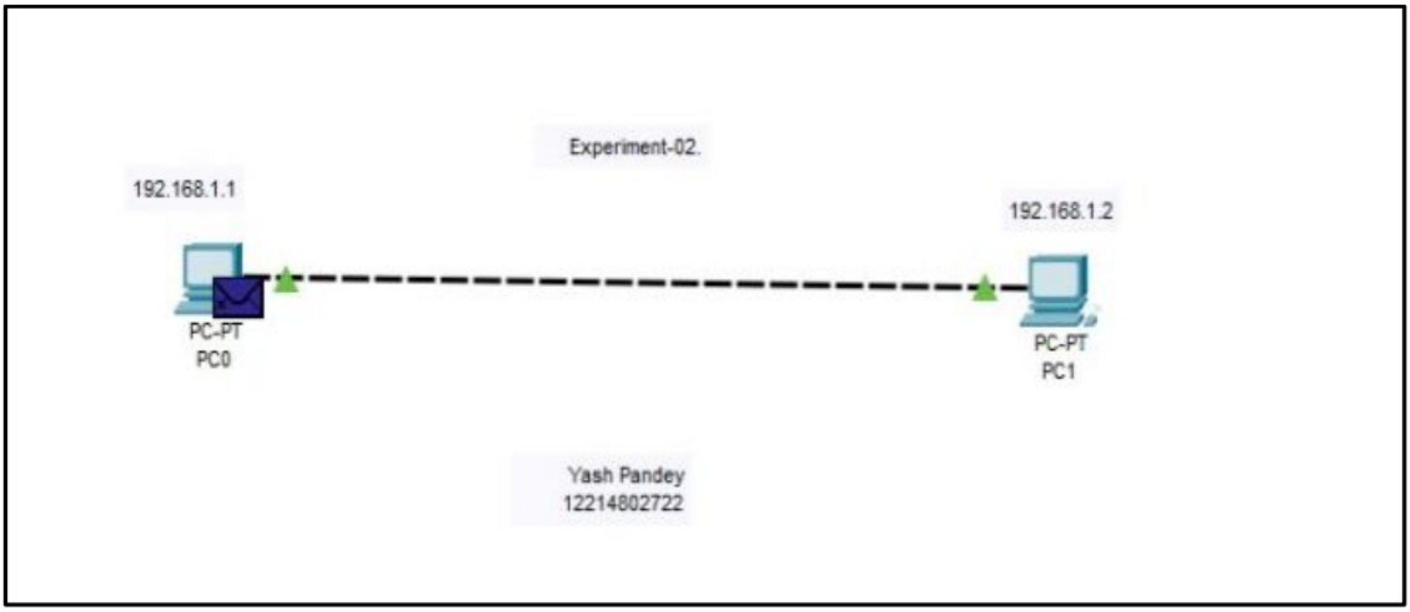
| Feature | Cisco Packet Tracer | Wireshark |
|---|---|---|
| Primary Function | Network simulation and configuration | Network packet capture and analysis |
| Use Case | Designing, configuring, and troubleshooting virtual networks | Analyzing real-time network traffic and protocols |
| Hardware/Software | Virtual network environment with simulated devices | Real network traffic capture tool |
| Protocol Simulation | Simulates network protocols and devices | Captures and decodes actual network traffic |
| Educational Focus | Networking fundamentals, device configuration | Protocol analysis, network troubleshooting |
| Configuration Practice | Extensive configuration of virtual devices | No configuration, focuses on traffic analysis |
| Real-time Analysis | Limited to simulation and emulation | Real-time packet capture and analysis |
| Cost | Free (with Cisco Networking Academy) | Free and open-source |

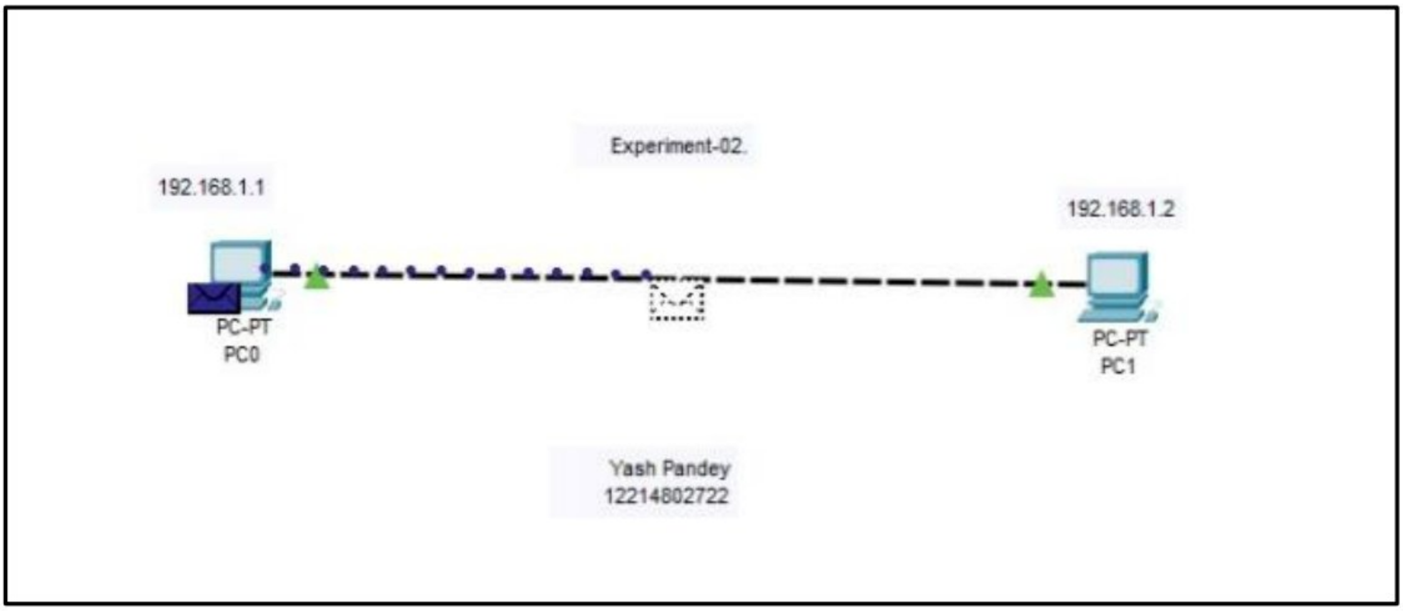
# <u>Experiment – 2.</u> *(21/08/2024)*

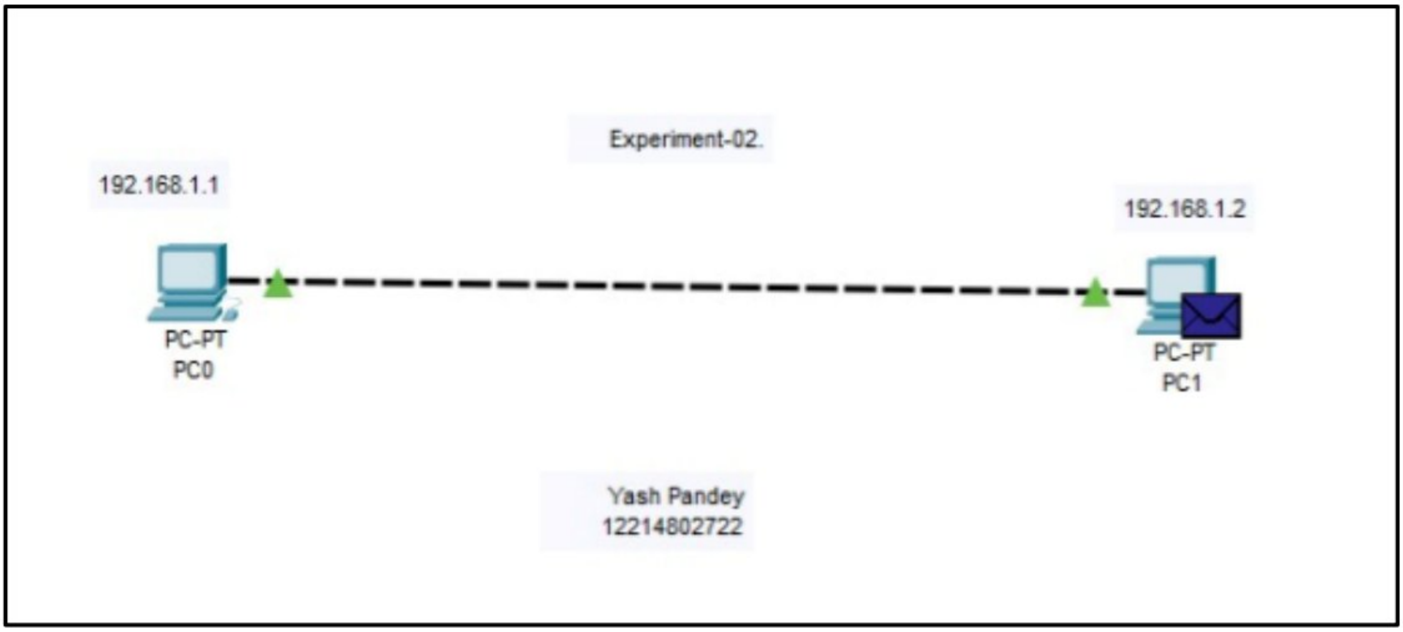
## Q :- <u>Implement Message Passing Between 2 Computers Having Consecutive IP – Addresses And Connected With cables ?</u>

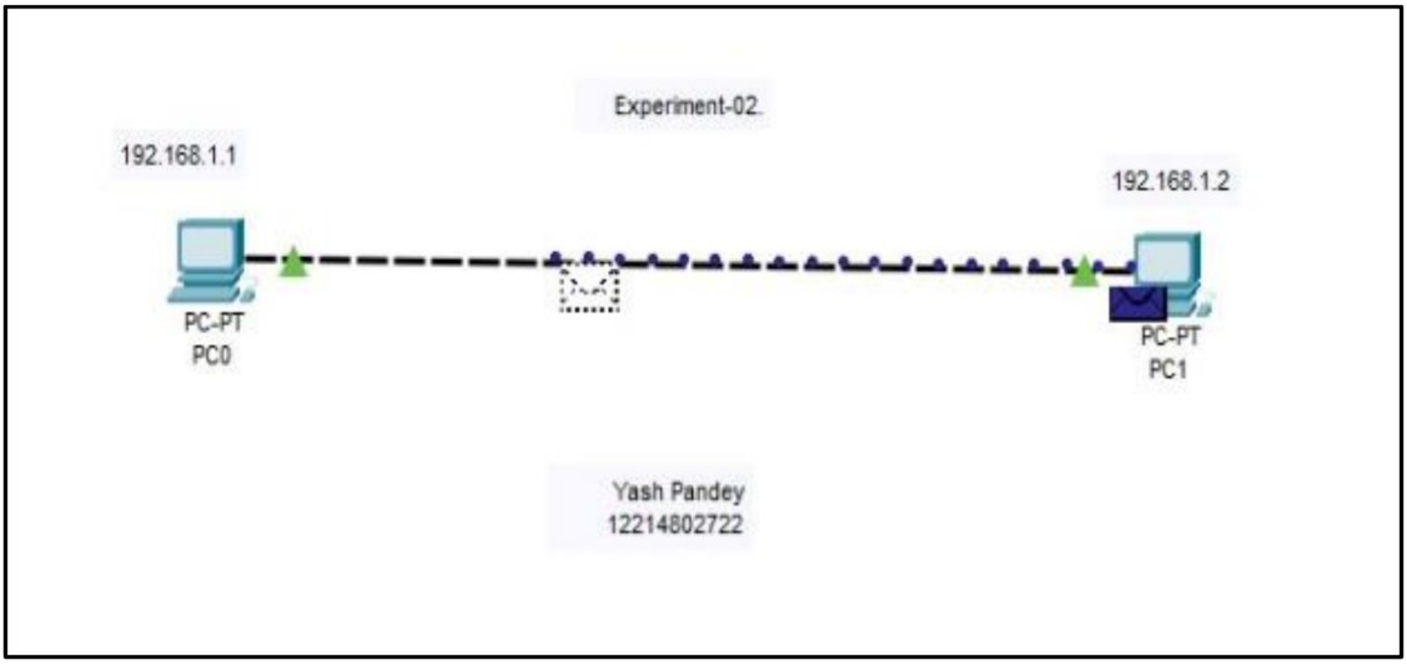
The message passing simulation illustrates how two computers communicate over a network. It shows the process of IP addressing, ARP, data encapsulation, and the transmission of data frames over the physical medium. Cisco Packet Tracer helps visualize each of these steps, allowing you to understand the flow of data between devices. We Uses two computers (PC1 and PC2) connected via a cable (typically a copper straight-through cable).

- PC1: IP Address: 192.168.1.1, Subnet Mask: 255.255.255.0.
- PC2: IP Address: 192.168.1.2, Subnet Mask: 255.255.255.0.




Experiment-02.
192.168.1.1
192.168.1.2
PC-PT
PC0
PC-PT
PC1
Yash Pandey
12214802722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Yash Pandey
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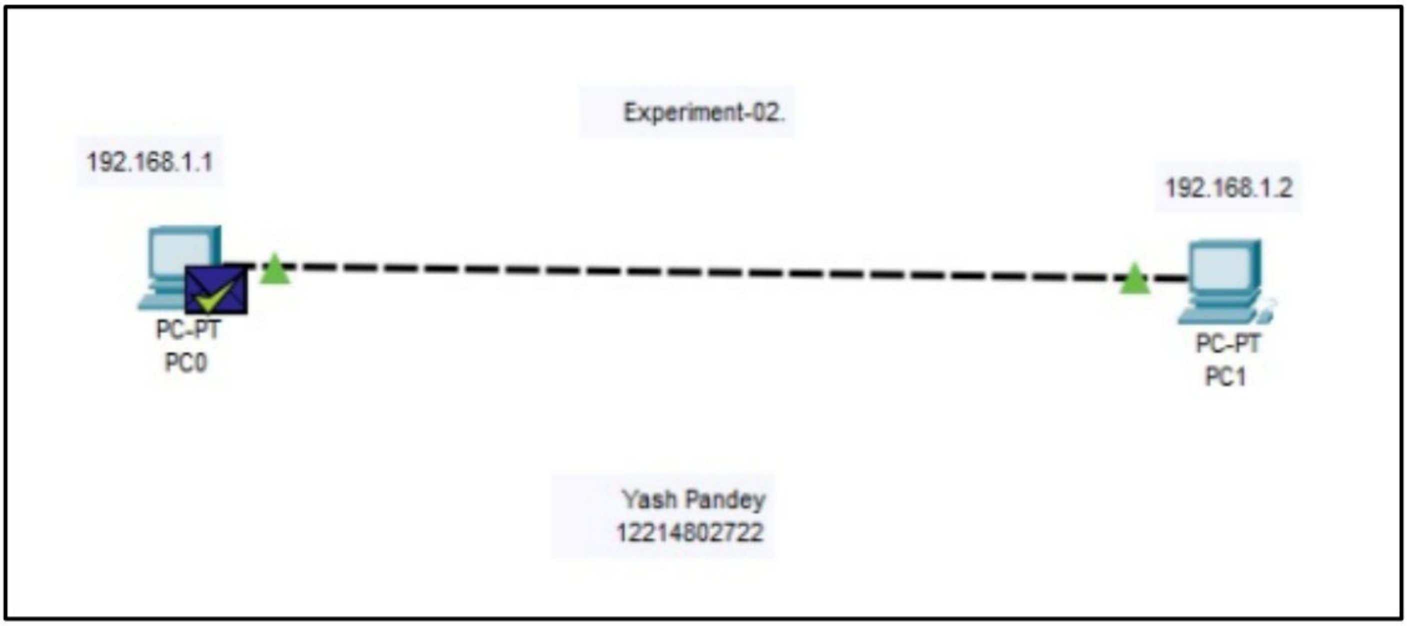

Experiment-02.
192.168.1.1
192.168.1.2
PC-PT
PC0
PC-PT
PC1
Yash Pandey
12214802722

# Experiment – 3. *(28/08/2024)*

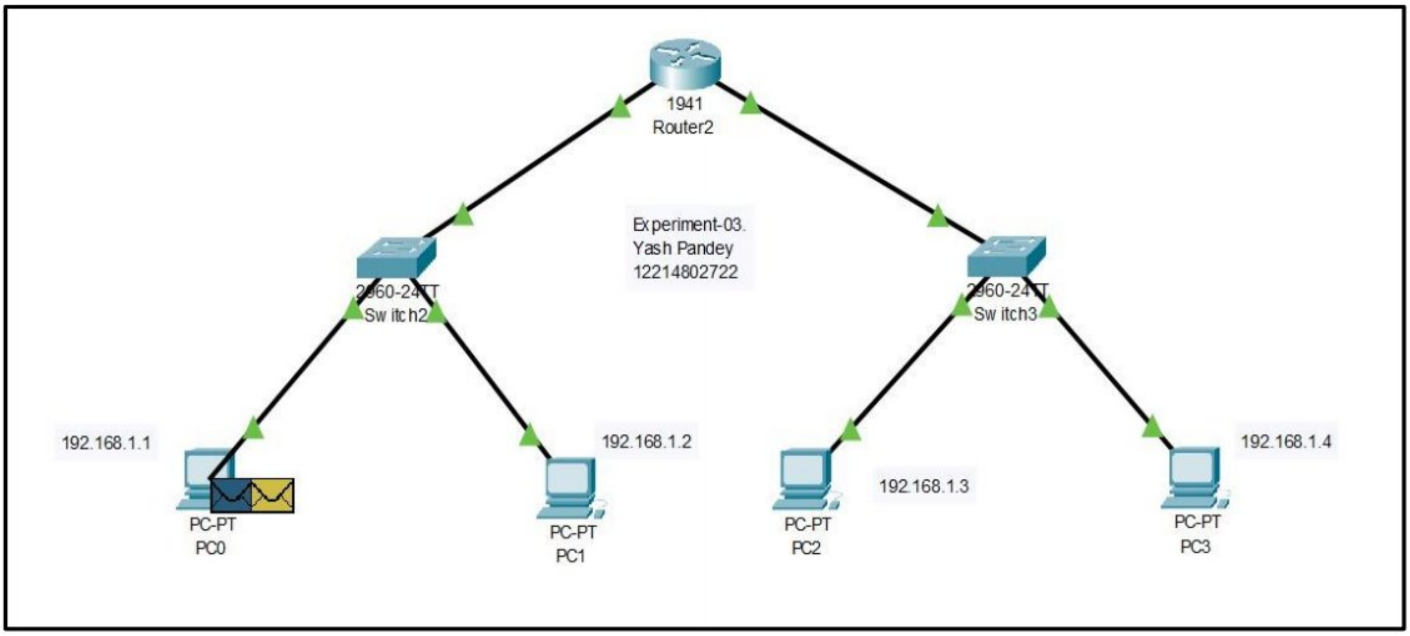
## Q :- Design An Network Having 4 PC's & 3 Switches which are Connected With cables To Implement Message Passing?

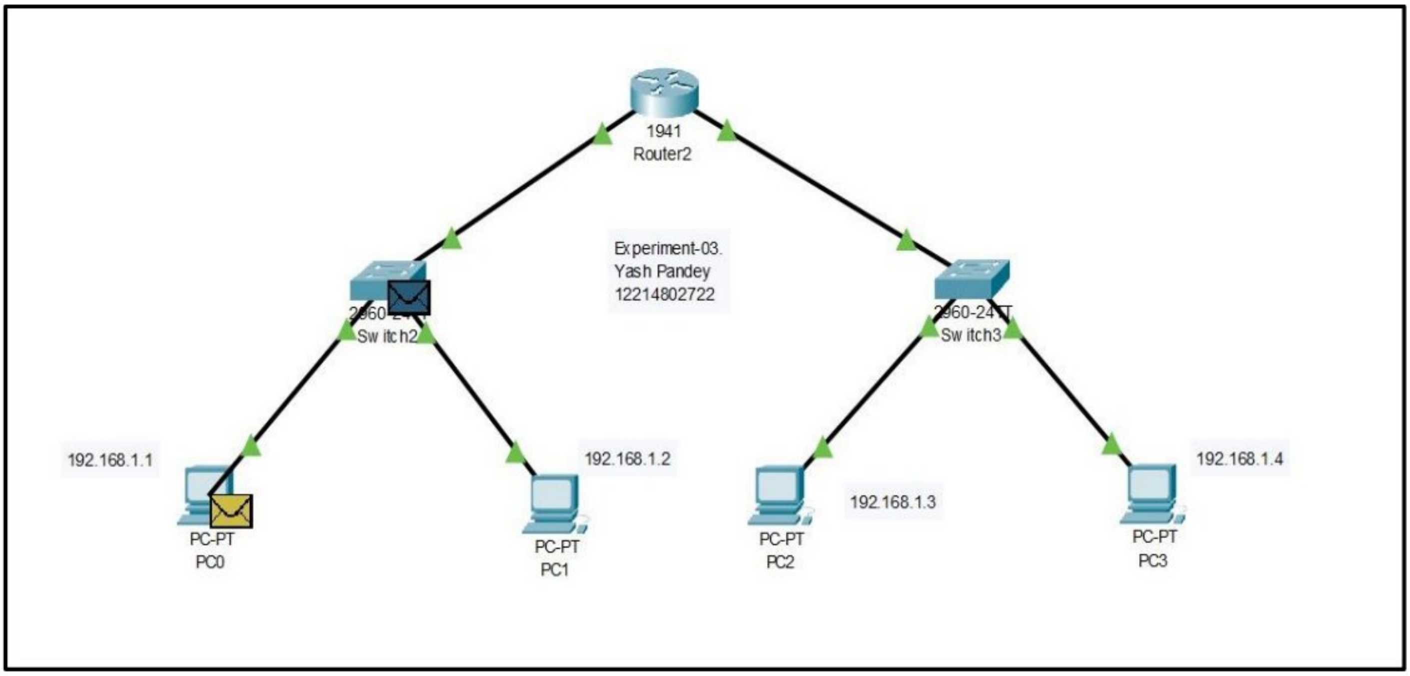
This Design In computer networks is a hierarchical structure that combines elements of both bus and star topologies. In this layout, nodes are arranged in a tree-like structure, with a central root node at the top, connected to one or more intermediate nodes, which in turn connect to other nodes further down the hierarchy. Each node in the tree topology has a single parent, except for the root node, which has none.

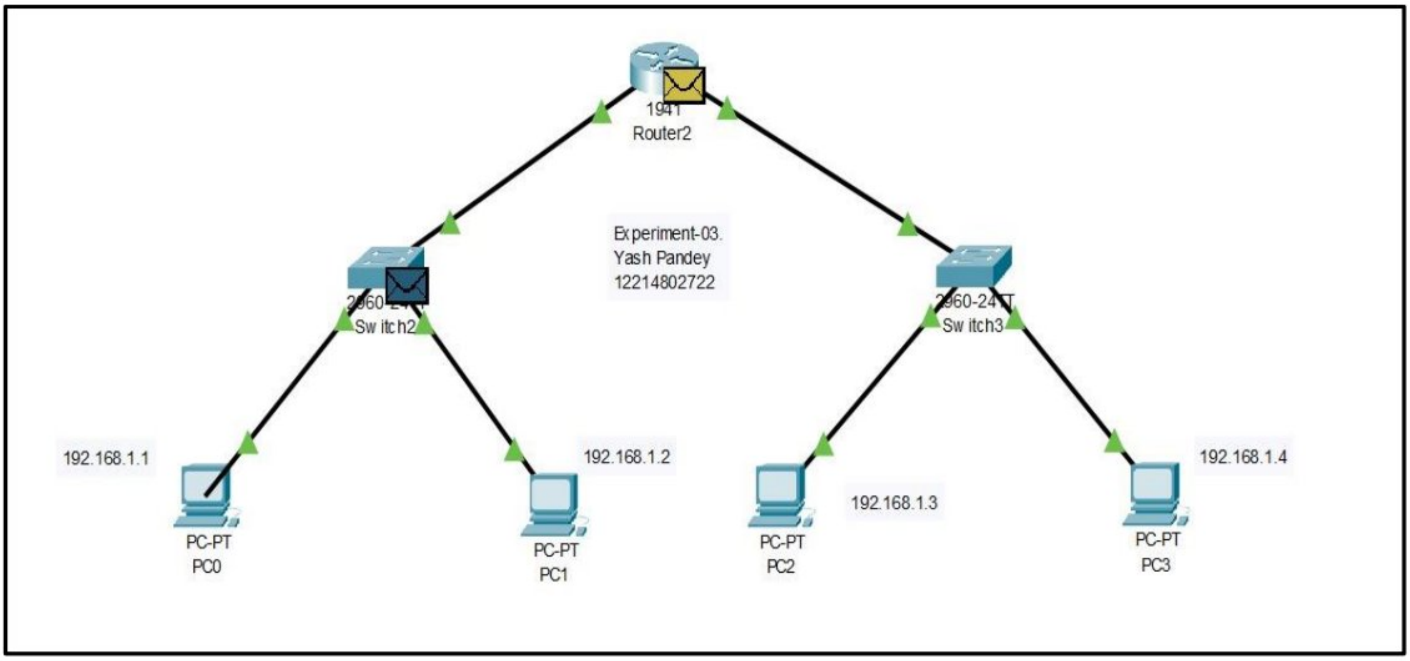
This structure is highly organized, making it easy to expand and manage, as new nodes can be added simply by connecting them to an appropriate parent node. Tree topology is commonly used in large networks like corporate or educational environments, where scalability and efficient data management are crucial. However, a drawback is that if a node higher in the hierarchy fails, it can disrupt communication for all the nodes dependent on it, making reliability a key consideration.

Tree topology is advantageous for large networks because it supports efficient and organized data management, allowing for easy troubleshooting and maintenance. The hierarchical structure ensures that data flow is well-controlled, with each level of the tree acting as a control point for traffic. This makes it suitable for environments requiring strict data segmentation, such as universities, government organizations, or large enterprises.

- PC0: IP Address: 192.168.1.1 , Subnet Mask: 255.255.255.0.
- PC1: IP Address: 192.168.1.2 , Subnet Mask: 255.255.255.0.
- PC2: IP Address: 192.168.1.3 , Subnet Mask: 255.255.255.0.
- PC3: IP Address: 192.168.1.4 , Subnet Mask: 255.255.255.0.




1941
Router2
Experiment-03.
Yash Pandey
12214802722
2960-24TT
Switch2
2960-24TT
Switch3
192.168.1.1
PC-PT
PC0
192.168.1.2
PC-PT
PC1
PC-PT
PC2
192.168.1.3
192.168.1.4
PC-PT
PC3


1941
Router2
Experiment-03.
Yash Pandey
12214802722
Switch2
2960-24TT
Switch3
192.168.1.1
PC-PT
PC0
192.168.1.2
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PC1
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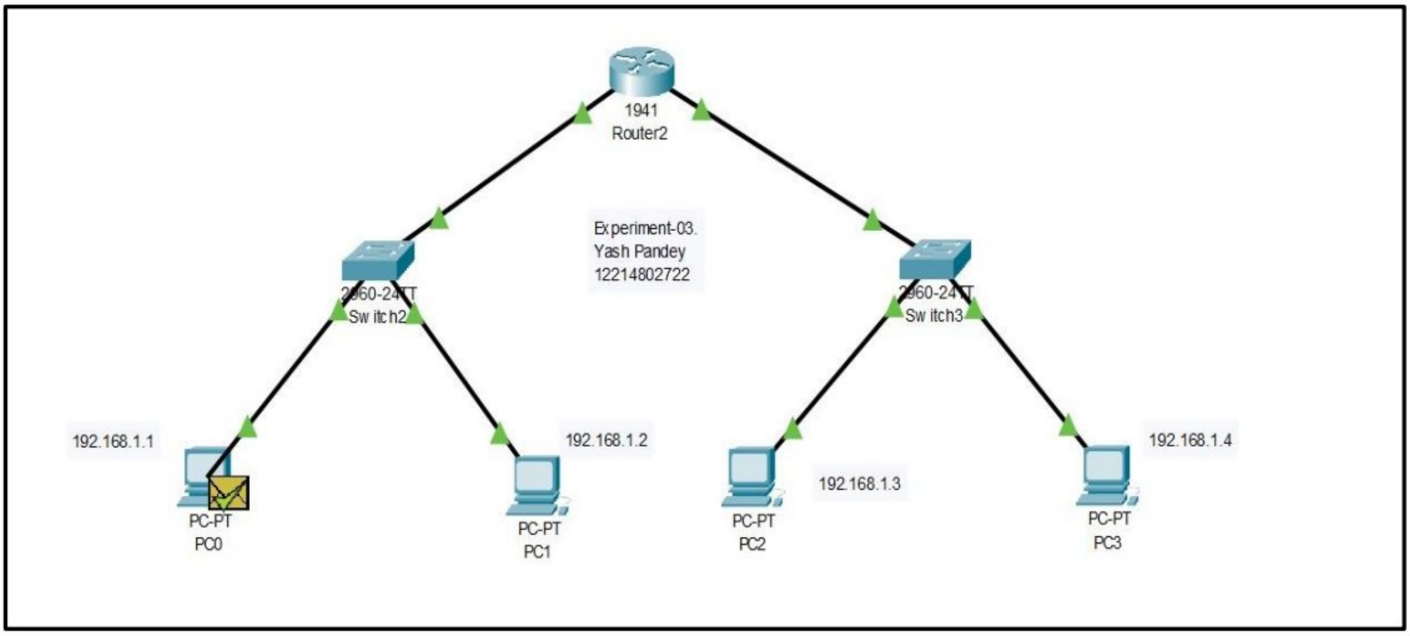
192.168.1.4
PC-PT
PC3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PC-PT
PC0
192.168.1.2
PC-PT
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PC-PT
PC2
192.168.1.3
192.168.1.4
PC-PT
PC3

## Ping Connection From Router -> Switch1 -> PC0

PC0

Physical | Config | Desktop

**Command Prompt**

```
Packet Tracer PC Command Line 1.0
PC>ping 192.168.5.132\
PC>ping 192.168.5.132

Pinging 192.168.5.132 with 32 bytes of data:

Request timed out.
Reply from 192.168.5.132: bytes=32 time=109ms TTL=127
Reply from 192.168.5.132: bytes=32 time=126ms TTL=127
Reply from 192.168.5.132: bytes=32 time=125ms TTL=127

Ping statistics for 192.168.5.132:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 109ms, Maximum = 126ms, Average = 120ms
```

## Ping Connection From Router -> Switch1 -> PC1

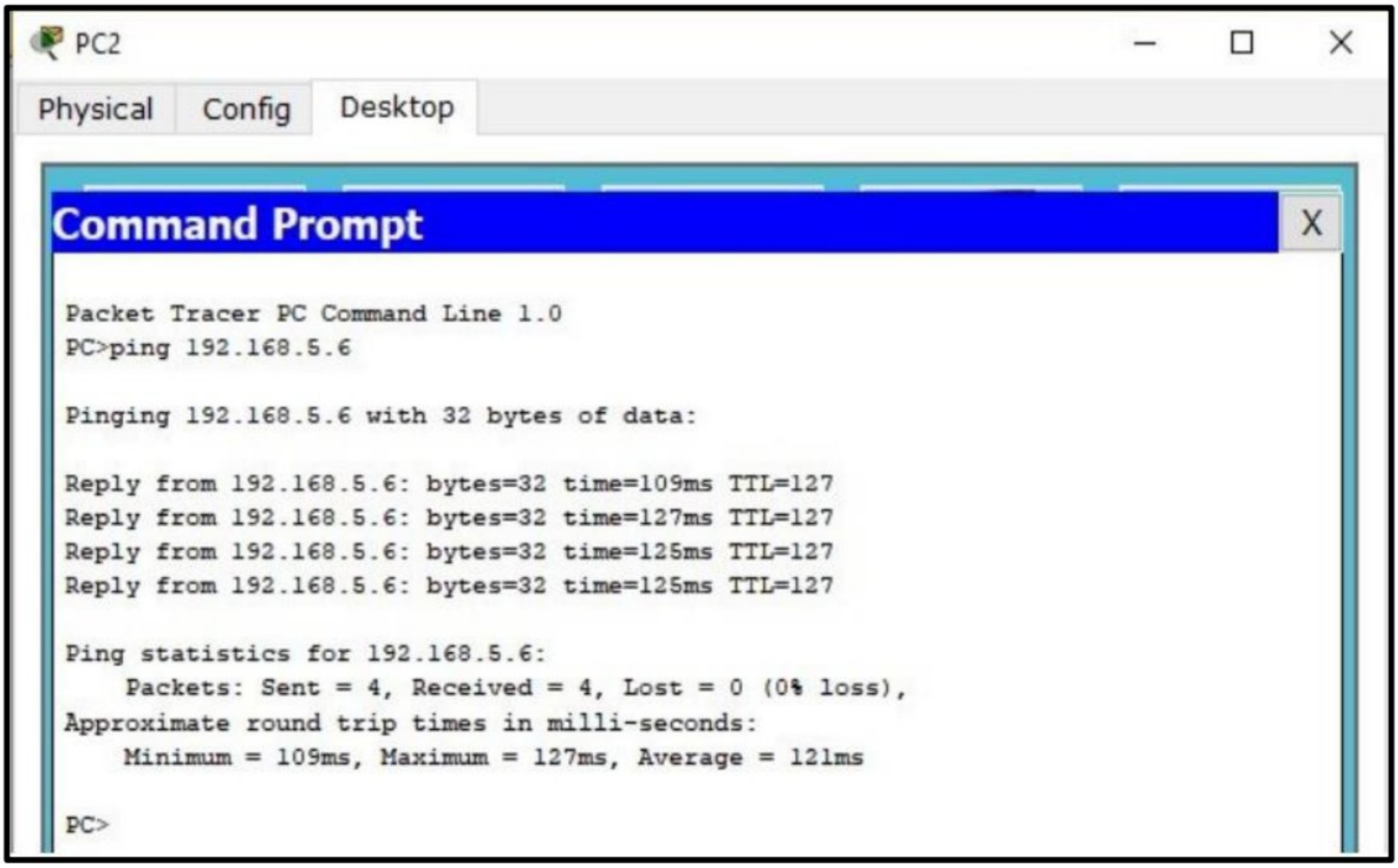
PC1

Physical | Config | Desktop

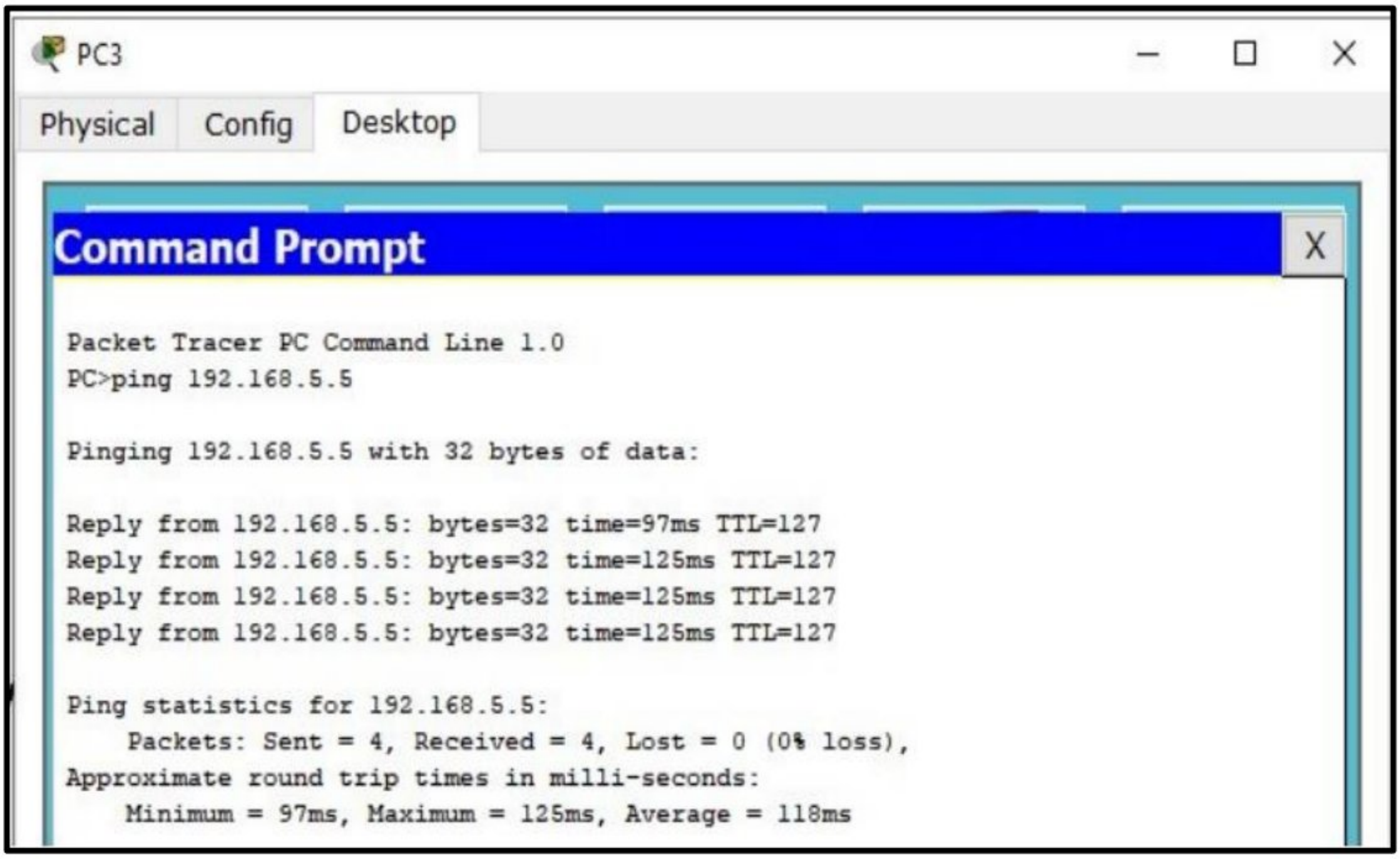
**Command Prompt**

```
Packet Tracer PC Command Line 1.0
PC>ping 192.168.5.133

Pinging 192.168.5.133 with 32 bytes of data:

Request timed out.
Reply from 192.168.5.133: bytes=32 time=109ms TTL=127
Reply from 192.168.5.133: bytes=32 time=109ms TTL=127
Reply from 192.168.5.133: bytes=32 time=125ms TTL=127

Ping statistics for 192.168.5.133:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 109ms, Maximum = 125ms, Average = 114ms

PC>
```

**Ping Connection From Router -> Switch2 -> PC2**




**Ping Connection From Router -> Switch2 -> PC3**



# Lab Exercise-1.

## Aim :- Introduction to Network Simulation Tools – Wireshark and Cisco Packet Tracer.

### Network Simulation

Network Simulation is a technique used to model and evaluate the behavior of various networks, including computer networks, telecommunications systems, and other communication infrastructures. The main goal of network simulation is to understand how different network components interact, predict overall performance, and experiment with different scenarios without the need for physical deployment.

In network simulation, virtual models of network devices such as routers, switches, servers, and links are created. These models can represent various types of equipment and communication protocols. Simulations can incorporate diverse network traffic and communication patterns to assess how the network handles different loads and conditions.

### Wireshark

Wireshark is a robust network protocol analyzer that allows users to inspect network traffic in detail. It captures packets as they travel through the network, providing comprehensive information about each packet, including headers and payloads. Wireshark is essential for network troubleshooting, performance analysis, and security auditing.

Wireshark is a powerful and widely-used network protocol analyzer that captures and inspects data packets in real-time, helping IT professionals, network administrators, and cybersecurity experts troubleshoot and analyze network issues. It provides deep visibility into the flow of network traffic, showing detailed information about protocols, data exchanges, and packet headers across a network. With its comprehensive filtering capabilities, users can isolate specific packet types, inspect protocol behavior, or monitor potential security risks.

### Current Version: 4.2.6

**Procedure to Install Wireshark:**

1. Visit the official Wireshark website using a web browser.
2. Click on the "Download" button to access the download page with various Wiresharks.
3. Download the suitable executable file and run it.
4. Proceed by clicking "Next" on the Setup and License Agreement screens.
5. Choose an installation location with adequate memory space.
6. Click "Next" to begin the installation; once it’s complete, click "Next" again.
7. Click "Finish" to complete the installation.

## Cisco Packet Tracer

Cisco Packet Tracer is a network simulation tool developed by Cisco Systems, which allows users to design, configure, and troubleshoot virtual networks. It supports a wide range of network protocols, including TCP. Packet Tracer enables the creation and simulation of network topologies using virtual devices like routers, switches, firewalls, PCs, and servers, with a user-friendly drag-and-drop interface for network design and experimentation.

**Current Version: 8.2.2**

**Procedure to Install Cisco Packet Tracer:**

1. Visit the official Netacad website using a web browser.
2. Click the "Login" button and select the appropriate login option.
3. On the next screen, click "Sign Up."
4. Enter your email, password, and other required details, then click "Register."
5. Log in using your password on the next screen.
6. Download the executable file and run it.
7. Click "Next" on the Setup and License Agreement screens.
8. Select an installation location with adequate memory space.
9. Click "Next" to start the installation; after it's done, click "Next" again.
10. Click "Finish" to complete the installation.

## Applications

### Wireshark:

- Provides detailed analysis of network packets.
- Helps detect unusual patterns and potential security breaches.

### Cisco Packet Tracer:

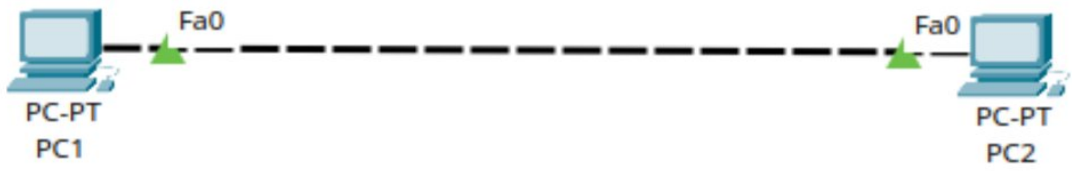
- Enables the creation of complex network topologies and configurations without physical hardware.
- Facilitates the simulation and troubleshooting of network issues to understand common problems and their solutions.

# Lab Exercise-2.

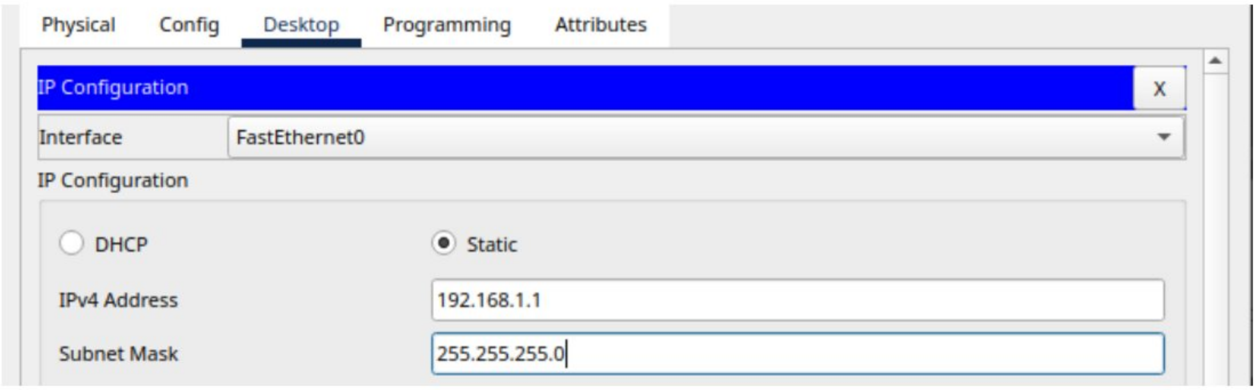
## Aim :- Implement message passing between 2 computers having consecutive IP-addresses & connected with cables.

1. **Open Cisco Packet Tracer**
   - Launch **Cisco Packet Tracer** on your computer. The main window should open with a blank workspace.
2. **Add Two PCs to the Workspace**
   - In the bottom left corner, you'll find a list of network devices.
   - Click on the **End Devices** icon (it looks like a PC).
   - Drag and drop **two PCs (PC-1 and PC-2)** into the workspace.

3. **Connect the PCs with a Copper Straight-Through Cable**
   - To connect the two PCs, click on the **Connections** icon (it looks like a lightning bolt).
   - Select the **Copper Cross-Over** cable (used to connect end devices to a switch or directly to each other if they are compatible).
   - Click on **PC-1** and connect the cable to the **FastEthernet0** port.
   - Then, click on **PC-2** and connect the other end of the cable to its **FastEthernet0** port.




4. **Assign IP Addresses to Both PCs**
   - Double-click on **PC-1** to open its configuration window.
   - Go to the **Desktop** tab and then click on **IP Configuration**.
   - Assign the following IP address details:
     - **IP Address**: 192.168.1.1
     - **Subnet Mask**: 255.255.255.0
   - Close the window after configuring PC-1.
   - Repeat the same process for **PC-2**, but assign the IP address:
   - **IP Address**: 192.168.1.2
   - **Subnet Mask**: 255.255.255.0

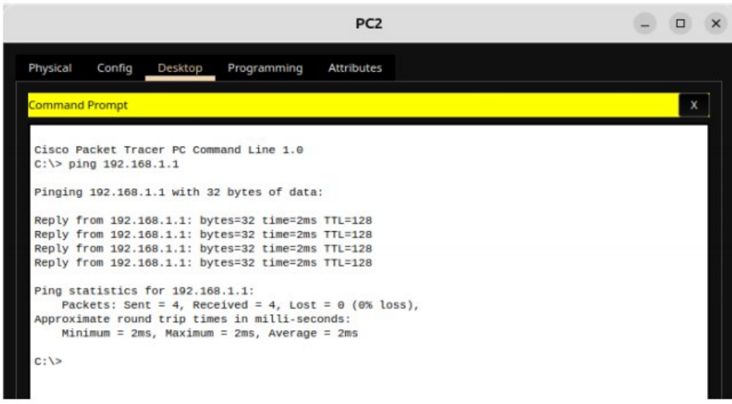


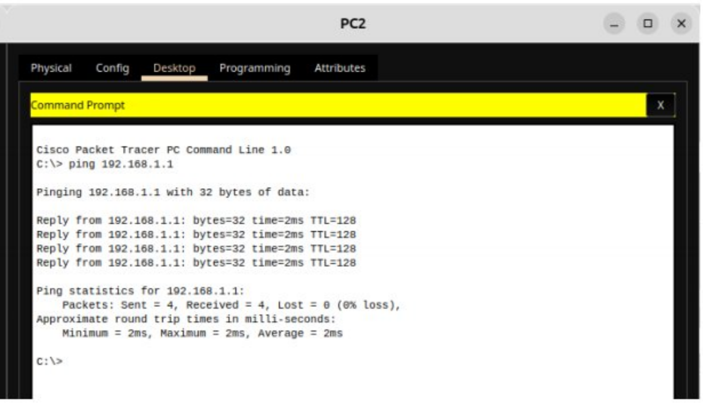

## 5. Check Connectivity Between the Two PCs

- Open the **Command Prompt** on **PC-1**:
    - Go to **PC-1**, open the **Desktop** tab, and select **Command Prompt**.
- Ping **PC-2**:
    - Type the following command in the command prompt:

**ping 192.168.1.2**

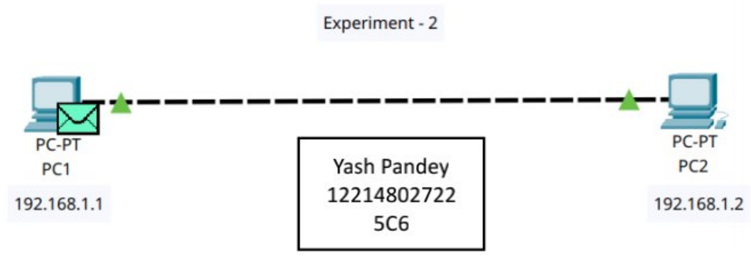
- This will check if the two computers are successfully connected.
- You should see **Reply from 192.168.1.2** if the connection is successful.

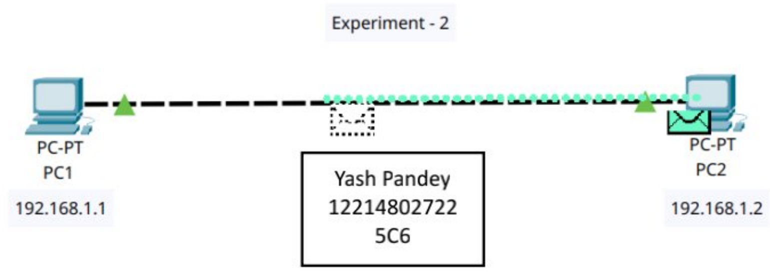


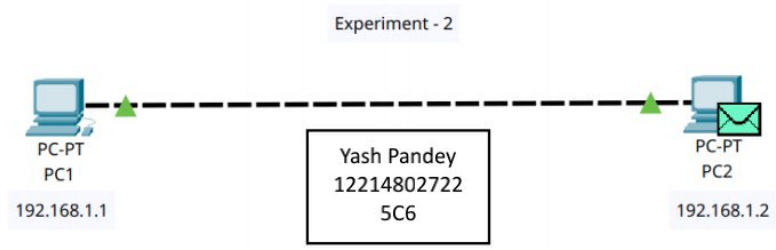



## 6. Simulate Message Passing

- Go to the **Simulation Mode** (from the bottom-right corner of the screen).
- Click on **Add Simple PDU** (looks like an envelope).
- First, click on **PC-1** and then on **PC-2** to simulate the message passing.
- The envelope icon will move from one PC to the other, indicating the message transfer.


Experiment - 2
PC-PT
PC1
192.168.1.1
PC-PT
PC2
192.168.1.2
Yash Pandey
12214802722
5C6


Experiment - 2
PC-PT
PC1
192.168.1.1
PC-PT
PC2
192.168.1.2
Yash Pandey
12214802722
5C6


Experiment - 2
PC-PT
PC1
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PC2
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Yash Pandey
12214802722
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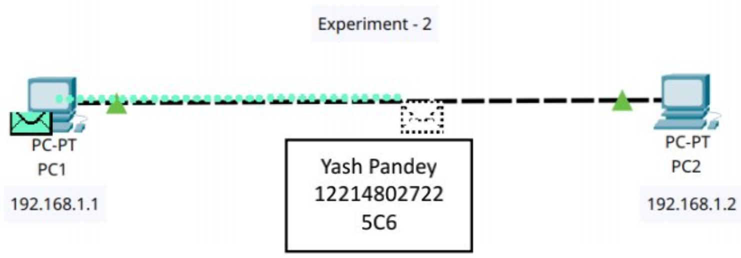
Experiment - 2
PC-PT
PC1
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PC2
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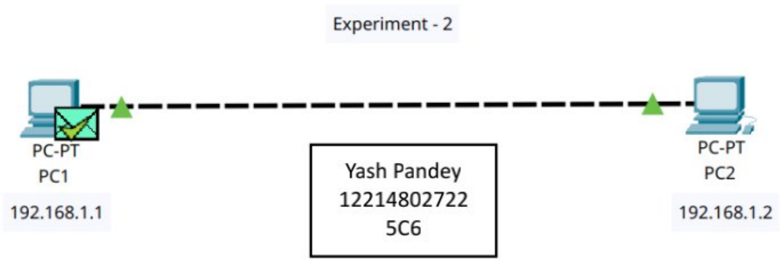
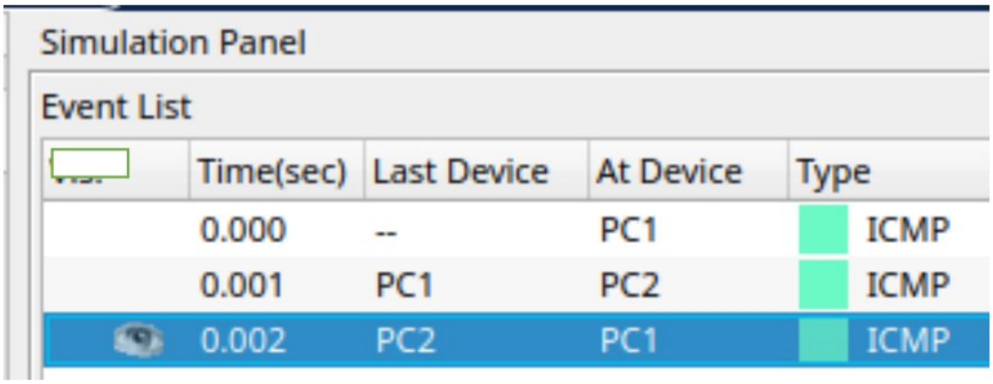
Yash Pandey
12214802722
5C6


Experiment - 2
PC-PT
PC1
192.168.1.1
PC-PT
PC2
192.168.1.2
Yash Pandey
12214802722
5C6

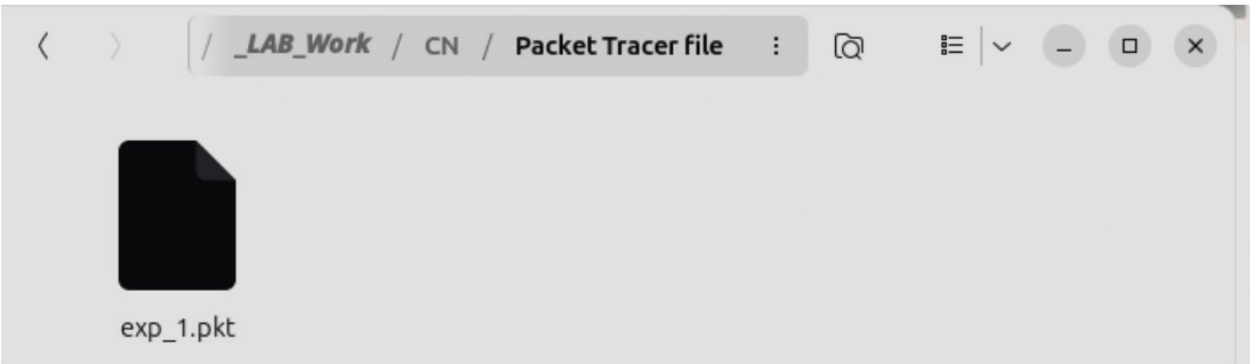
7. **Observe the Message Flow**

- In the **Event List**, you can observe how the message (packet) flows between the two computers step-by-step.
- You can speed up or slow down the simulation to better understand how the data is transferred.




8. **Save the Project**

- Once you're done, save your simulation by going to **File > Save As** and choosing a suitable location and file name.



# Lab Exercise-3.

## Aim :- Design Tree Topology Network having 4 PCs , 3 Switches which are connected with cables.

### Tree Topology Network

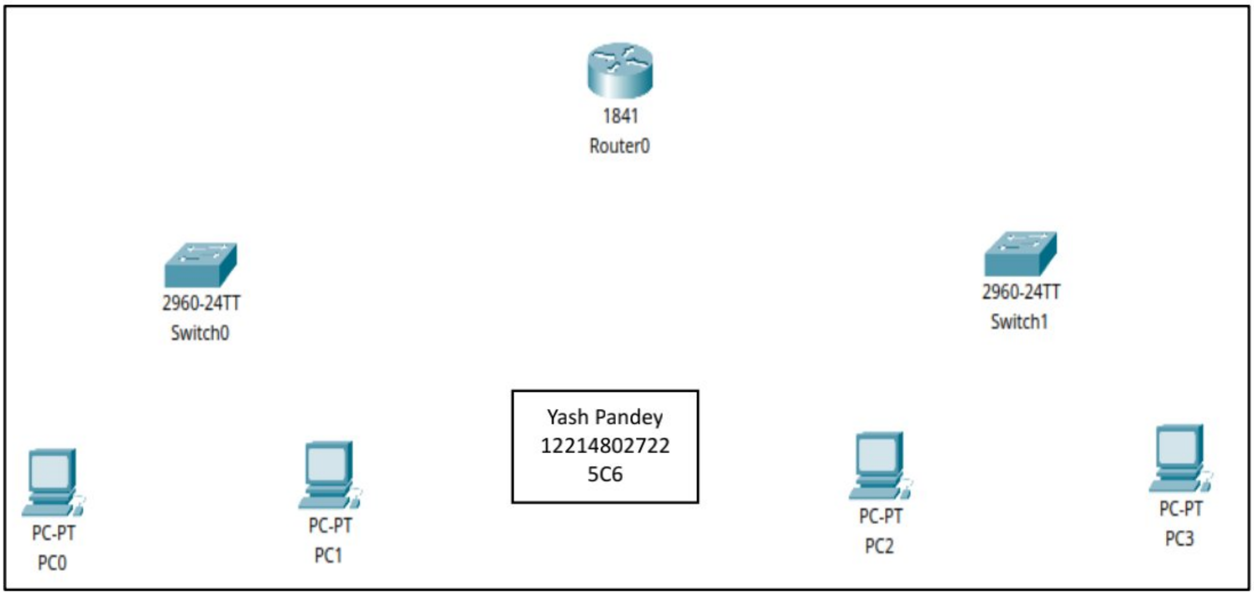
Tree topology, also called a star-bus topology, is a hybrid network topology in which multiple star networks are connected together in a hierarchical manner. At the core of the tree topology is a central node (typically a router), which branches out to other nodes, typically switches, which further connect to end devices such as computers. This type of topology provides efficient scalability, making it ideal for large networks. It enables structured communication, central control, and easy troubleshooting.

### *1. Open Cisco Packet Tracer*

- Launch **Cisco Packet Tracer** on your system and open a new workspace.

### *2. Add Network Devices*

- **Router**: From the bottom-left panel, select the **Routers** icon and drag a **1841 router** to the workspace.
- **Switches**: Select the **Switches** icon and drag two **2960 switches** onto the workspace.
- **PCs**: Select the **End Devices** icon and drag **four PCs (PC-PT)** to the workspace.

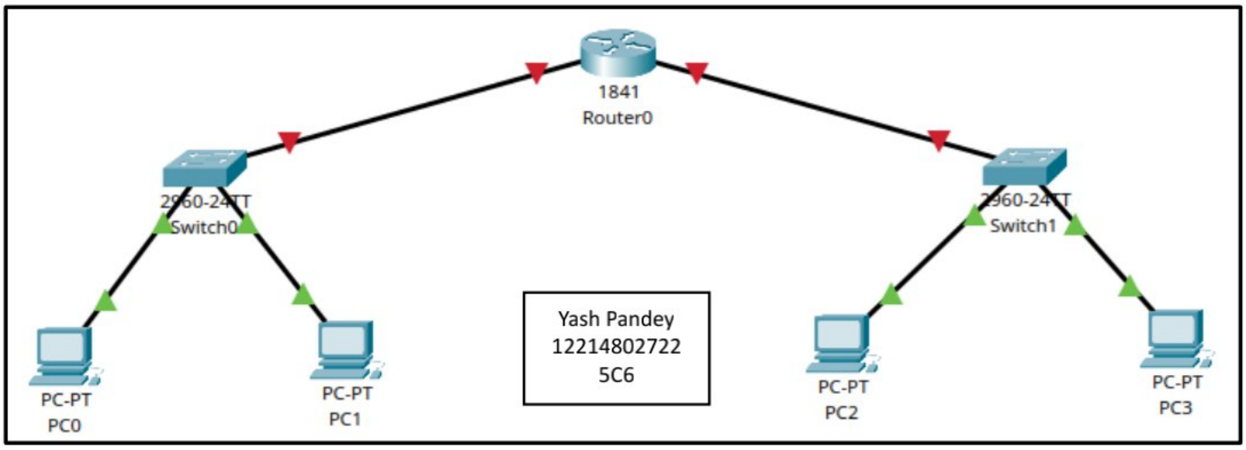


### *3. Connect the Devices with Cables*

- Use **Copper Straight-Through** cables for connections between devices.
- **Connect Router to Switch 1**:
    - Click the **Connections** icon.
    - Select **Copper Straight-Through** cable.
    - Connect the **FastEthernet0/0** port of the **Router0** to the **FastEthernet0/1** port of **Switch0** (left switch).
- **Connect Router to Switch 2**:
    - Use another **Copper Straight-Through** cable to connect the **FastEthernet0/1** port of the **Router0** to the **FastEthernet0/1** port of **Switch1** (right switch).
- **Connect Switches to PCs**:
    - For **Switch0** (left):
        - Connect **FastEthernet0/2** of the switch to **FastEthernet0** of **PC0**.
        - Connect **FastEthernet0/3** of the switch to **FastEthernet0** of **PC1**.
    - For **Switch1** (right):
        - Connect **FastEthernet0/2** of the switch to **FastEthernet0** of **PC2**.
        - Connect **FastEthernet0/3** of the switch to **FastEthernet0** of **PC3**.

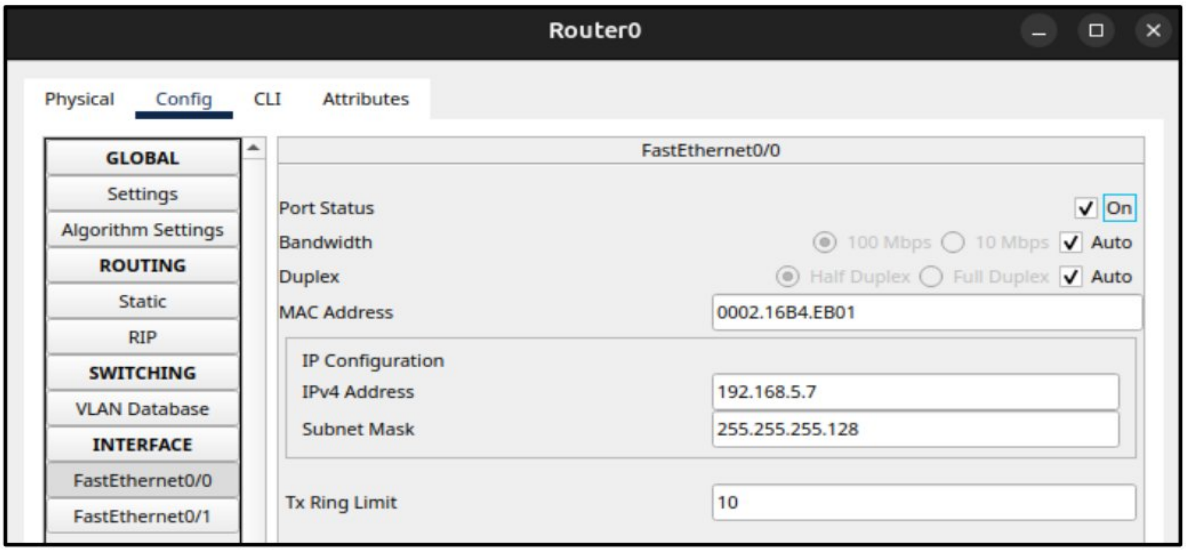
### *4. Assign IP Addresses to Each Device*

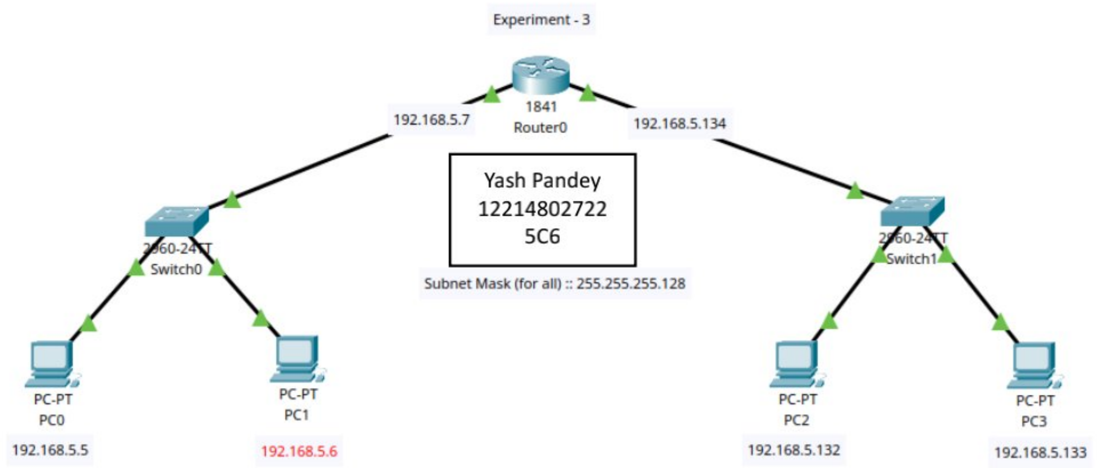
- **PC0**:
    - Go to the **Desktop** tab.
    - Click on **IP Configuration**.
    - Set **IP Address**: 192.168.5.5
    - Set **Subnet Mask**: 255.255.255.0
- **PC1**:
    - Set **IP Address**: 192.168.5.6
    - Set **Subnet Mask**: 255.255.255.0
- **PC2**:
    - Set **IP Address**: 192.168.5.132
    - Set **Subnet Mask**: 255.255.255.0
- **PC3**:
    - Set **IP Address**: 192.168.5.133
    - Set **Subnet Mask**: 255.255.255.0




### 5. *Configure Router0 (IP Address Assignment Using Config Tab)*

- After adding all the devices and cables, now configure the router's interfaces.
- **Step 1: Configure FastEthernet0/0**
  - Click on **Router0** to open its configuration window.
  - Go to the **Config** tab.
  - In the left pane, click on **FastEthernet0/0** under the **Interface** subsection.
  - Set the **IP Address** to 192.168.5.7.
  - Set the **Subnet Mask** to 255.255.255.0.
  - Turn on the **Port Status** by clicking the **ON/OFF** switch.
- **Step 2: Configure FastEthernet0/1**
  - In the same window, select **FastEthernet0/1**.
  - Set the **IP Address** to 192.168.5.134.
  - Set the **Subnet Mask** to 255.255.255.0.
  - Turn on the **Port Status** by clicking the **ON/OFF** switch.
- Once both interfaces are configured and the router is powered on, exit the window.





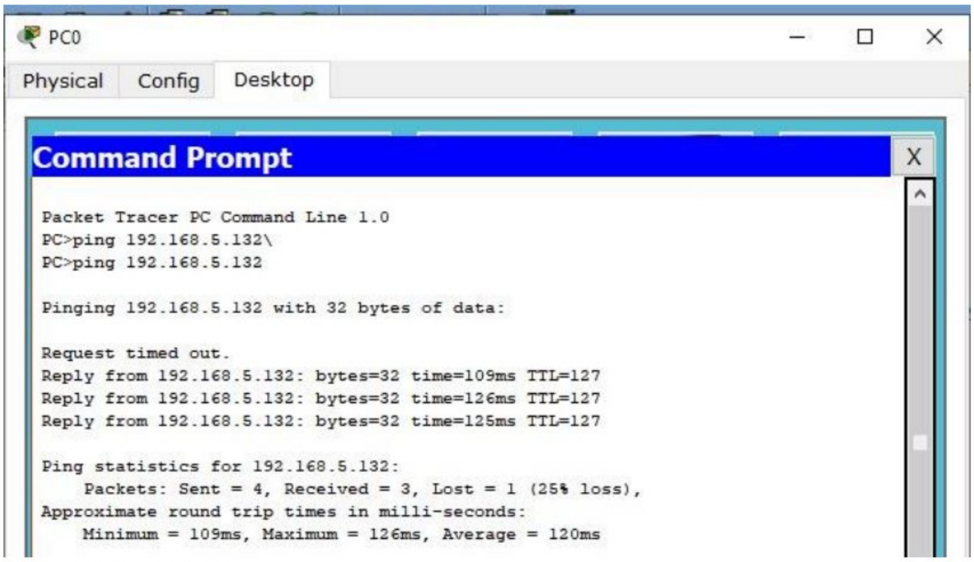

## *6.Verifying Network Connectivity*

After setting up the IP addresses and ensuring the router and switches are correctly connected, verify the connection between the PCs by pinging them in the following sequence:

- **Ping from PC0 to PC2**:
    - On **PC0**, go to the **Desktop** tab and open the **Command Prompt**.
    - Type the command:

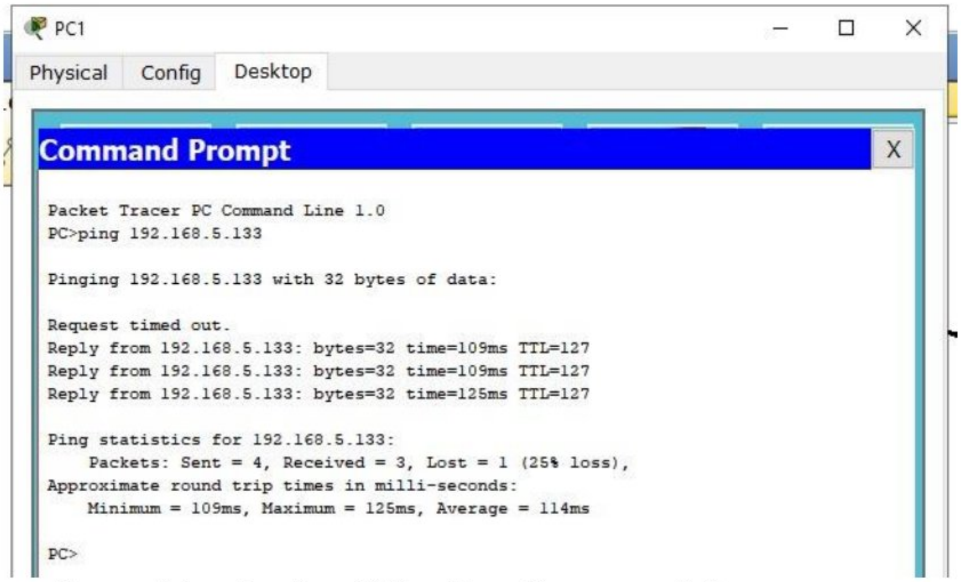
ping 192.168.5.132

- If the configuration is correct, you should see successful replies from **PC2**.

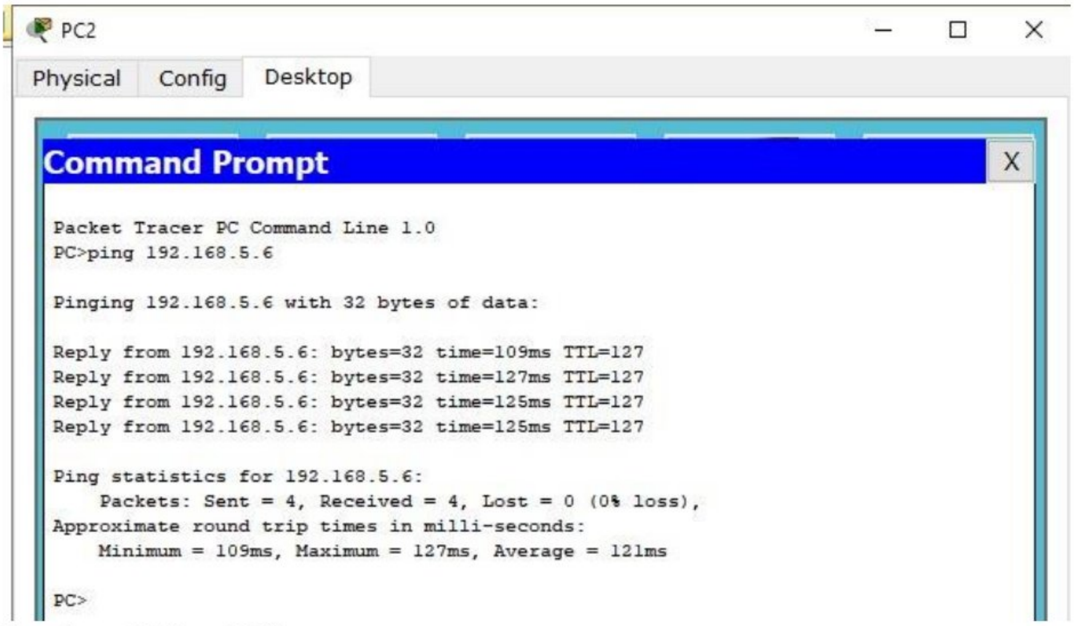



- **Ping from PC1 to PC3**:
    - On **PC1**, go to the **Command Prompt** and type: ping 192.168.5.133
    - You should receive replies from **PC3** if the connection is established.

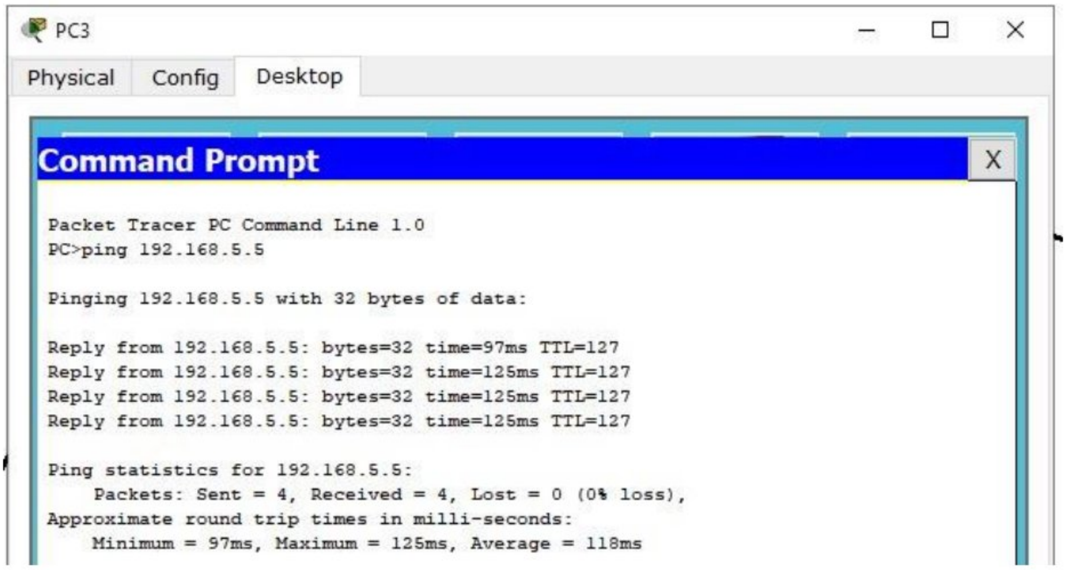
- **Ping from PC2 to PC1**:
  - On **PC2**, open the **Command Prompt** and type: ping 192.168.5.6




  - Successful replies from **PC1** will confirm connectivity.

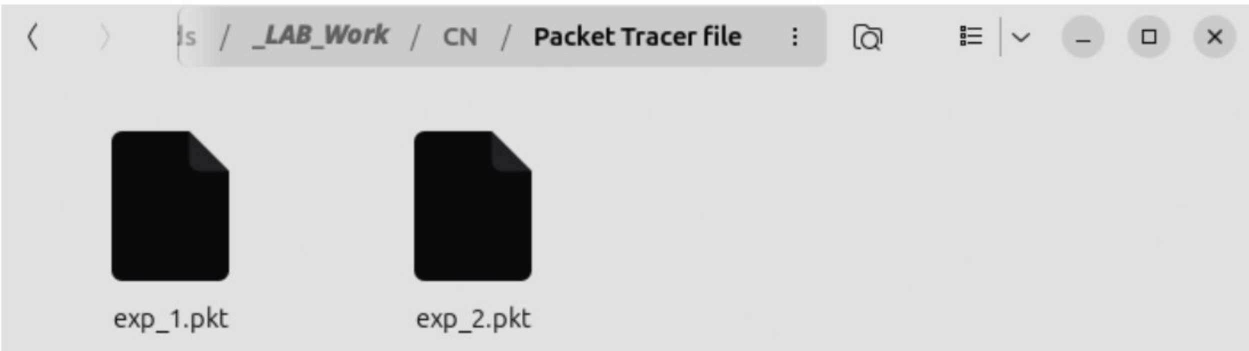



- **Ping from PC3 to PC0**:
  - On **PC3**, go to the **Command Prompt** and type: **ping 192.168.5.5**
  - If successful, the replies will indicate that **PC3** can communicate with **PC0**.




### *7. Save the Configuration*

- After successful verification of connectivity between all the PCs, save your project:
  - Go to **File** > **Save As** and choose a suitable location and file name.




## Conclusion

In this tree topology simulation, two switches (each connected to a set of PCs) communicate via a central router. The IP addresses are distributed across two subnets, and the router is configured to route traffic between the devices. This structure demonstrates the hierarchy and scalability of a tree topology, which is often used in enterprise networks to manage a large number of devices efficiently.

# Lab Exercise-4.

## Aim :- Implement the static routing using Cisco Packet Tracer

Routing is the process of path selection in any network. A computer network is made of many machines, called *nodes*, and paths or links that connect those nodes. Communication between two nodes in an interconnected network can take place through many different paths. Routing is the process of selecting the best path using some predetermined rules. Routing is important as routing creates efficiency in network communication. Network communication failures result in long wait times for website pages to load for users. It can also cause website servers to crash because they can't handle a large number of users. Routing helps minimize network failure by managing data traffic so that a network can use as much of its capacity as possible without creating congestion. Data moves along any network in the form of data packets. Each data packet has a header that contains information about the packet's intended destination. As a packet travels to its destination, several routers might route it multiple times. Routers perform this process millions of times each second with millions of packets. When a data packet arrives, the router first looks up its address in a routing table. This is similar to a passenger consulting a bus timetable to find the best bus route to their destination. Then the router forwards or moves the packet onward to the next point in the network.

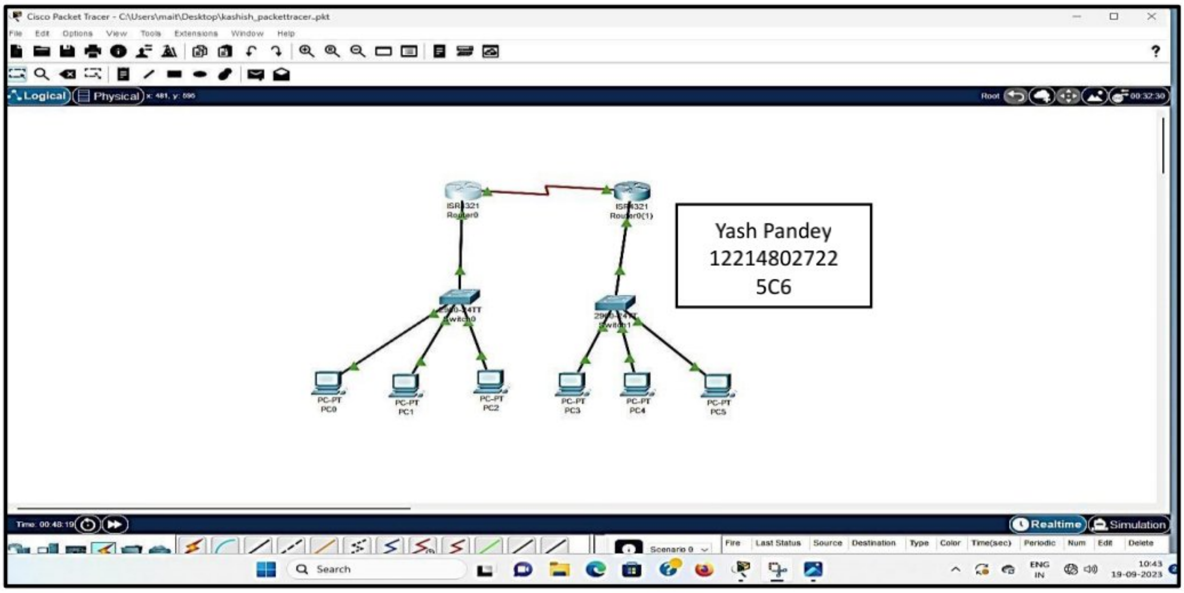
### Types of routing

**Static routing:** Network administrators use static routing, or *nonadaptive routing*, to define a route when there is a single route or a preferred route for traffic to reach a destination. Static routing uses small routing tables with only one entry for each destination. It also requires less computation time than dynamic routing because each route is preconfigured. Because static routes are preconfigured, administrators must manually reconfigure routes to adapt to changes in the network when they occur. Static routes are generally used in networks where administrators don't expect any changes.

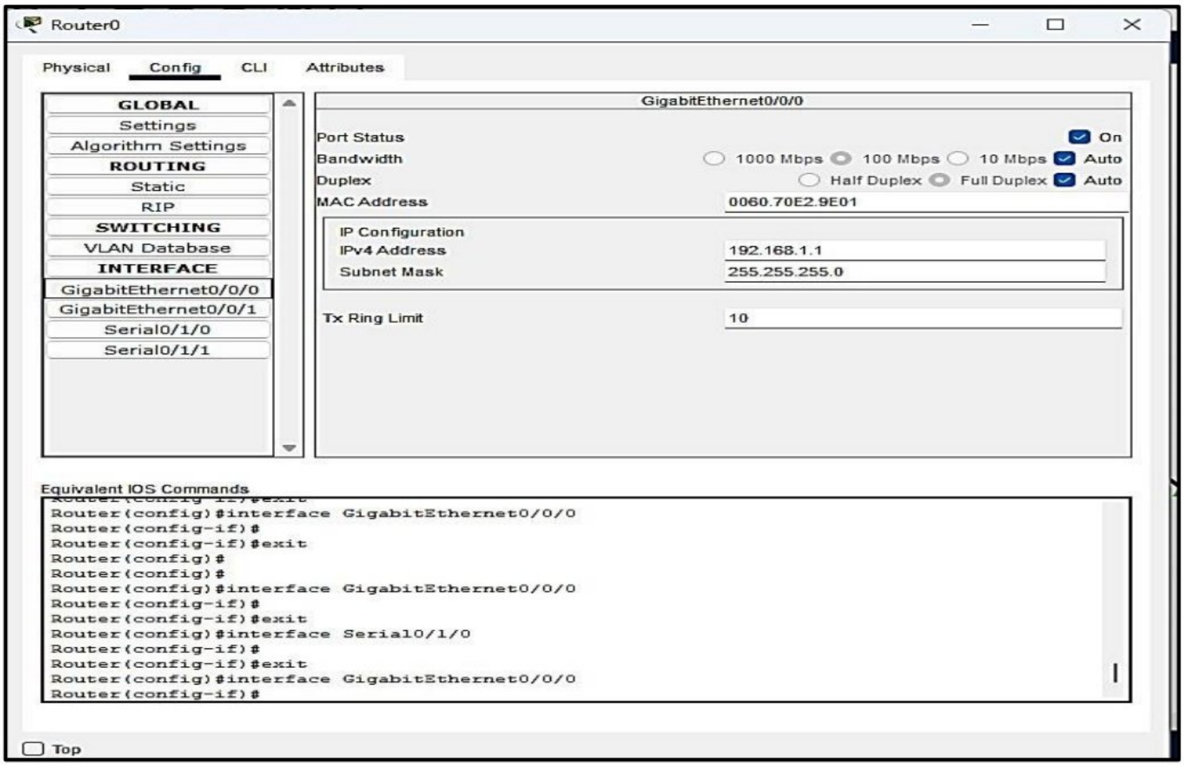
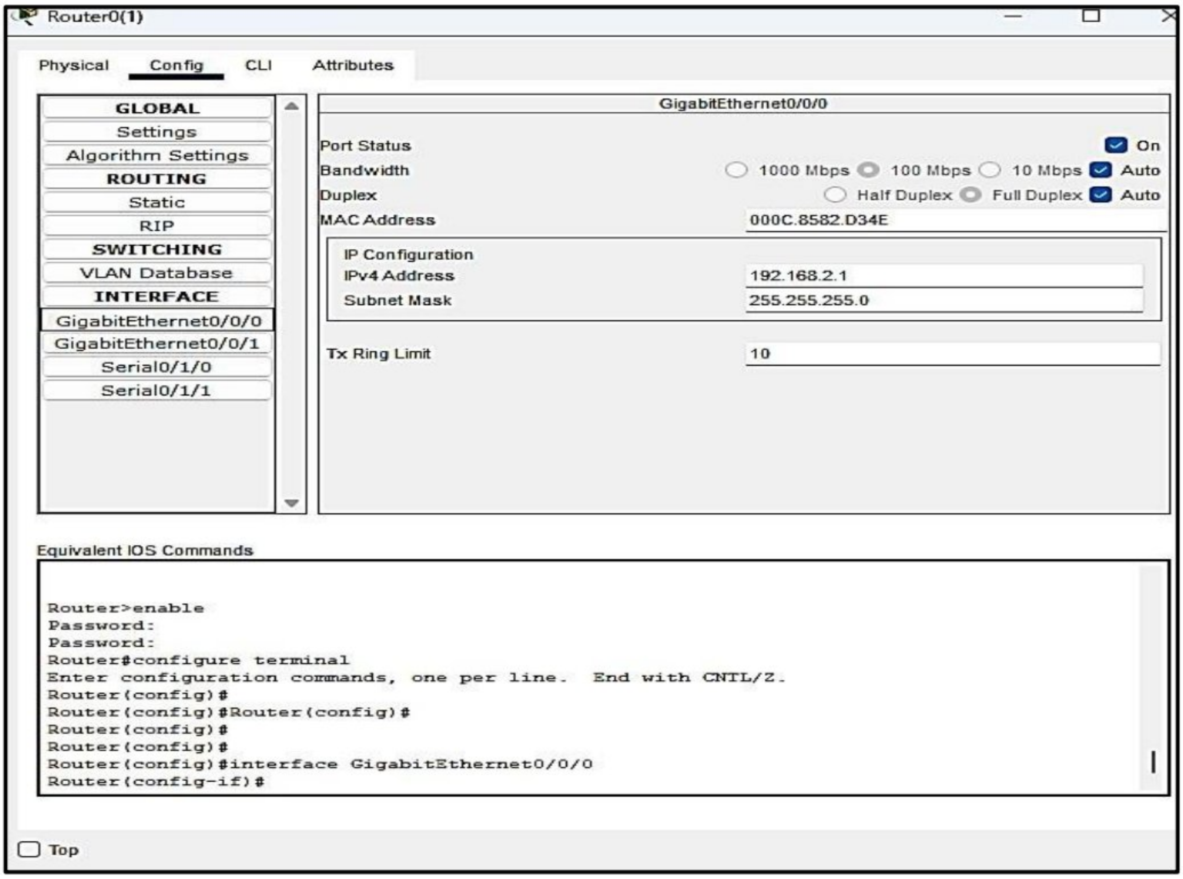
**Dynamic routing:** It is also called **adaptive routing**, is a process where a router can forward data via a different route for a given destination based on the current conditions of the communication circuits within a system. The term is most commonly associated with data networking to describe the capability of a network to 'route around' damage, such as loss of a node or a connection between nodes, as long as other path choices are available. Dynamic routing allows as many routes as possible to remain valid in response to the change.

## Implementation of Static Routing using Cisco Packet Tracer

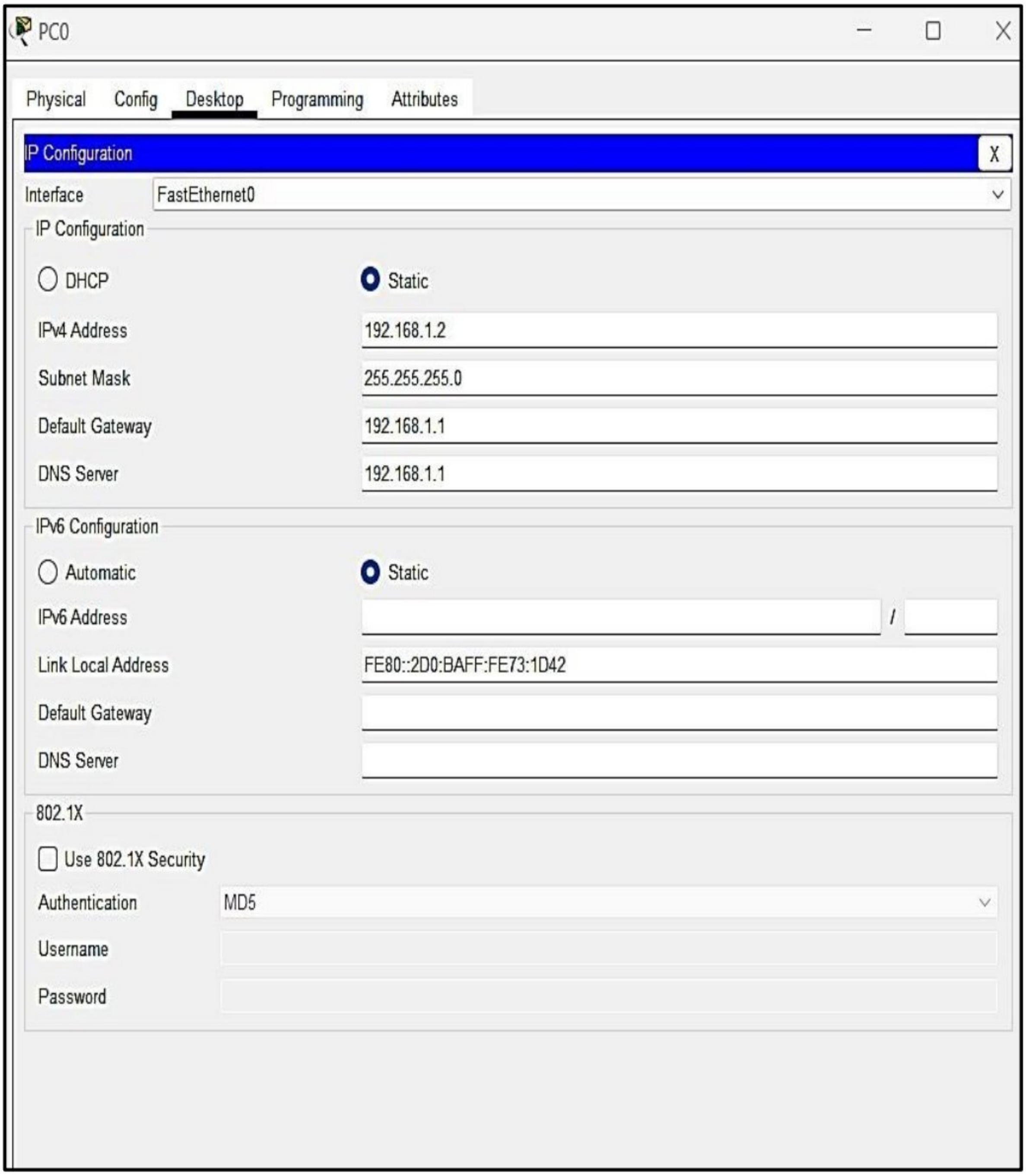
Design a network as shown below in which we built 3 different networks with two routers connected to each other. Both routers each connected to one switch which are further connected to nodes.



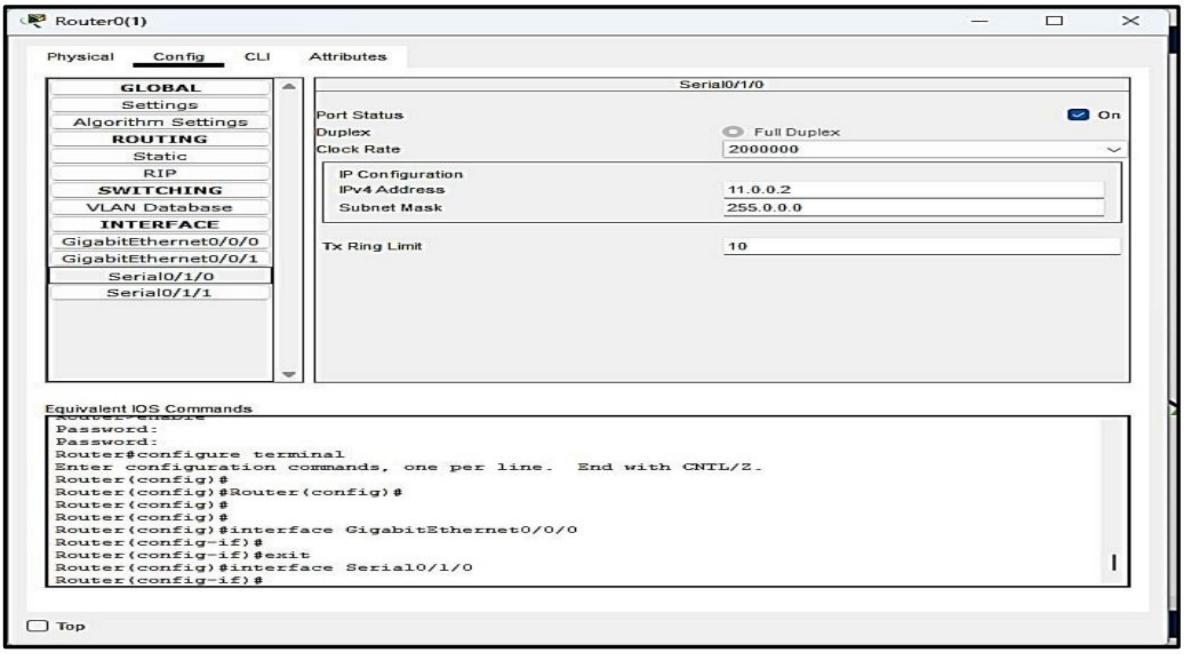

**Configure both the routers using "Gigabit Ethernet 0/0/0" to connect them to the switches.**

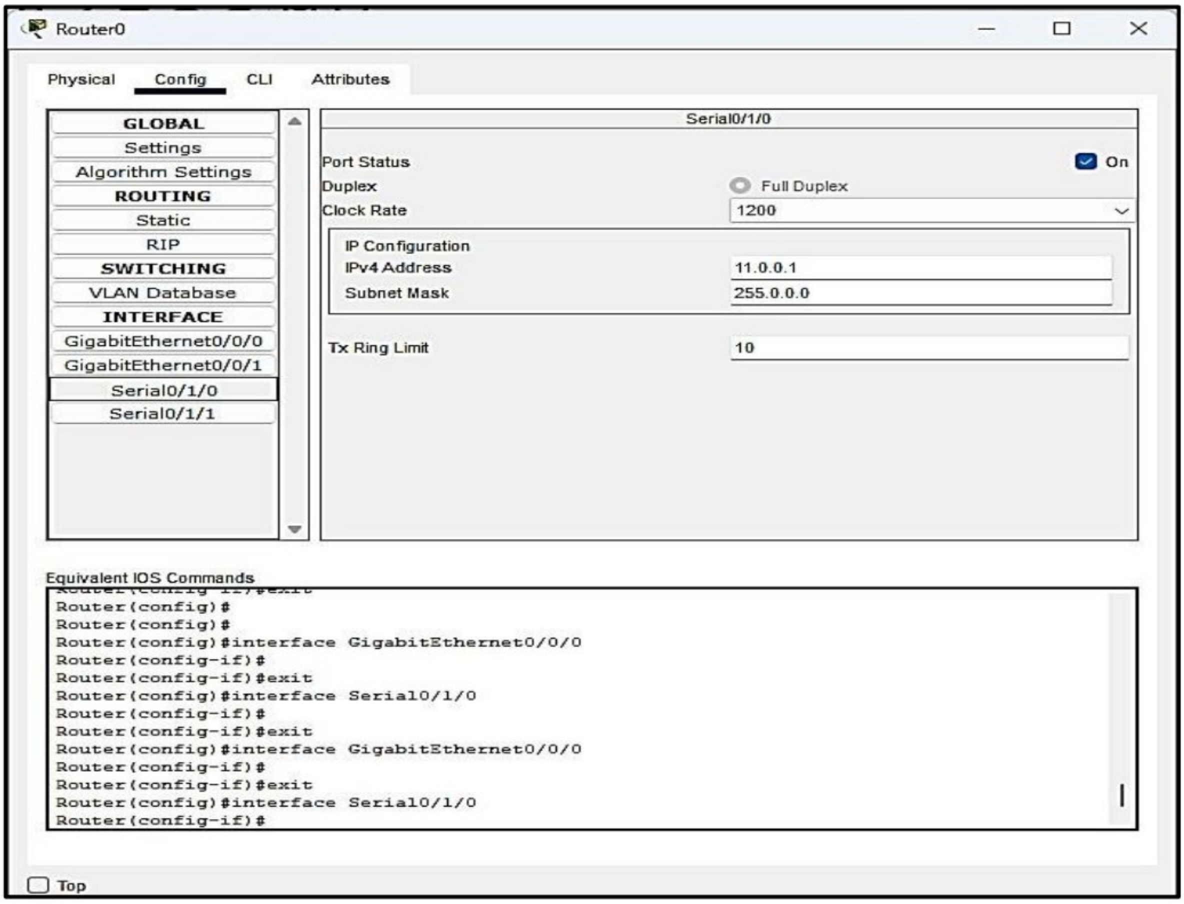

Router0
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0/0
GigabitEthernet0/0/1
Serial0/1/0
Serial0/1/1
GigabitEthernet0/0/0
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0060.70E2.9E01
IP Configuration
IPv4 Address
192.168.1.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10
Equivalent IOS Commands
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/1/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#
Top

Router0(1)
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0/0
GigabitEthernet0/0/1
Serial0/1/0
Serial0/1/1
GigabitEthernet0/0/0
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
000C.8582.D34E
IP Configuration
IPv4 Address
192.168.2.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10
Equivalent IOS Commands
Router>enable
Password:
Password:
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#Router(config)#
Router(config)#
Router(config)#
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#
Top

Then we configure all the nodes of the network as below for the PC's connected to the Switch0 we gave the IP's 192.168.1.2, 192.168.1.3 respectively and for the PCs connected to Switch1 we gave the IPs 192.168.2.2, 192.168.2.3 respectively.

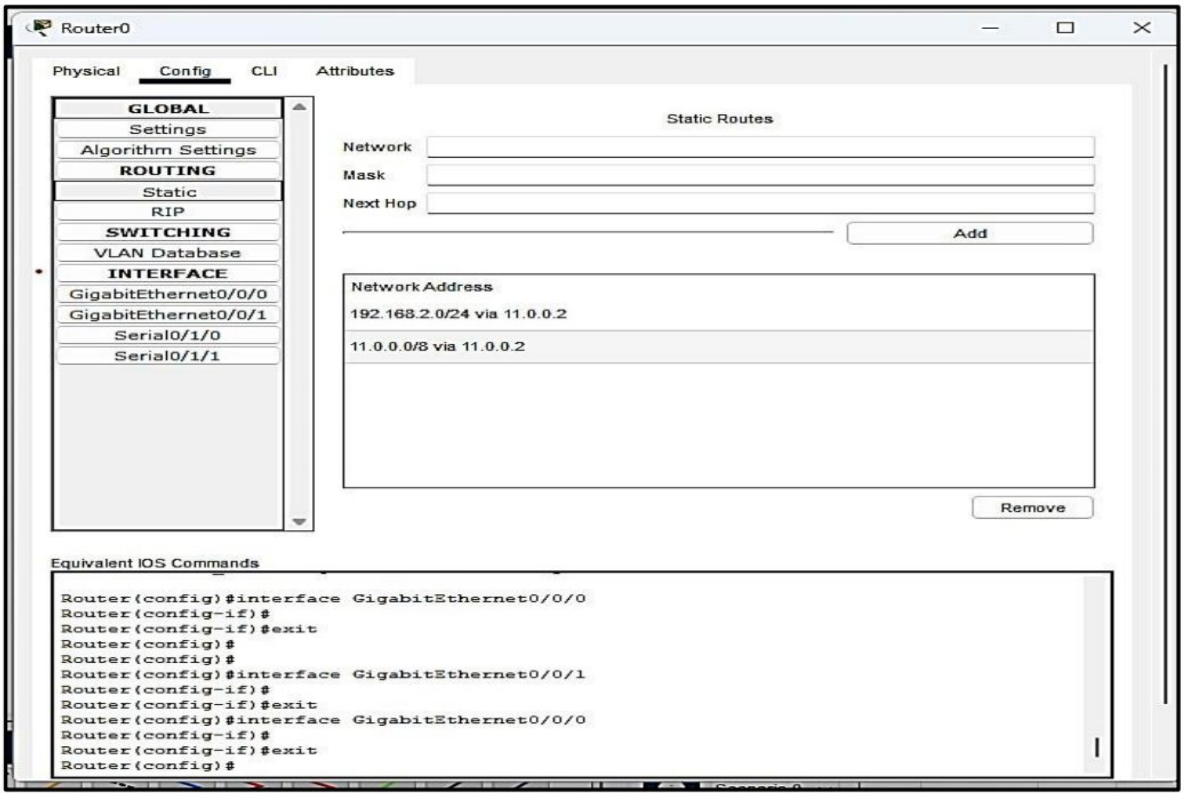


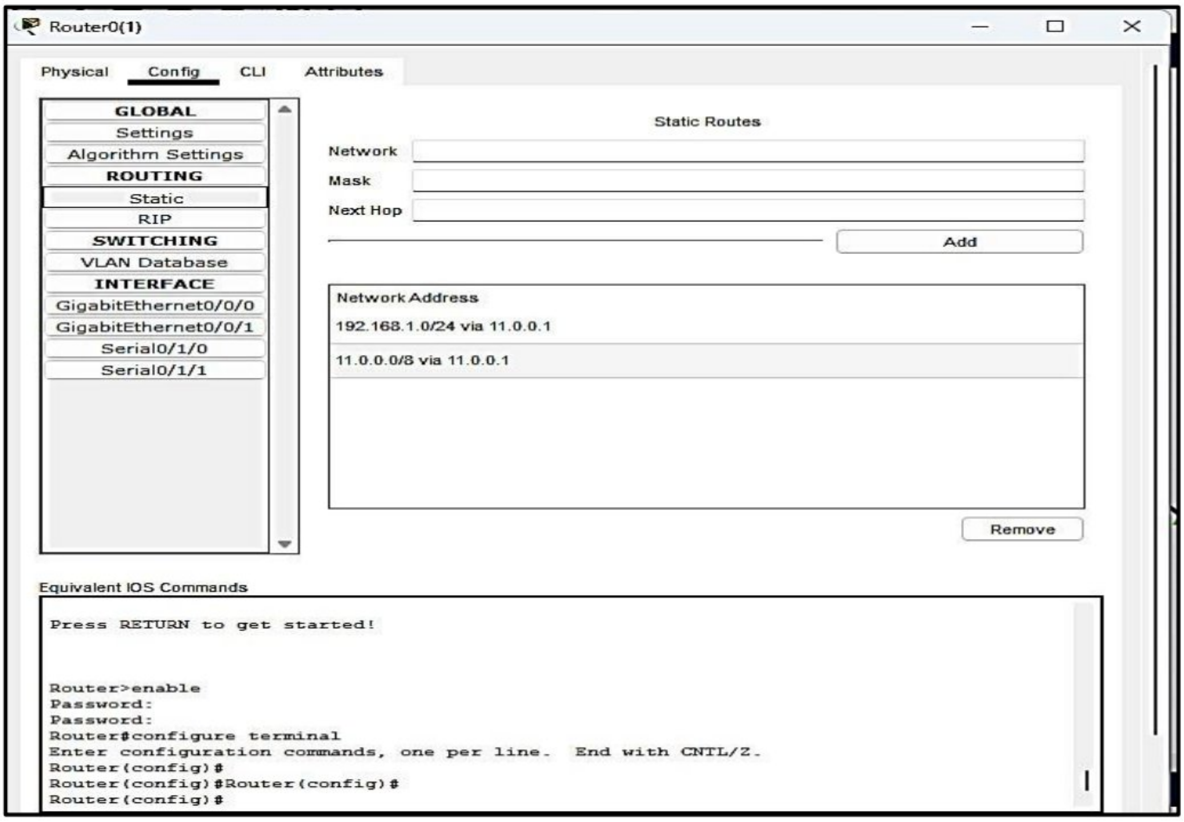
After configuring the PC's we configure the Serial port of both the router to establish a network between the other two networks.

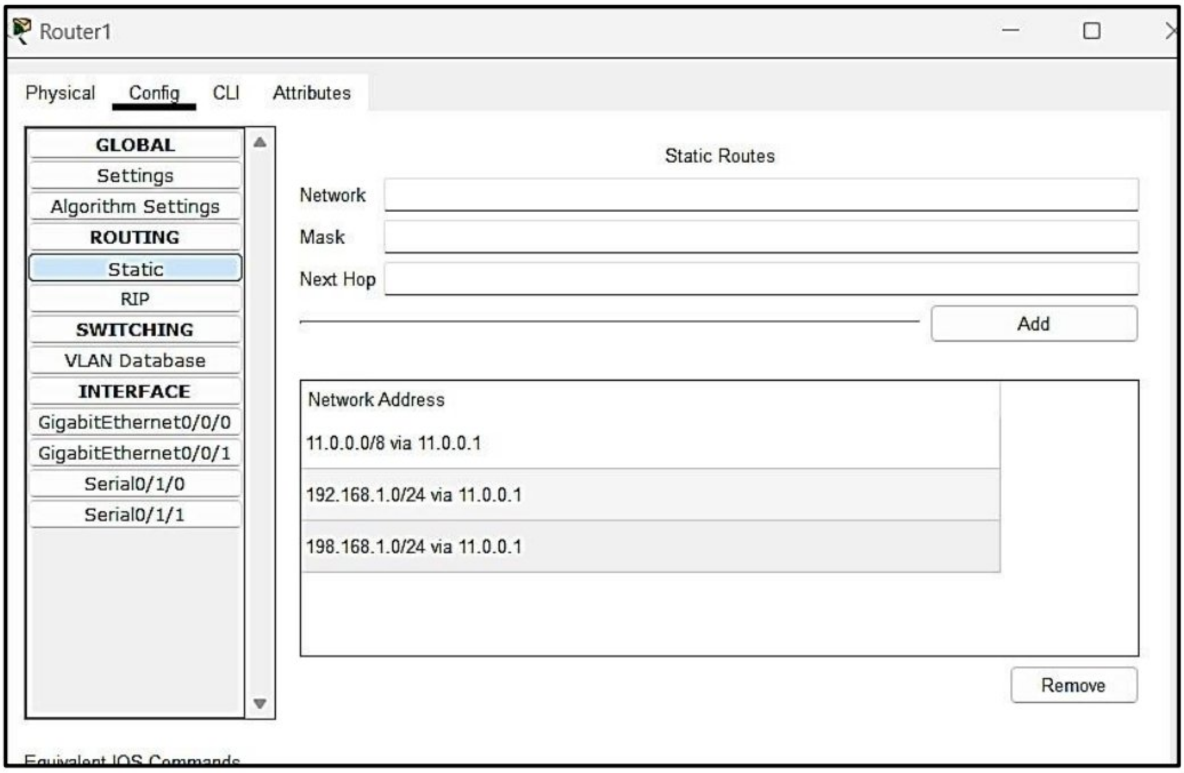


Now we route the packets coming from the one network to the Router to the end device of the other network using static routing.




Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0/0
GigabitEthernet0/0/1
Serial0/1/0
Serial0/1/1
Static Routes
Network
Mask
Next Hop
Add
Network Address
11.0.0.0/8 via 11.0.0.1
192.168.1.0/24 via 11.0.0.1
198.168.1.0/24 via 11.0.0.1
Remove

# Lab Exercise-5.

## Aim :- Implement the DHCP Protocol using Cisco Packet Tracer

The DHCP stands for **Dynamic Host Configuration Protocol**. It is a network management protocol used on IP networks. A DHCP server is used to assign an IP address and other configurations to the connected devices on the network to communicate with others. DHCP server allows a system to request IP addresses and other networking parameters automatically from the internet service providers. It reduces the network administrator's work. When the DHCP server is absent, the IP address for a computer or other device needs to be manually assigned. But, later, these devices cannot be connected outside the local subnet.

A DHCP can be implemented on home networks to wide area networks and region ISP networks. Most of the home networks receive a globally unique id within the ISP ( Internet Service Provider) networks. For the local networks, DHCP assigns a local IP address to every connected device within the network. We can also use the routers as a DHCP server. There are millions of devices in the world, and each individual device needs a unique IP address. The TCP/IP protocol supports a built-in DHCP protocol. So, it automatically assigns a unique IP address to each connected device and keeps tabs of them.

### Almost all IP addresses are dynamic :-

- **DHCP Server:** A device or software that runs the DHCP service, responsible for managing the pool of available IP addresses and assigning them to clients.
- **DHCP Client:** A device that requests and receives IP configuration information from a DHCP server.
- **DHCP Lease:** The period for which an IP address is assigned to a client. The client must renew the lease before it expires to continue using the IP address.

## Advantages of DHCP server

The DHCP server simplified device management. In addition to device management, it provides the following benefits:

- **Accurate IP configuration:** It provides an easy way to troubleshoot and use the DHCP server. Thus, it minimizes the risk of invalid IP configuration parameters.
- **Reduced IP address conflicts:** The connected devices have a unique IP address. The DHCP server ensures the one IP address is used only once. Thus, it reduces the IP address conflict.
- **Automation of IP address administration:** The DHCP server automatically assigns an IP address to each device. In the absence of a DHCP server, we have to manually assign the IP address. It keeps track of every IP address. So it is easy to manage all the devices from one point.
- **Efficient change management:** The DHCP provides an easy way to change the addresses, endpoints, and scopes. For example, if we want to change the IP address scheme for the entire organization; it will allow us to configure the system with the new scheme easily. Similarly, in the case of a new device, no configuration will be needed.

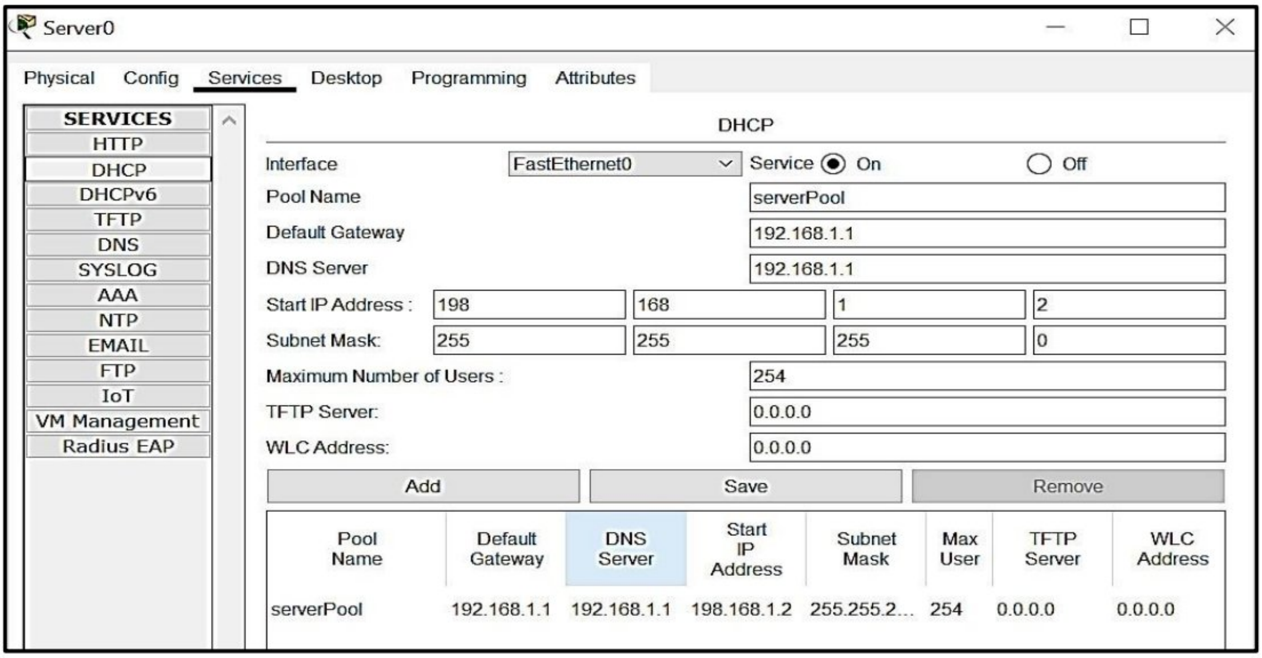
## Server Configuration

**Procedure:**

1. **Network Topology Design:**
   - o Open Cisco Packet Tracer and create a new project.
   - o Design a simple network topology consisting of:
     - **One Router:** To provide connectivity to an external network (simulating the Internet).
     - **One Switch:** To connect multiple devices within the LAN.
     - **Multiple PCs:** Representing the DHCP clients that will receive IP configuration.
     - **One Server:** To act as the DHCP server.
   - o Connect the devices using appropriate cables (e.g., Ethernet cables).
2. **DHCP Server Configuration:**
   - o Select the server device in Packet Tracer. o Go to the "Services" tab and select "DHCP." o Enable the DHCP service. o Configure the following parameters:

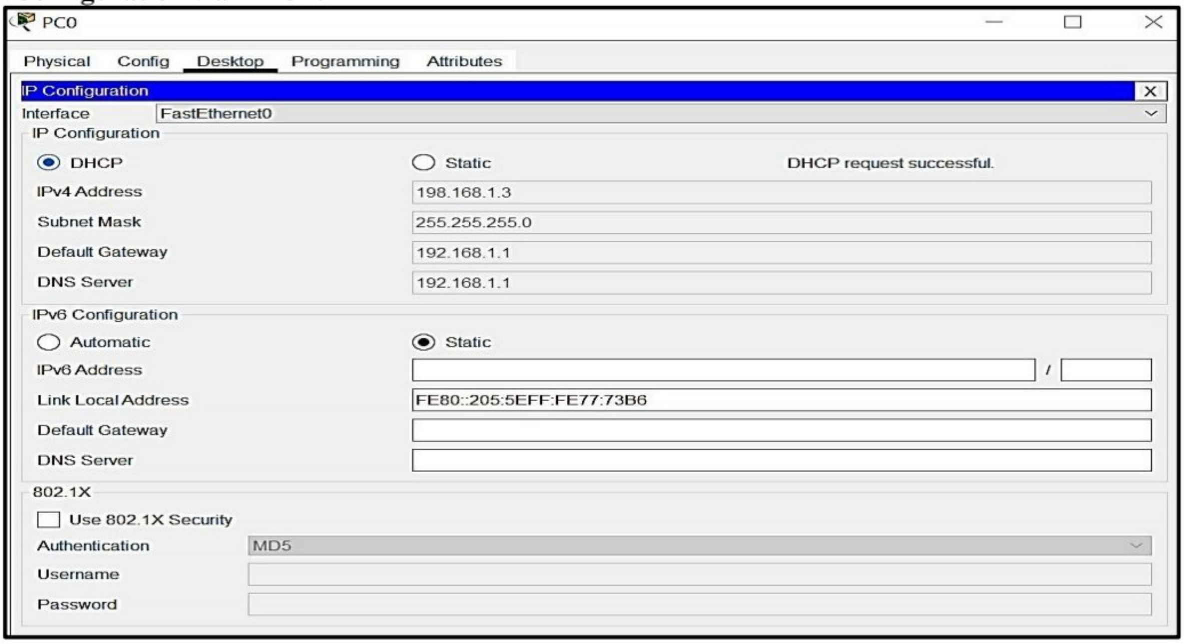
- **Default Gateway:** The IP address of the router interface connected to the LAN.
- **DNS Server:** The IP address of a DNS server (you can use a public DNS server like 8.8.8.8).
    - **Start IP Address:** The first IP address in the pool of addresses to be assigned.
    - **Maximum Number of Users:** The total number of IP addresses in the pool.
    - **Subnet Mask:** The subnet mask for the network.
  - Save the configuration.

3. **PC Configuration:**
    - Select each PC in the network.
    - Go to the "Desktop" tab and select "IP Configuration." o Set the IP configuration to "DHCP."
    - Observe that the PC automatically receives an IP address, subnet mask, default gateway, and DNS server address from the DHCP server.
4. **Testing Connectivity:**
    - Use the "Command Prompt" on each PC to ping other devices on the network and the default gateway.
    - Verify that the devices can communicate with each other.
    - Try accessing a website from one of the PCs to test Internet connectivity.

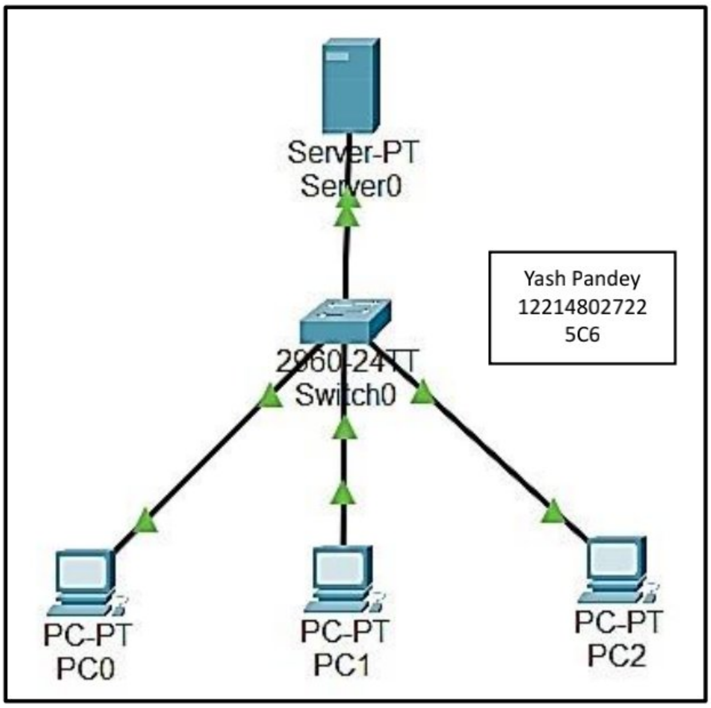



IP

**Configuration via DHCP:**




# Topology:



# Lab Exercise-6.

## Aim :- To Implement DNS , Email Services In The Network Using The Cisco Packet Tracer .

### Domain Name System

DNS, or Domain Name System, is a crucial component of the internet that functions as a distributed naming system. Its primary purpose is to translate user-friendly domain names into numerical IP (Internet Protocol) addresses that computers and networking equipment use to identify each other on the internet. Here's a more detailed explanation of DNS:

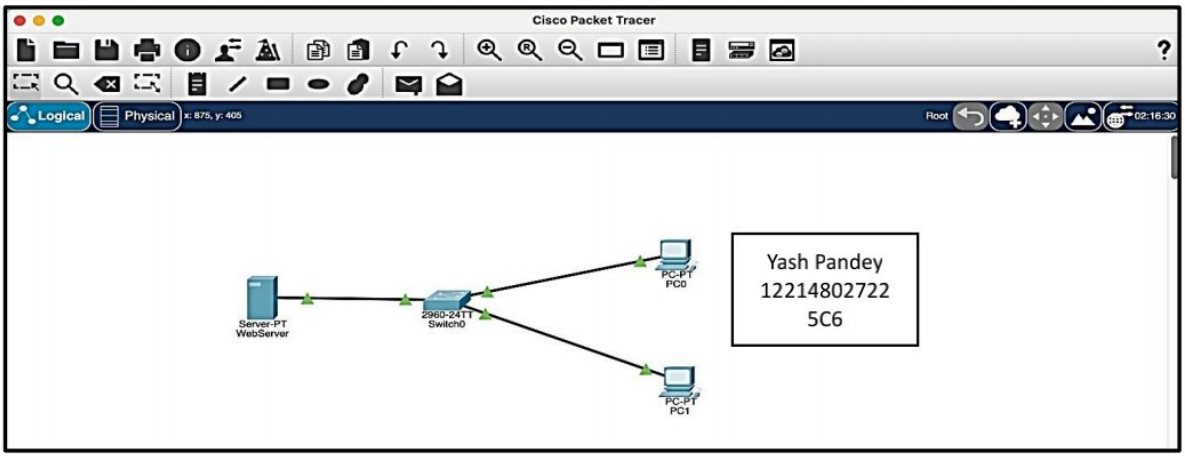
1. **Domain Names:** On the internet, websites, servers, and other networked devices are identified by numerical IP addresses (e.g., 192.168.1.1). However, these IP addresses are challenging for humans to remember. DNS provides a way to associate easily memorable domain names (e.g., www.example.com) with IP addresses.
2. **Hierarchical Structure:** DNS uses a hierarchical structure, resembling a tree, with a root domain at the top (.), followed by top-level domains (TLDs) like .com, .org, and country-code TLDs like .uk or .jp. Beneath TLDs are second-level domains (SLDs) and subdomains, forming a structured naming system.
3. **DNS Servers:** DNS operates through a distributed network of DNS servers. These servers are organized into different categories:
   - **Root Servers:** These are the foundational servers at the top of the DNS hierarchy. They hold information about the top-level domains (.com, .org, etc.) and their authoritative name servers.
   - **Top-Level Domain (TLD) Servers:** Each TLD (like .com) has its authoritative name servers, responsible for managing domain registrations and delegating authority to lower-level domain servers.
   - **Authoritative Name Servers:** These servers store DNS records for specific domains or subdomains. They provide authoritative responses for DNS queries about the domains they manage.
   - **Recursive DNS Servers:** Also known as resolvers, these servers are typically operated by Internet Service Providers (ISPs) or third-party DNS service providers. They

resolve DNS queries on behalf of clients by recursively querying other DNS servers to find the final IP address associated with a domain name.

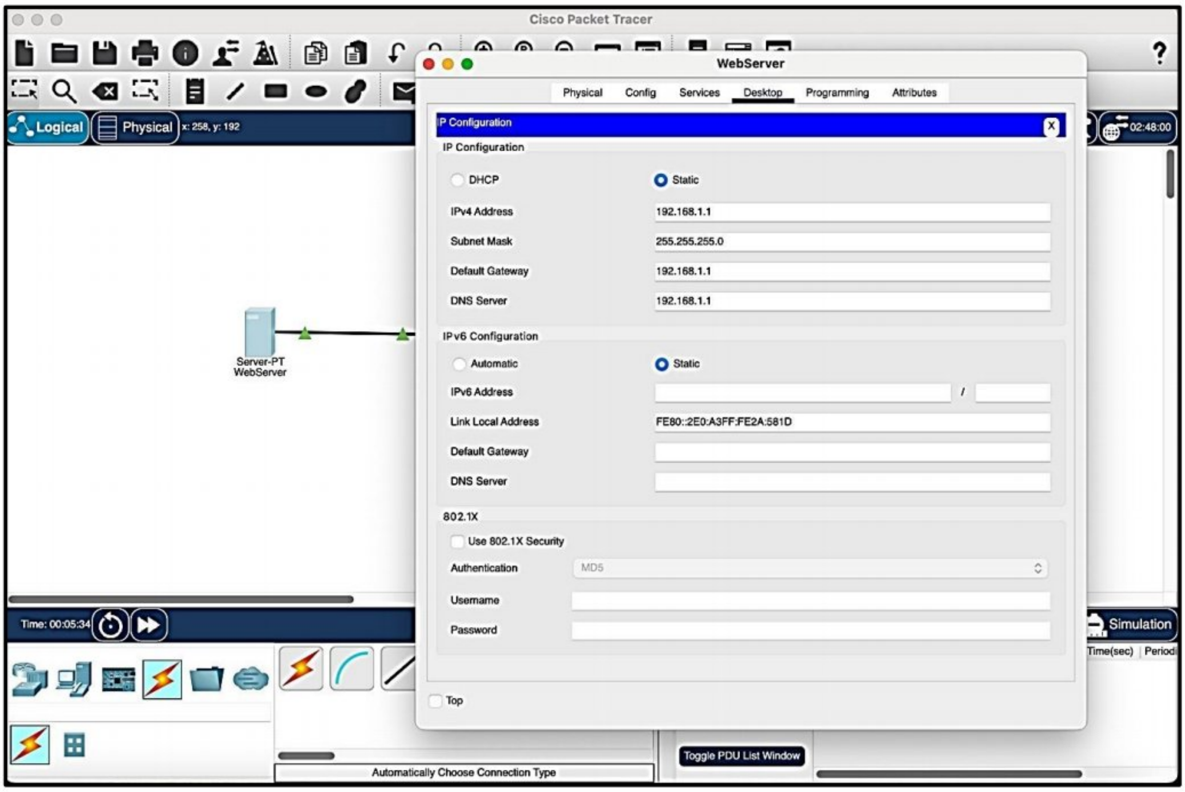
4. **DNS Resolution Process:** When a user enters a domain name in a web browser, their computer sends a DNS query to a recursive DNS server. The recursive server iteratively queries the DNS hierarchy until it finds the authoritative name server for the requested domain. The authoritative server provides the IP address associated with the domain, which is then returned to the user's computer, allowing it to connect to the desired website or service.
5. **DNS Records:** DNS records are data entries that contain information about a domain.
   - **A Record:** Associates a domain name with an IPv4 address.
   - **AAAA Record:** Associates a domain name with an IPv6 address.
   - **CNAME Record:** Creates an alias for a domain (canonical name).
   - **MX Record:** Specifies mail servers responsible for receiving email for a domain.
   - **TXT Record:** Holds text-based information, often used for various purposes, including domain verification and email authentication.

In summary, DNS is a critical system that enables users to access websites and services on the internet using human-readable domain names. It works by translating these domain names into IP addresses through a distributed network of DNS servers, allowing computers and devices to communicate and find each other on the internet.

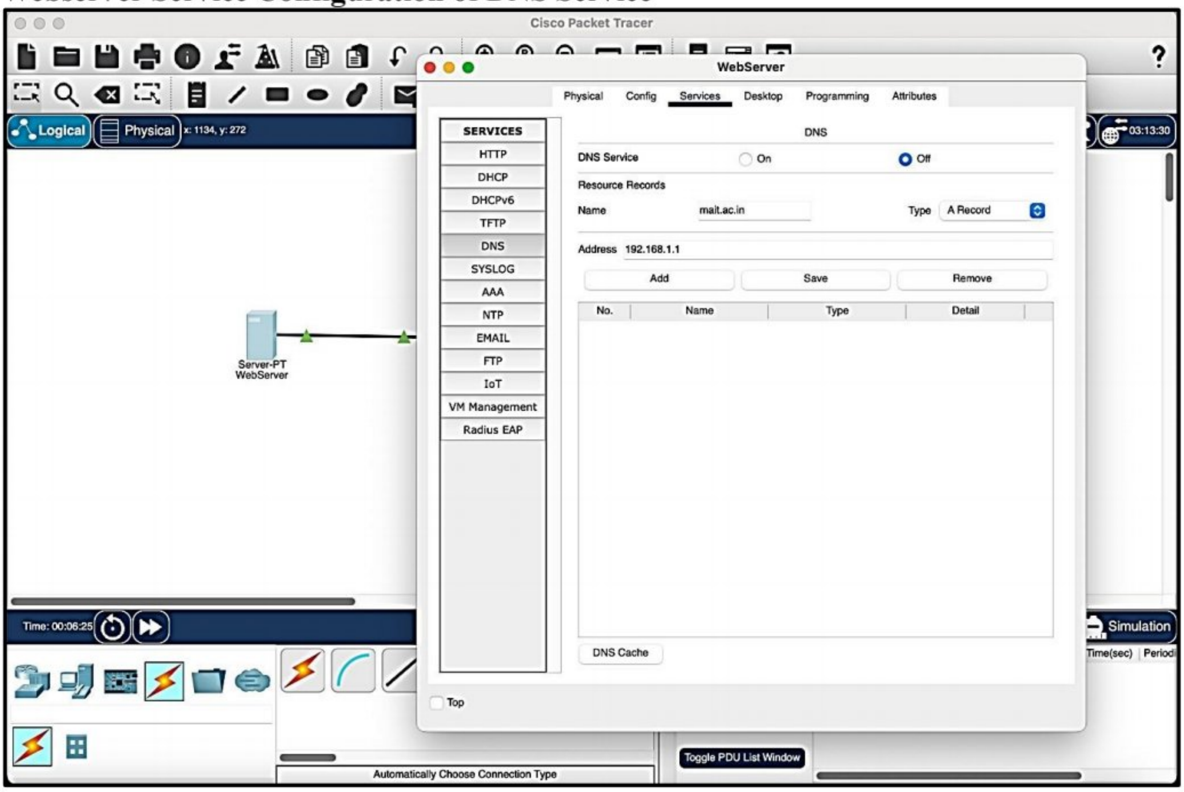
**Topology**



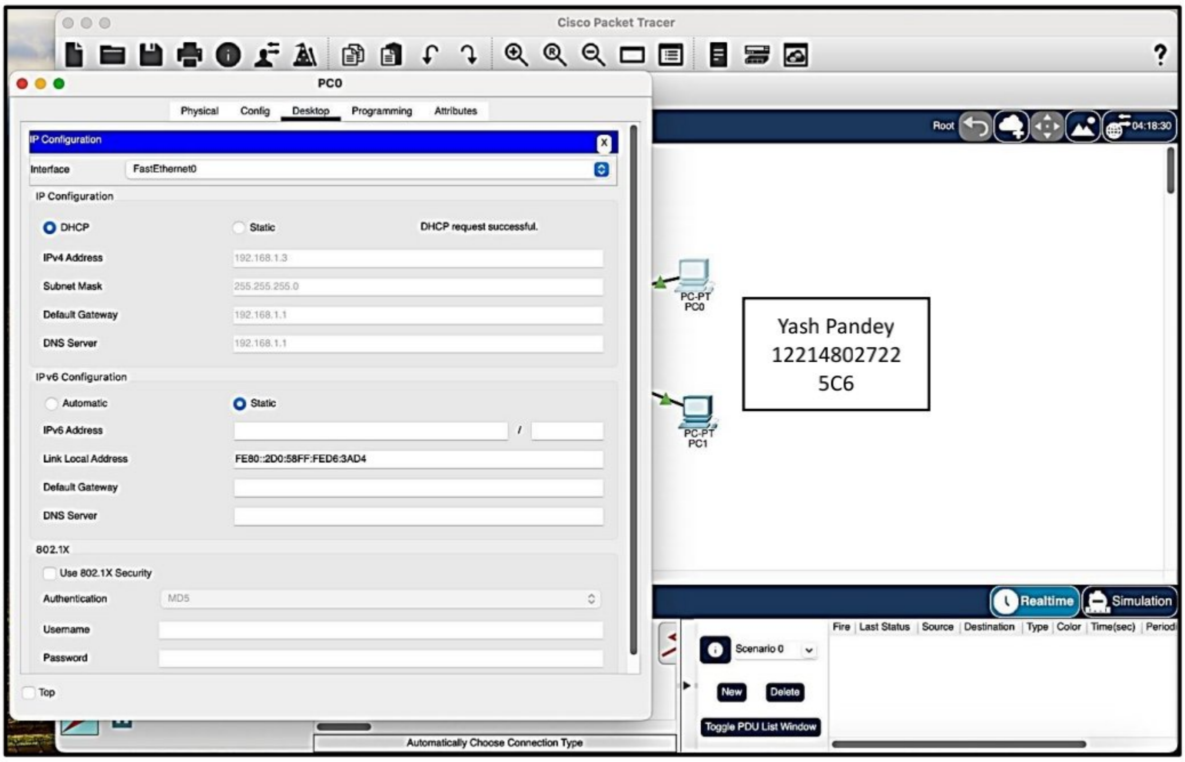
## Web Server IP Configuration



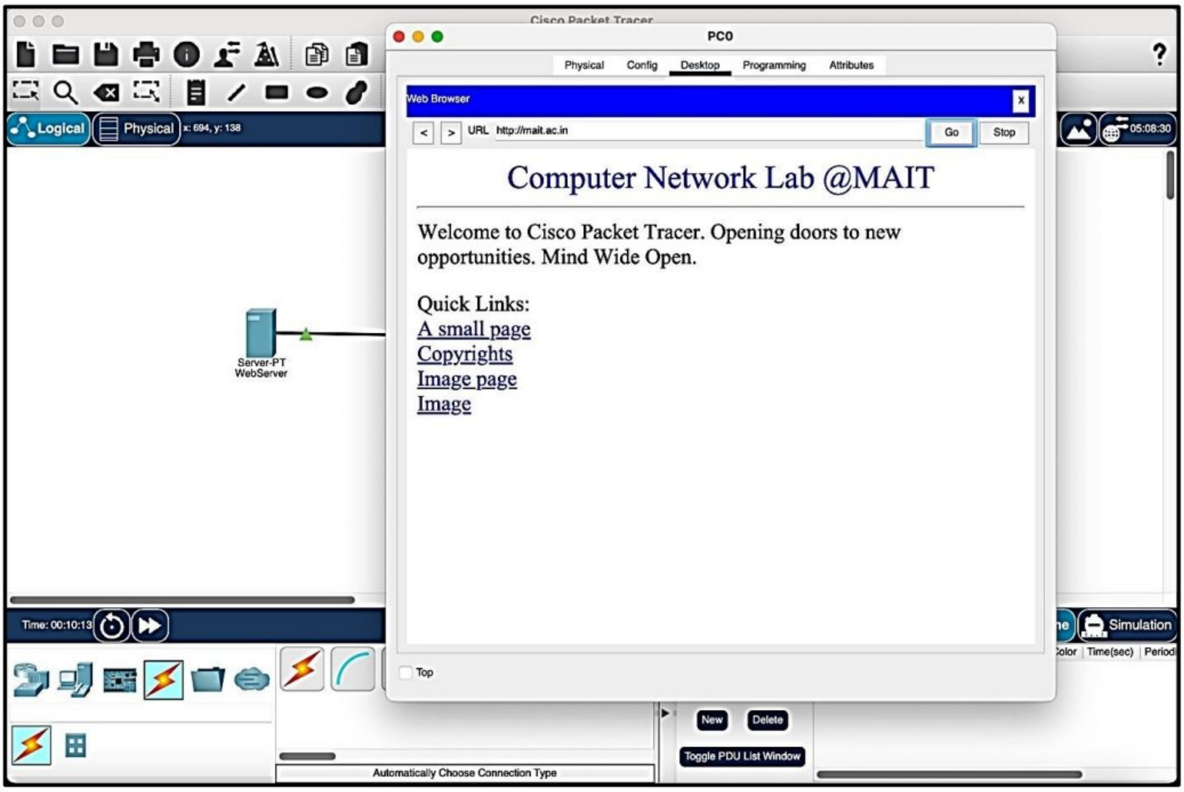

## Webserver Service Configuration of DNS Service

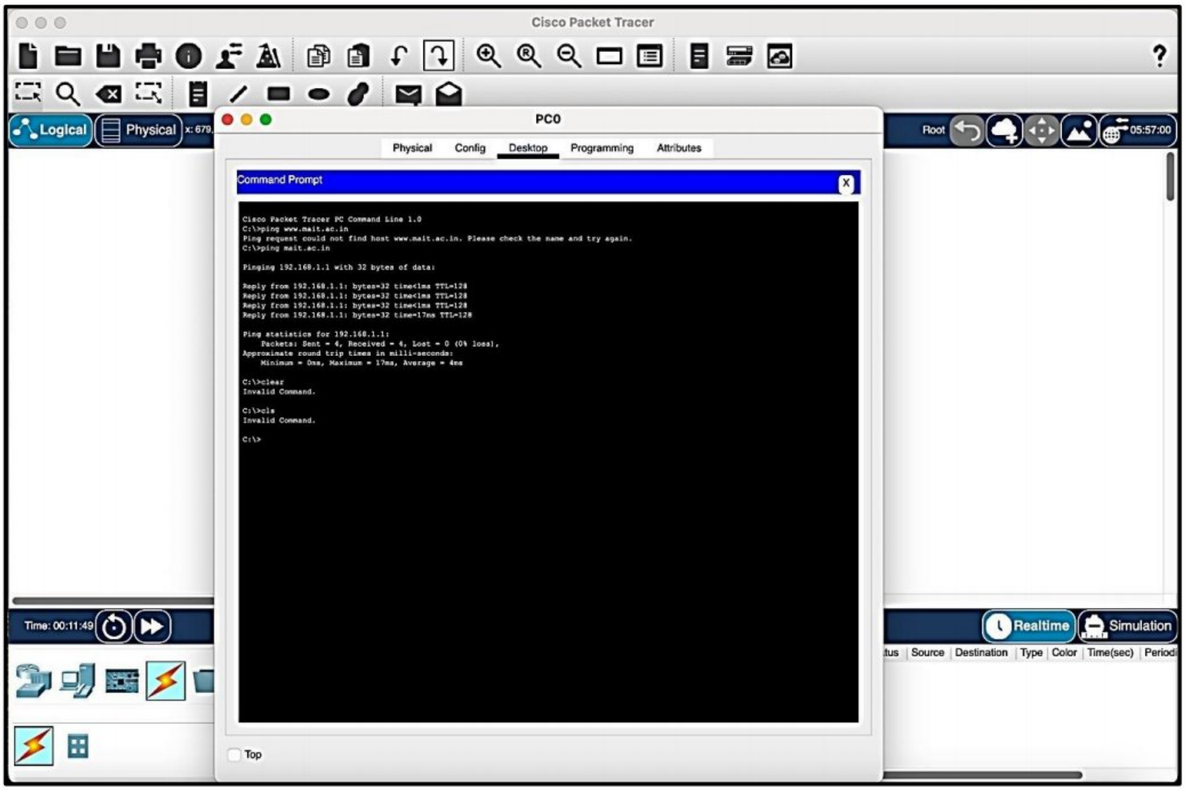


**PC IP Configuration via DHCP or Static**




Final output at client of DNS resolve on web browser






## Email Server SMTP

SMTP stands for Simple Mail Transfer Protocol. It is a standard protocol used for sending and receiving email messages over the internet. SMTP is primarily responsible for the transmission of outgoing emails from a sender's email client or server to the recipient's email server.

Here are some key points about SMTP:

1. **Sending Emails:** SMTP is the protocol used when you send an email from your email client (such as Outlook, Gmail, or Apple Mail) to a mail server. The server then routes the email to its destination based on the recipient's email address.
2. **Port 25:** SMTP typically operates on port 25 for unencrypted communication. However, encrypted variants like SMTP over TLS (SMTPS) on port 465 and SMTP STARTTLS on port 587 are commonly used to enhance email security and privacy.
3. **Text-Based Protocol:** SMTP is a text-based protocol, which means that communication between email servers and clients occurs using plain text commands and responses. While this makes it relatively simple, it also means that SMTP messages can be intercepted if not secured with encryption.

4. **Message Format:** SMTP defines the format and structure of email messages. It includes fields for the sender's address, recipient's address, subject, and the message body.
5. **Relaying:** SMTP servers are categorized into two types: outgoing (SMTP client) servers and incoming (SMTP server) servers. Outgoing servers are responsible for relaying emails from the sender to the recipient's server. Incoming servers handle the delivery of emails to the recipient's inbox.
6. **Authentication:** To prevent unauthorized use of SMTP servers, many servers require authentication before allowing users to send emails. This involves providing a username and password to prove the sender's identity.
7. **Error Codes:** SMTP uses a set of error codes (e.g., 550 for mailbox unavailable) to indicate the status of email delivery attempts. These codes help diagnose issues with email delivery.
8. **SMTP Servers:** Email service providers, businesses, and organizations operate SMTP servers to manage outgoing email traffic. Popular SMTP server software includes Microsoft Exchange, Postfix, and Sendmail.

SMTP is a fundamental protocol that underpins email communication on the internet. While it is essential for sending emails, it primarily focuses on routing and delivering messages. Other protocols, such as POP3 and IMAP, handle the retrieval of emails from a recipient's mailbox.

POP3

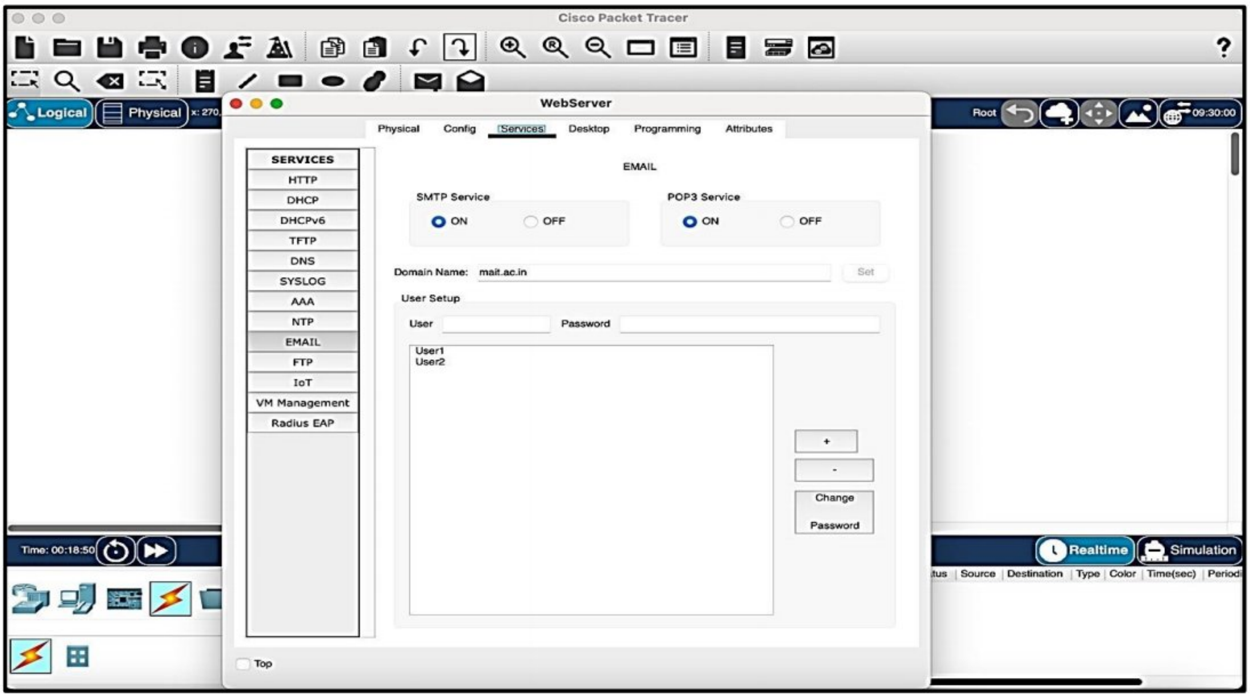
POP3, or Post Office Protocol version 3, is a standard internet protocol used for retrieving email messages from a mail server to a local email client or email application. It is one of the most widely used protocols for receiving emails. Here are the key points about POP3:

1. **Email Retrieval:** POP3 is primarily used to retrieve emails from a mail server to a user's local device or email client (e.g., Outlook, Thunderbird). It allows users to download emails from their mailbox to their computer or mobile device.
2. **Port 110:** POP3 typically operates on port 110 for unencrypted communication. However, POP3 over TLS (POP3S) on port 995 is commonly used to encrypt the communication between the email client and the server.
3. **Storage and Management:** Unlike some other email protocols, such as IMAP (Internet Message Access Protocol), which keep emails on the server and allow users to organize and manage them remotely, POP3 downloads emails to the local device, and they are typically removed from the server after retrieval. However, there are settings in some email clients that can be configured to leave copies of emails on the server.

4. **Stateless:** POP3 is a stateless protocol, meaning it does not keep track of the messages that have been previously downloaded. It relies on the email client to manage and keep track of downloaded emails.
5. **Single Device Access:** POP3 is often used when users access their emails from a single device or want to keep local copies of their messages. It may not be the best choice for users who need to access their emails from multiple devices and keep them synchronized.
6. **Authentication:** To retrieve emails using POP3, users typically need to provide their username and password to authenticate themselves with the email server.
7. **Message Deletion:** In the default POP3 configuration, once an email is downloaded to a local device, it is deleted from the server. However, there are settings in some email clients that can be configured to leave copies of emails on the server for a specified period.
8. **No Folder Structure:** POP3 does not support server-side folder structures or email organization. The email client manages email folders and organization locally.

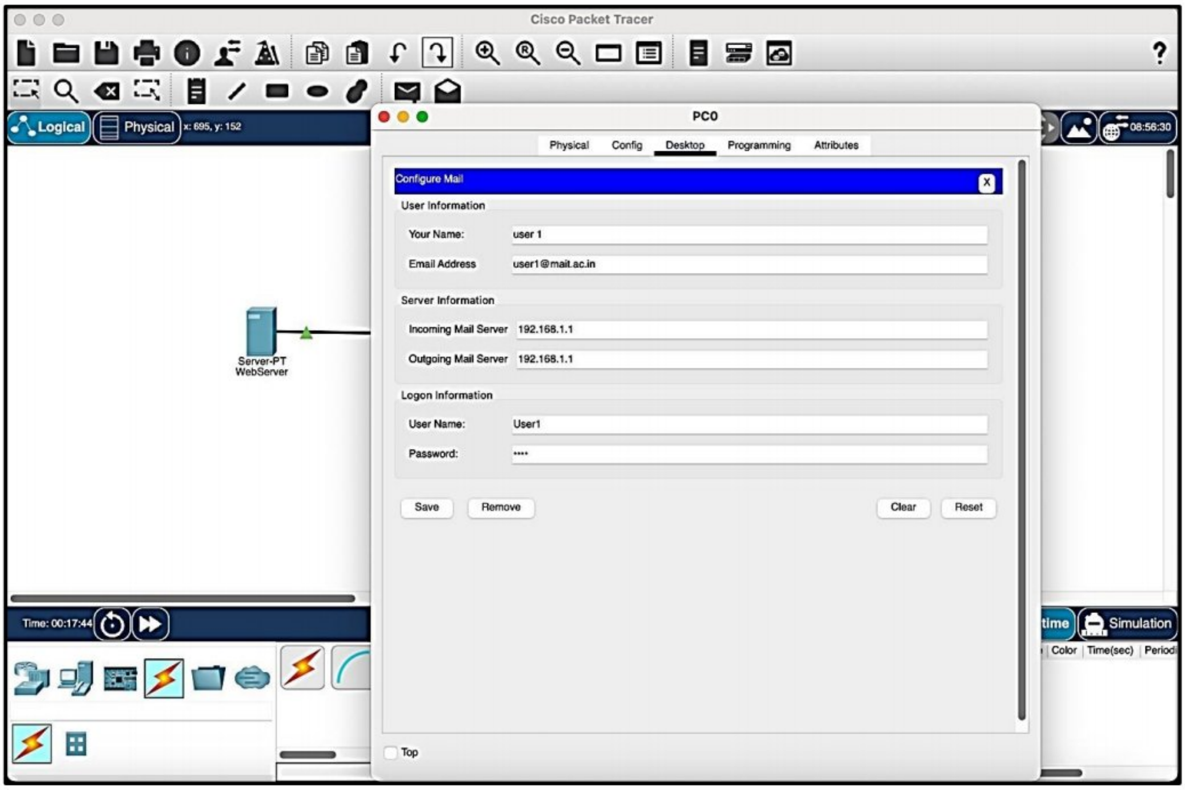
In summary, POP3 is a protocol used for downloading email messages from a mail server to a local email client. It is suitable for users who prefer to keep local copies of their emails and access them from a single device. However, it may not be the best choice for users who require access to their emails from multiple devices with synchronization and server-based organization.

**Server configuration**

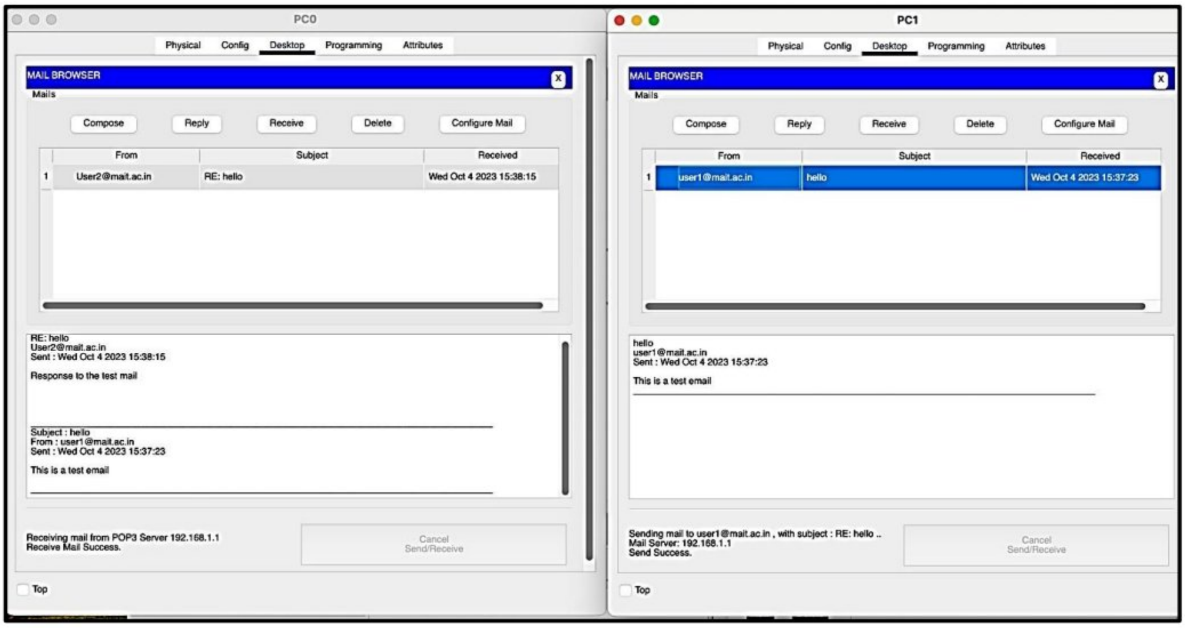


Add two users with their respective credentials on the webserver with the domain set as mait.ac.in

**Configuration on users PCs**




## Final Output :-



# Lab Exercise-7.

## Aim :- To Implement RIP & IGRP Protocol .

### RIP: Routing Information Protocol

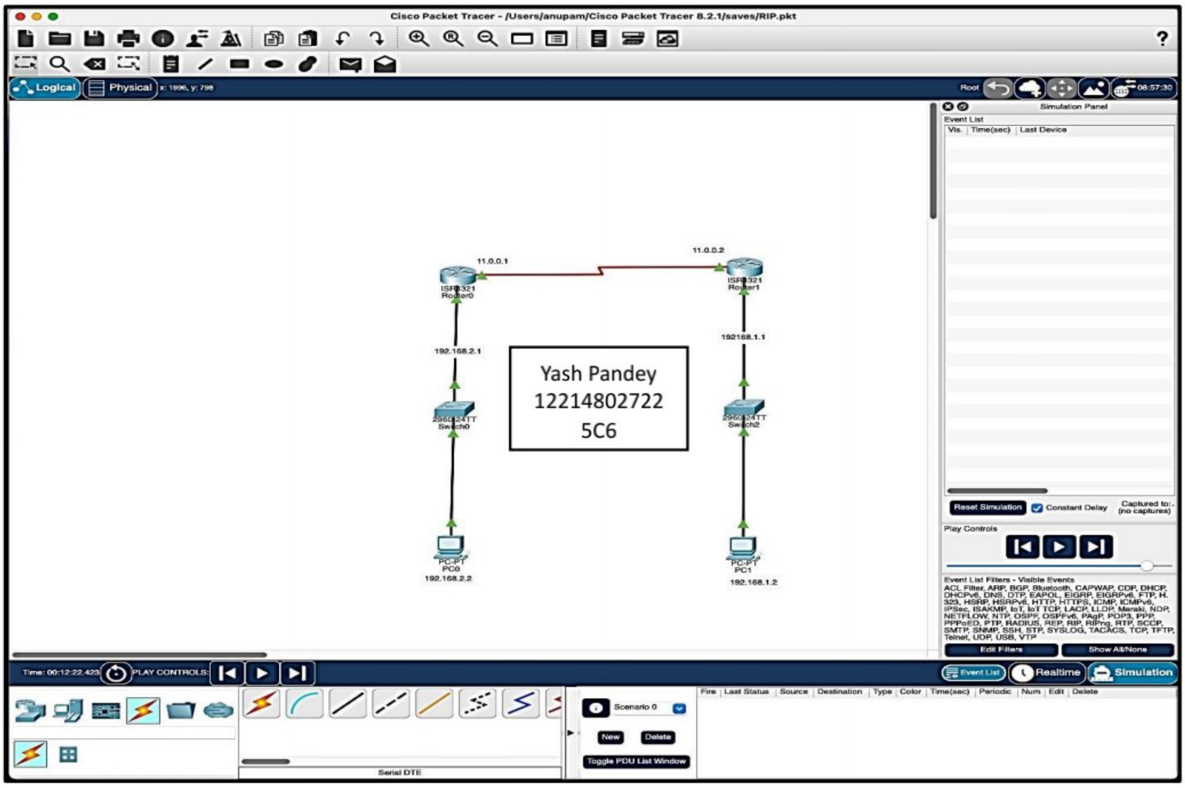
RIP, or Routing Information Protocol, is one of the oldest and simplest interior gateway protocols (IGP) used in computer networking. It is specifically categorized as a distance-vector routing protocol. Here are the key points about RIP:

1. **Routing Protocol:** RIP is used to determine the best path for routing data packets within an autonomous system (AS). An AS is a collection of IP networks and routers under the control of a single organization or administration.
2. **Distance-Vector Protocol:** RIP is a distance-vector protocol, which means that routers exchange routing information with their neighbors to determine the best path to reach a destination network. Each router maintains a routing table that contains information about the distance (number of hops) to reach various networks.
3. **Hop Count:** RIP uses hop count as its metric to measure the distance to a destination network. Each router increments the hop count by one when it forwards a packet to another router. RIP routers aim to find the path with the fewest hops to reach a destination.
4. **Routing Updates:** RIP routers periodically send routing updates to their neighboring routers. These updates contain information about the networks they know about and the number of hops to reach them. Routers use these updates to build and update their routing tables.
5. **Loop Prevention:** To prevent routing loops, RIP includes a feature called split horizon, where a router does not advertise routes back to the neighbor from which it learned them.

   Additionally, RIP uses a maximum hop count of 15 to avoid routing loops.
6. **Convergence Time:** RIP has a relatively slow convergence time. When network topology changes occur, it may take some time for RIP routers to converge and update their routing tables.
7. **Versions:** There are two versions of RIP: RIP version 1 (RIPv1) and RIP version 2 (RIPv2). RIPv1 has limitations, such as not supporting subnet information and using broadcast for routing updates. RIPv2 addresses these limitations and includes support for subnet masks and multicast routing updates.

8. **Usage:** RIP is generally considered an older and less sophisticated routing protocol compared to modern alternatives like OSPF (Open Shortest Path First) and BGP (Border Gateway Protocol). As a result, it is less commonly used in large or complex networks but may still be found in smaller networks or for educational purposes.

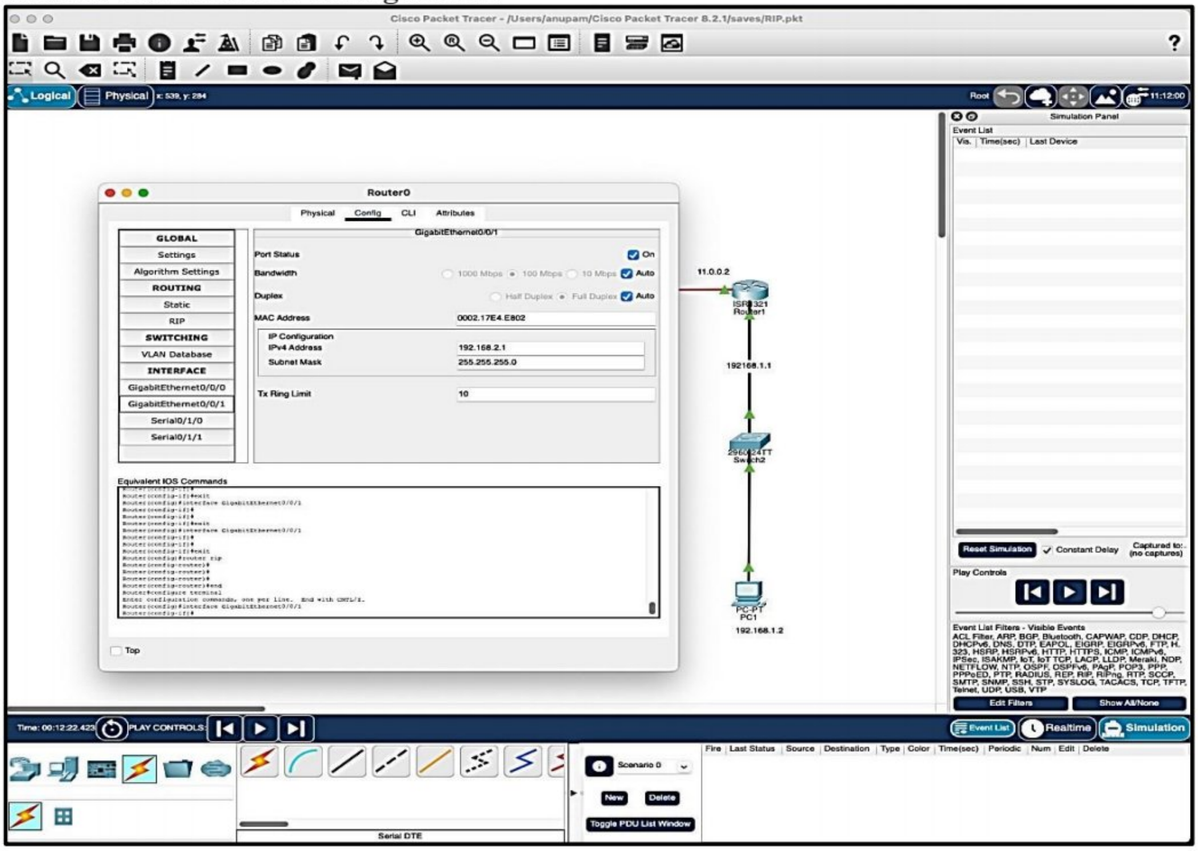
**Network Topology**



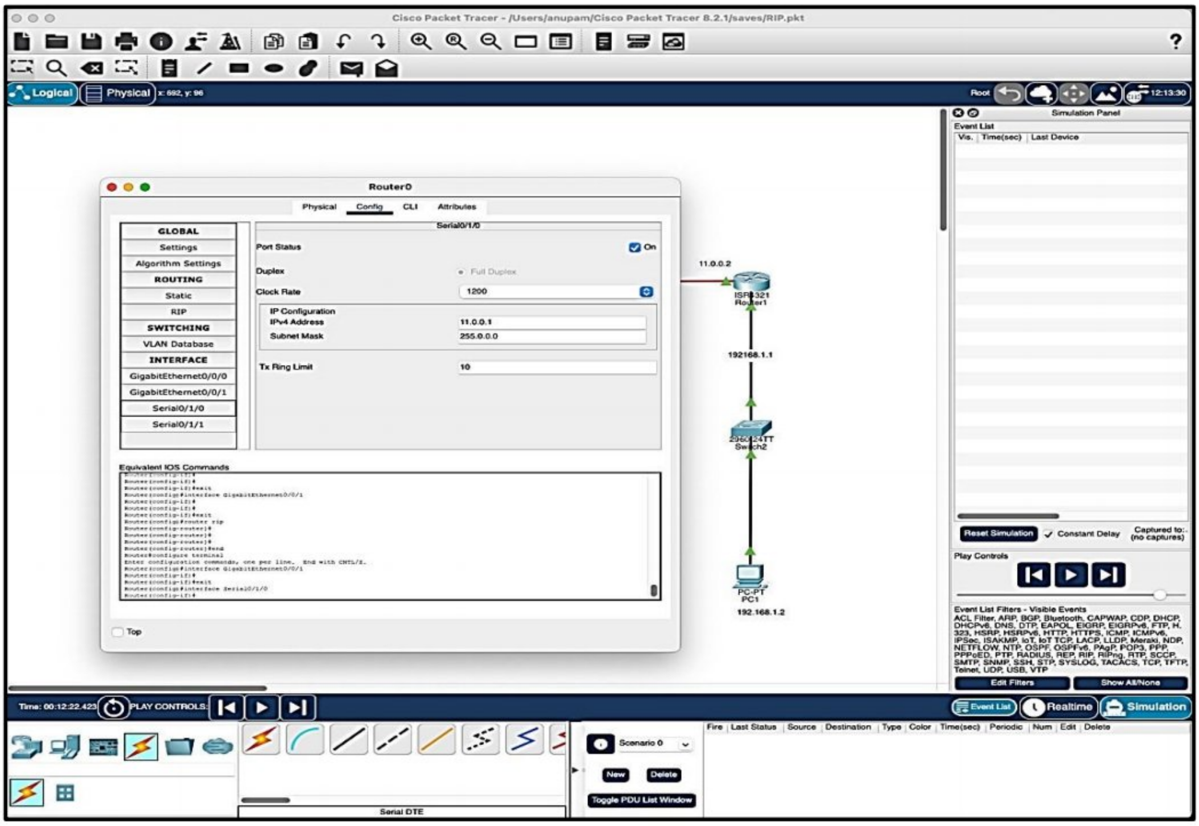

Drag and drop all the network components.

Connect the devices using straight through cables and router via serial cable.

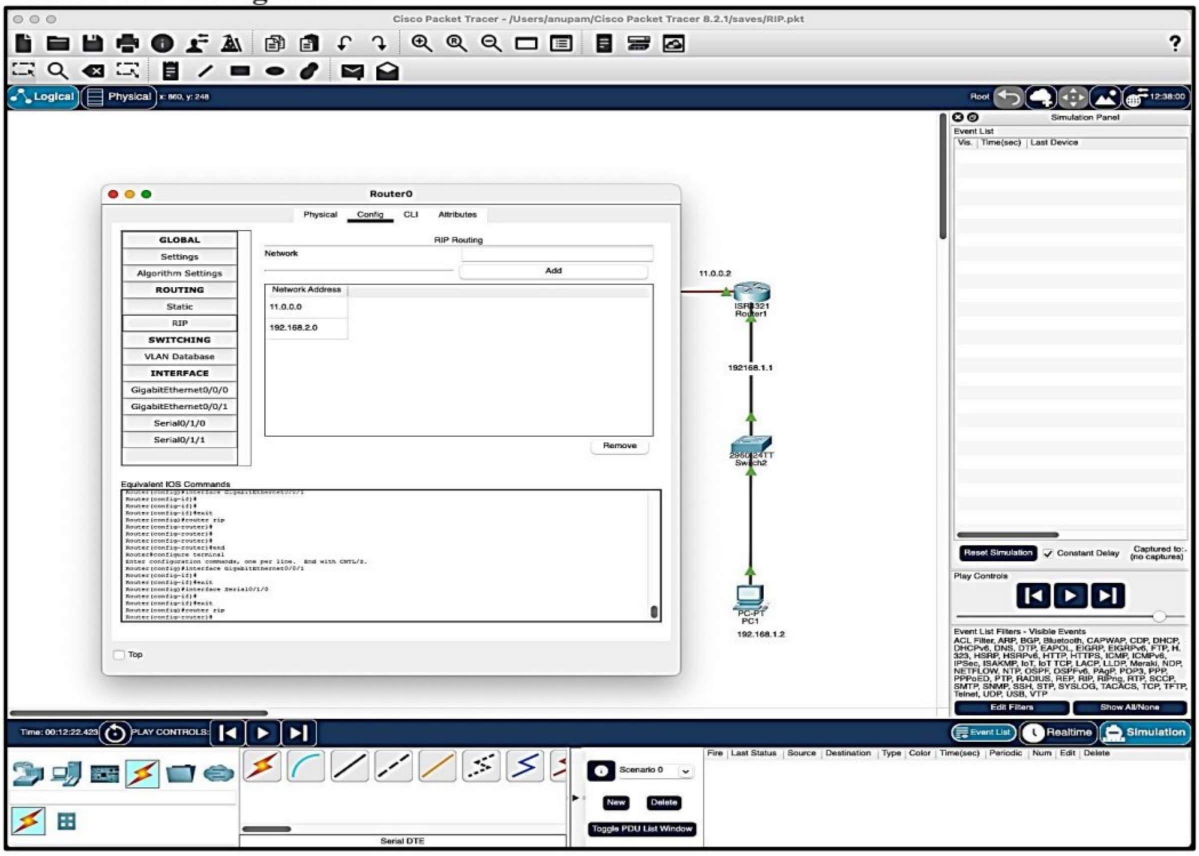
## Router0 Fast Ethernet Configuration



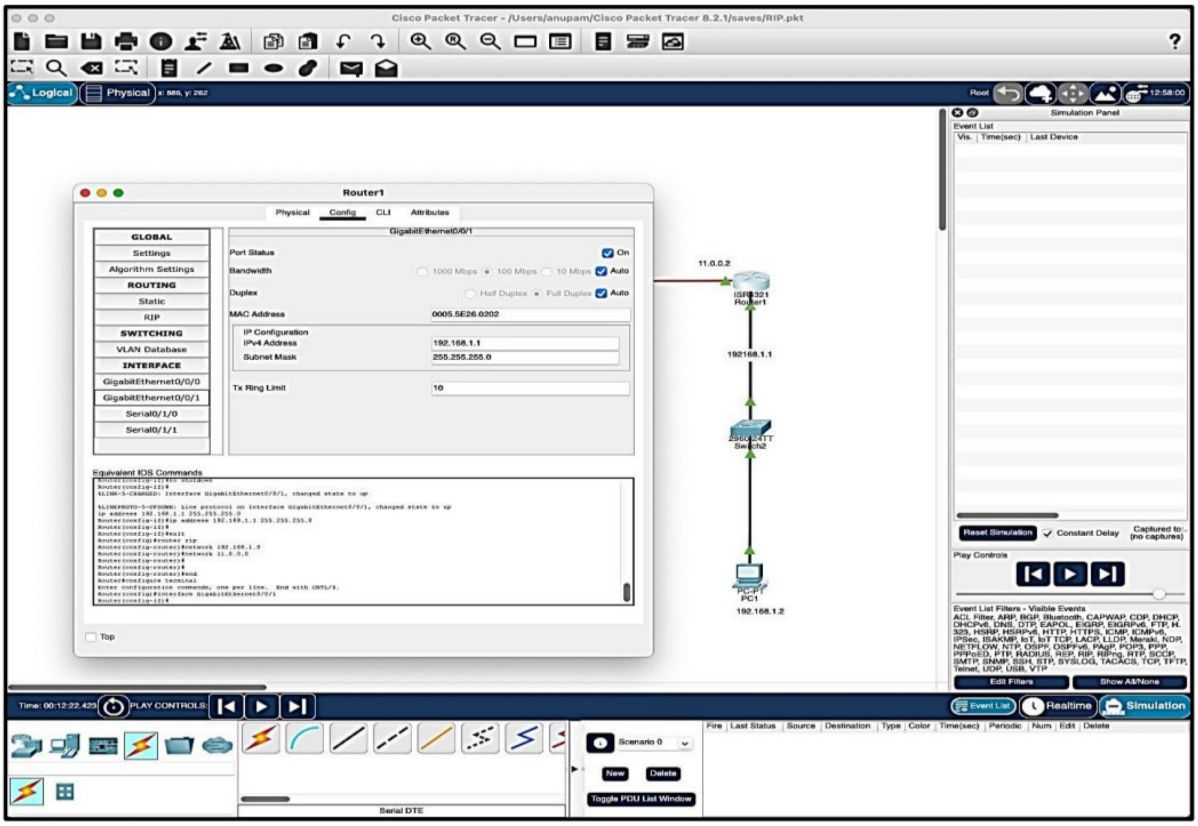

## Router0 Serial Port Configuration



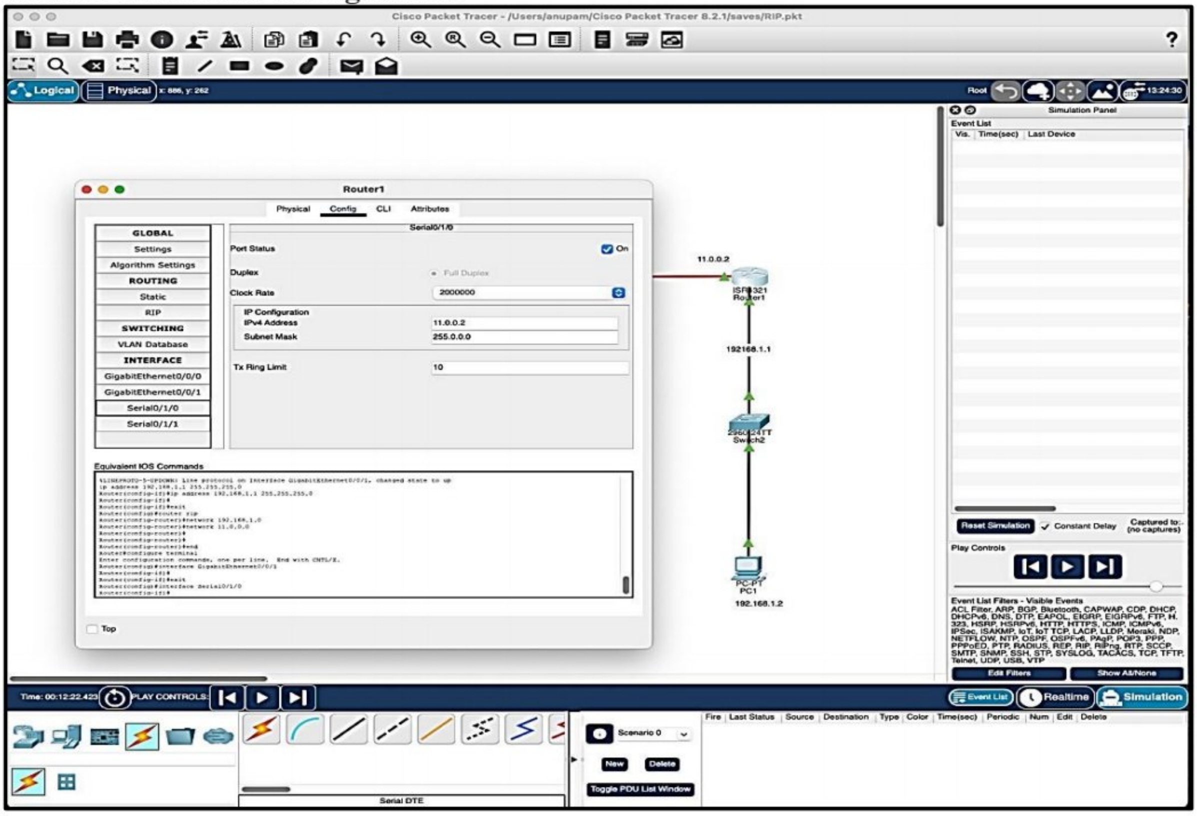
## Router0 RIP Configuration



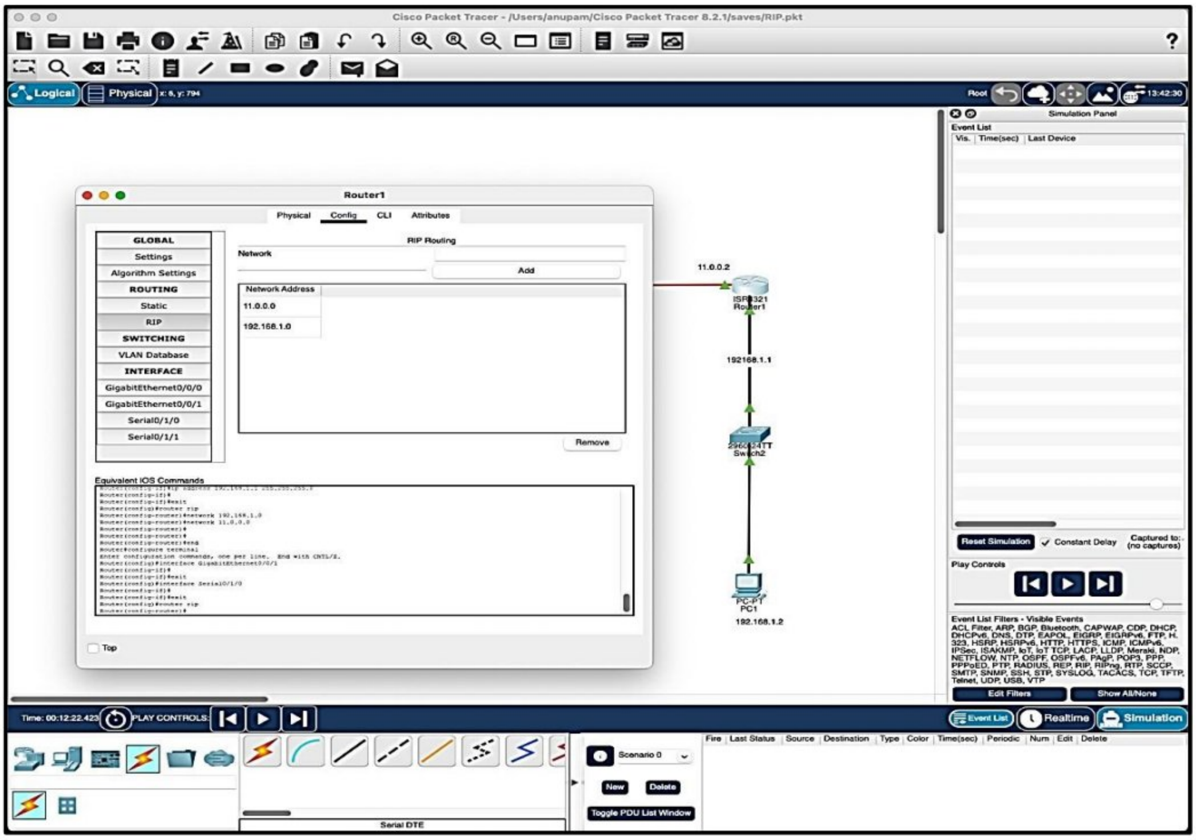

## Router1 Fast Ethernet Configuration



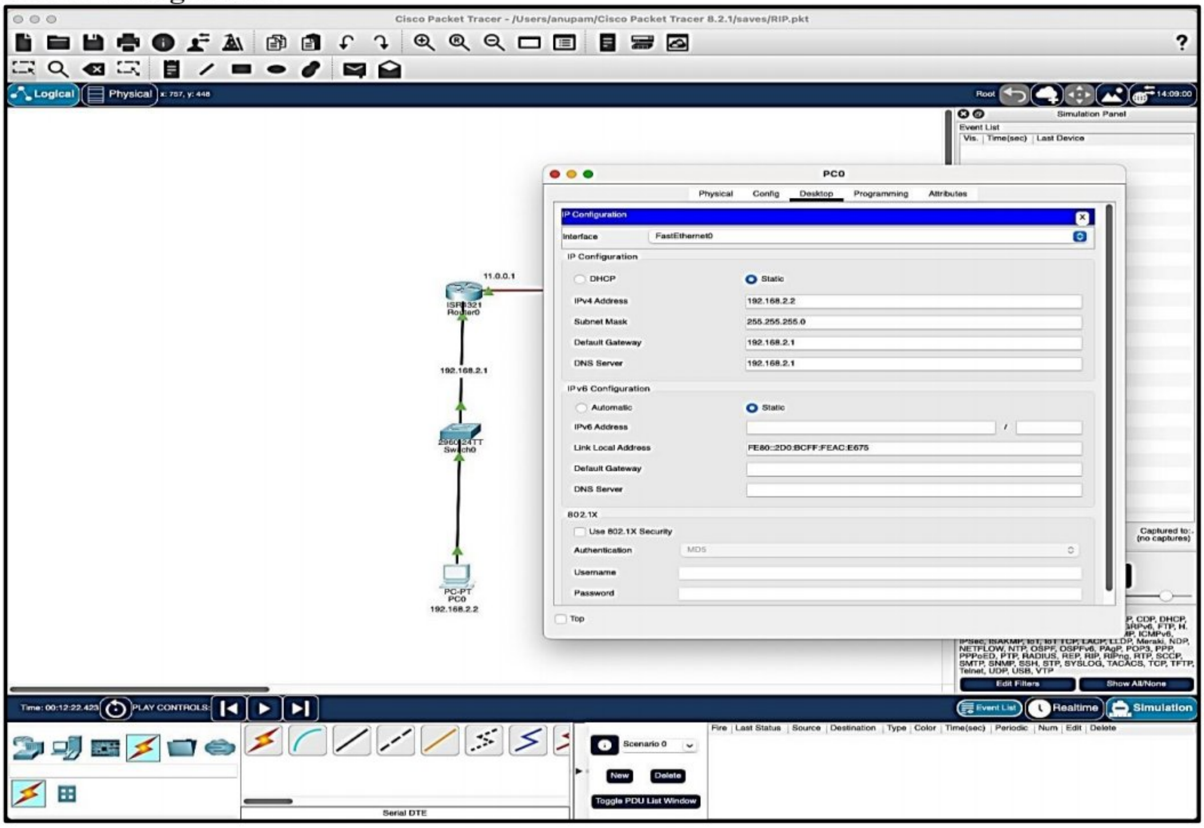
## Router1 Serial Port Configuration



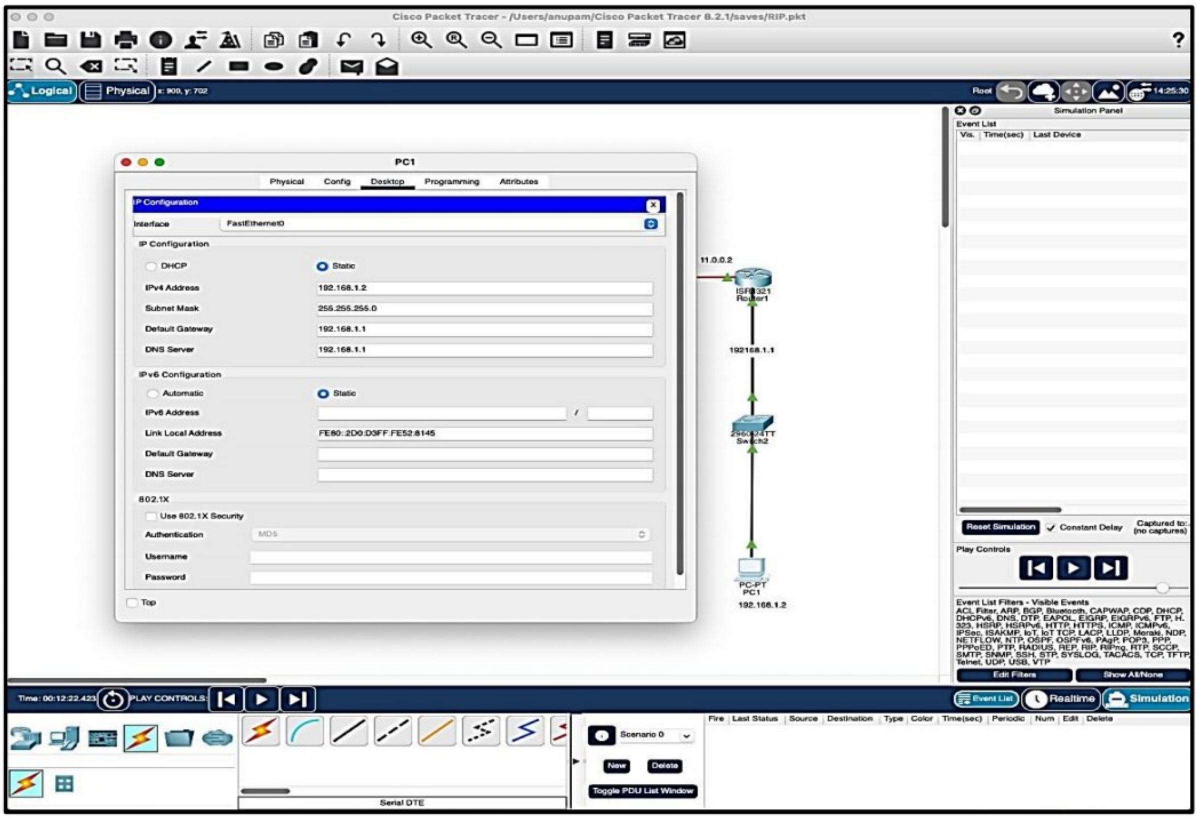

## Router1 RIP Configuration



## PC0 IP configuration

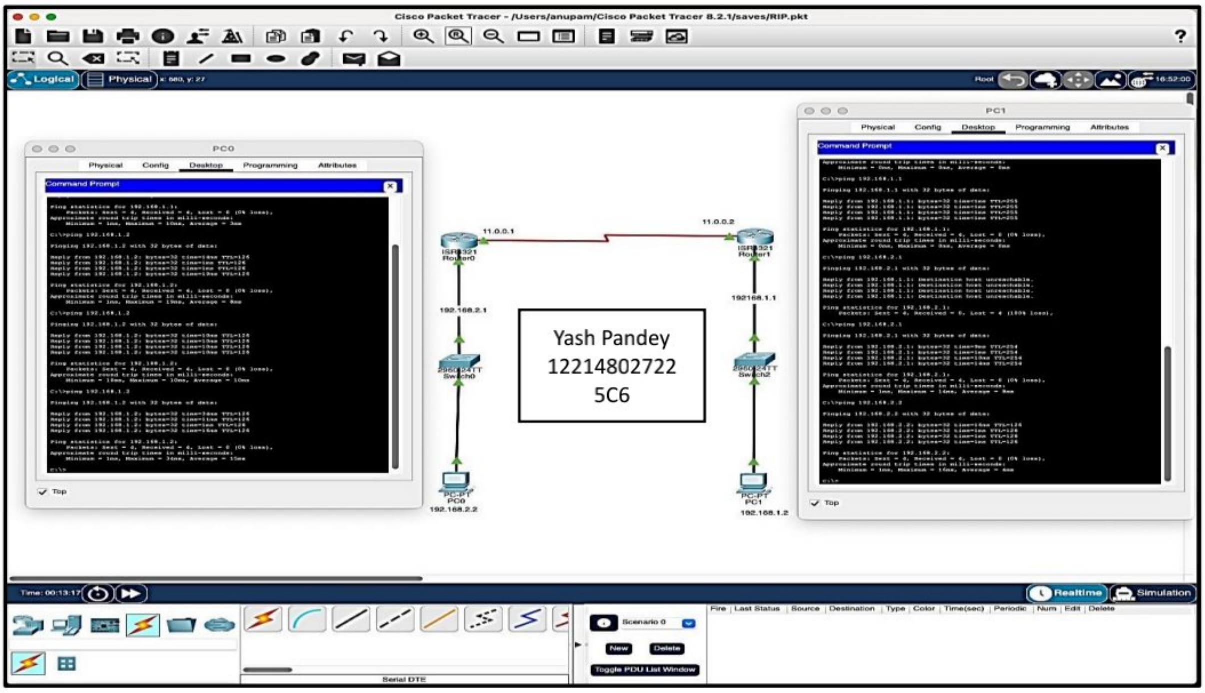



## PC1 IP Configuration



### Ping to Test PC0 to PC1

We will now use ping command to test if the RIP is working and we are able to connect to the PC1 on Router1 from PC0 on Router0




PING were successful from both the devices.

## IGRP, or Interior Gateway Routing Protocol

It is a Cisco-proprietary interior gateway protocol used for routing within a single autonomous system (AS) in computer networking. Here are the key points about IGRP:

1. **Cisco Proprietary:** IGRP was developed by Cisco and is a proprietary routing protocol, meaning it is specific to Cisco devices and is not an open standard like OSPF (Open Shortest Path First) or RIP (Routing Information Protocol).
2. **Distance-Vector Protocol:** IGRP is a distance-vector routing protocol, similar to RIP. It calculates the best path to a destination network based on a metric, but unlike RIP, IGRP uses a more complex metric known as a composite metric.
3. **Composite Metric:** IGRP's metric, known as a composite metric, takes into account factors like bandwidth, delay, reliability, and load. This makes IGRP more capable of selecting the best path based on network conditions and attributes.
4. **Routing Updates:** IGRP routers exchange routing updates with their neighboring routers to share information about the network topology. These updates contain information about networks and their associated metrics.

5. **Autonomous System (AS):** IGRP is used to route data within a single autonomous system, which is a collection of interconnected networks and routers under the control of a single organization or administration.
6. **Loop Prevention:** IGRP includes mechanisms to prevent routing loops, such as route poisoning, where a router advertises a failed route with an infinite metric to inform other routers of the failure.
7. **Compatibility:** IGRP is specific to Cisco routers and is not interoperable with other routing protocols. It is typically used in networks where all routers are Cisco devices.
8. **Convergence Time:** IGRP has a faster convergence time compared to older distance-vector protocols like RIP, but it may still take some time for the network to converge after topology changes.
9. **Replacement by EIGRP:** IGRP has largely been replaced by Cisco's Enhanced Interior Gateway Routing Protocol (EIGRP), which offers more advanced features and capabilities. EIGRP is also a Cisco-proprietary routing protocol but provides better scalability and faster convergence.

# Lab Exercise-8.

## Aim :- To Implement Multiple Router N/w & EIGRP Protocol .

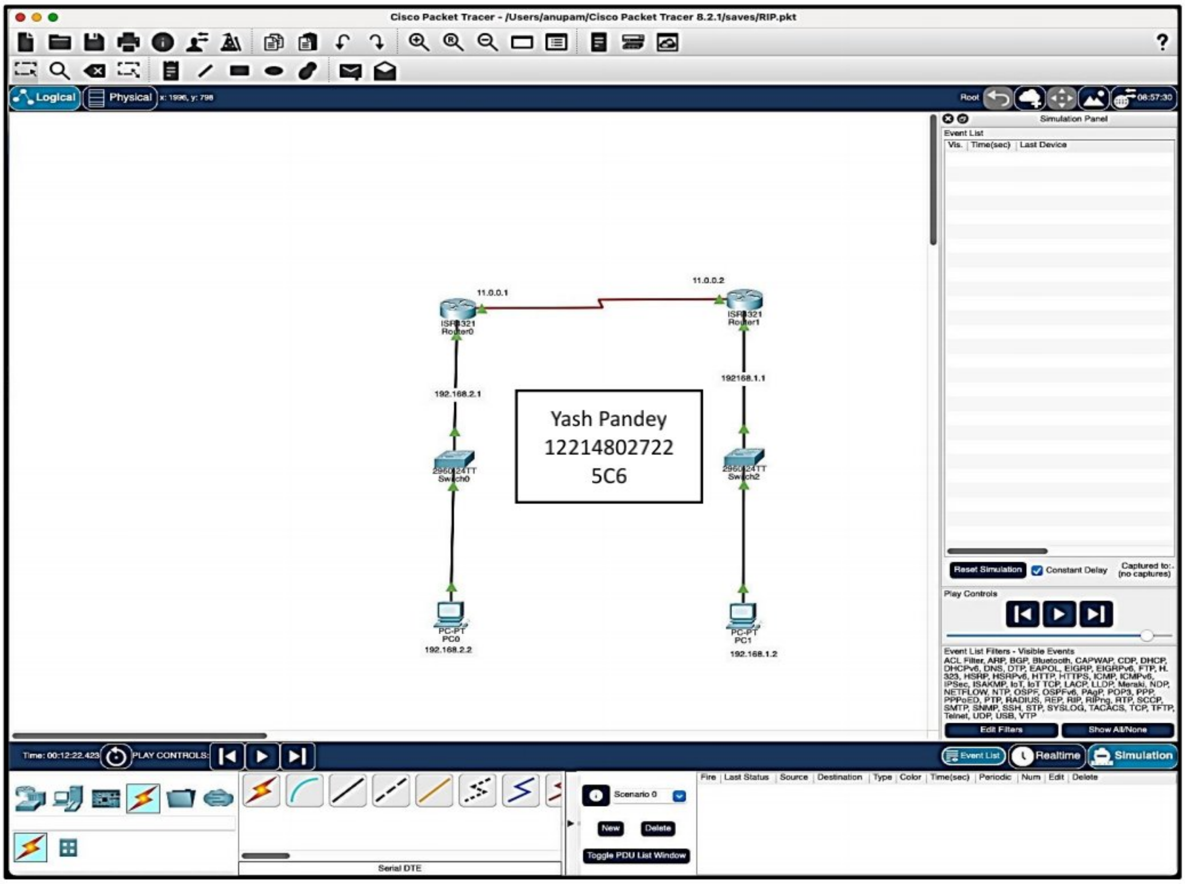
### EIGRP, or Enhanced Interior Gateway Routing Protocol

It is a routing protocol used in computer networking to efficiently route data packets within a network. It is classified as an advanced distance-vector routing protocol but incorporates elements of both distance-vector and link-state protocols. Here are key points of EIGRP:

1. **Cisco Proprietary:** EIGRP was initially developed by Cisco Systems and is a proprietary routing protocol. However, Cisco released EIGRP as an open standard, making it available for use in multi-vendor network environments through the Informational RFC 7868.
2. **Hybrid Protocol:** EIGRP is often referred to as a hybrid protocol because it combines features of distance-vector and link-state routing protocols. It uses the Diffusing Update Algorithm (DUAL) to calculate the best routes based on both distance and bandwidth.
3. **Metric Calculation:** EIGRP uses a composite metric called "metric" or "K-values" to calculate the best path to a destination. This metric considers factors such as bandwidth, delay, reliability, and load when determining the optimal route.
4. **Neighbor Discovery:** EIGRP routers establish neighbor relationships with directly connected routers to exchange routing information. This information exchange helps routers build and maintain their routing tables.
5. **Partial Updates:** EIGRP employs partial updates, meaning that when a change in the network occurs, it only sends updates about the specific change rather than the entire routing table. This reduces network traffic and speeds up convergence.
6. **Loop-Free Topology:** EIGRP uses the DUAL algorithm to ensure a loop-free topology. It does this by maintaining a feasible successor route in addition to the primary route, which can be quickly used if the primary route fails.
7. **VLSM and CIDR Support:** EIGRP supports Variable Length Subnet Masks (VLSM) and Classless Inter-Domain Routing (CIDR), making it suitable for networks with complex addressing schemes.

8. **Authentication:** EIGRP supports authentication to secure routing updates and prevent unauthorized routers from participating in the routing process.
9. **Wide Adoption:** While EIGRP is a Cisco proprietary protocol, it is widely used in Ciscobased networks and is known for its fast convergence and efficient use of network resources.
10. **Scalability:** EIGRP is suitable for both small and large networks. It scales well and can handle networks of various sizes.
11. **Compatibility:** EIGRP can coexist with other routing protocols, making it possible to integrate Cisco-based networks with non-Cisco networks.

**Network Topology**




We will be using the same network topology as RIP with same configuration of ports and its respective IP addresses.

**Router0 EIGRP Configuration**

```
Router(config)#router eigrp 1
Router(config-router)#network 192.168.1.0
Router(config-router)#network 11.0.0.0
Router(config-router)#
Router(config-router)#
Router(config-router)#quit
                        ^
% Invalid input detected at '^' marker.

Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router(config)#router eigrp 1

Router(config-router)#network 11.0.0.0

Router(config-router)#network 192.168.2.0

Router(config-router)#

%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 11.0.0.2 (Serial0/1/0) is up: new adjacency

Router(config-router)#

Router(config-router)#exit

Router(config)#exit

Router#

**Router1 EIGRP Configuration**

```
Router(config)#router eigrp 1
Router(config-router)#network 192.168.1.0
Router(config-router)#
Router(config-router)#network 11.0.0.0
Router(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 11.0.0.1 (Serial0/1/0) is up: new adjacency

Router(config-router)#network 192.168.1.0
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router(config)#router eigrp 1

Router(config-router)#network 11.0.0.0

Router(config-router)#network 192.168.2.0

Router(config-router)#

%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 11.0.0.2 (Serial0/1/0) is up: new adjacency

Router(config-router)#

Router(config-router)#exit

Router(config)#exit

Router#

### Checking the EIGRP routes

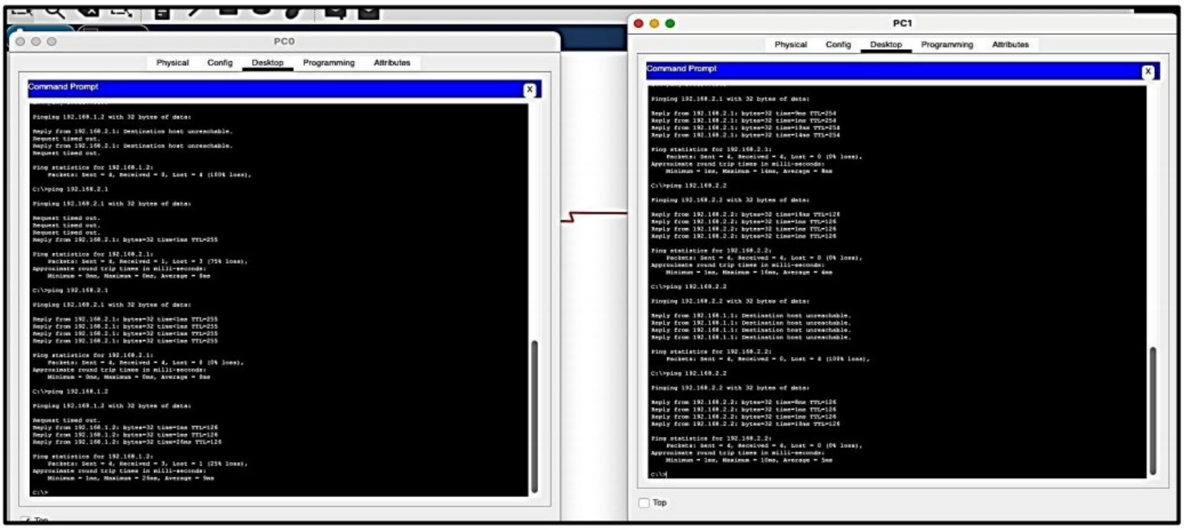
The command is **sh ip route**

```
Router#
Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     11.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       11.0.0.0/8 is directly connected, Serial0/1/0
L       11.0.0.2/32 is directly connected, Serial0/1/0
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.2.1/32 is directly connected, GigabitEthernet0/0/0
```

**Ping Test**



# Lab Exercise-9.

## Aim :- To Implement NAT Using Cisco Packet Tracer .

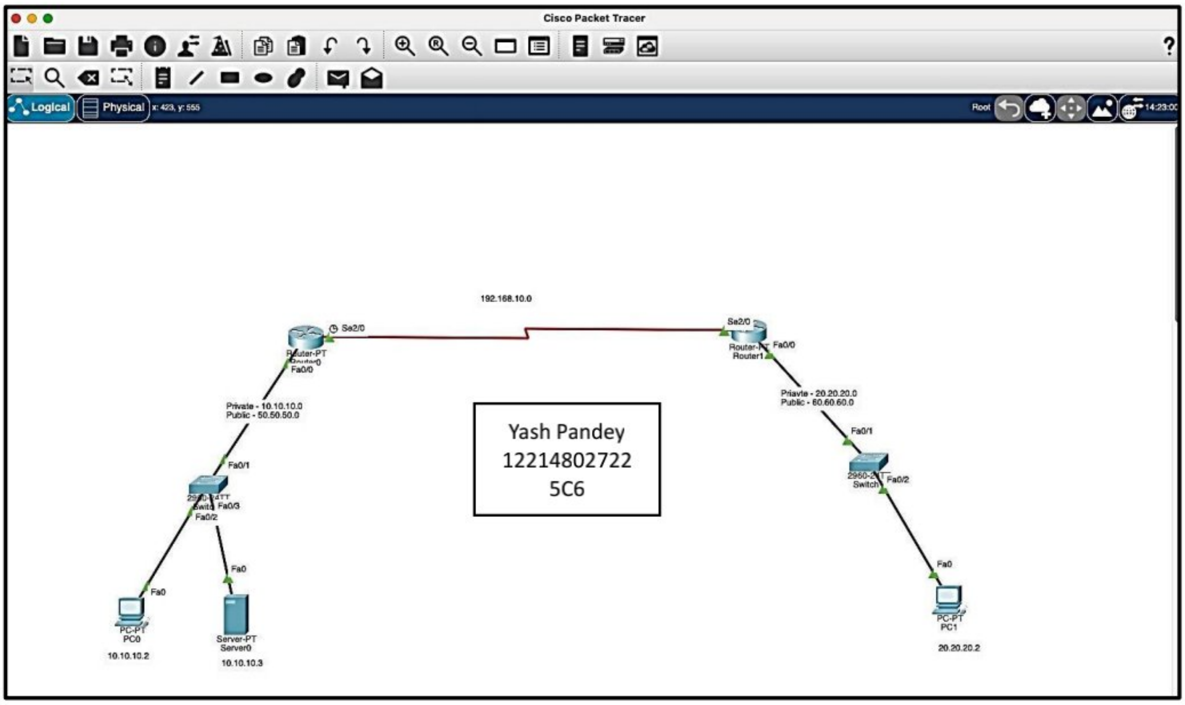
### Network Address Translation

Network Address Translation (NAT) is a networking technique used to modify network address information in packet headers while in transit through a router, firewall, or gateway device. NAT serves several important purposes in network communication:

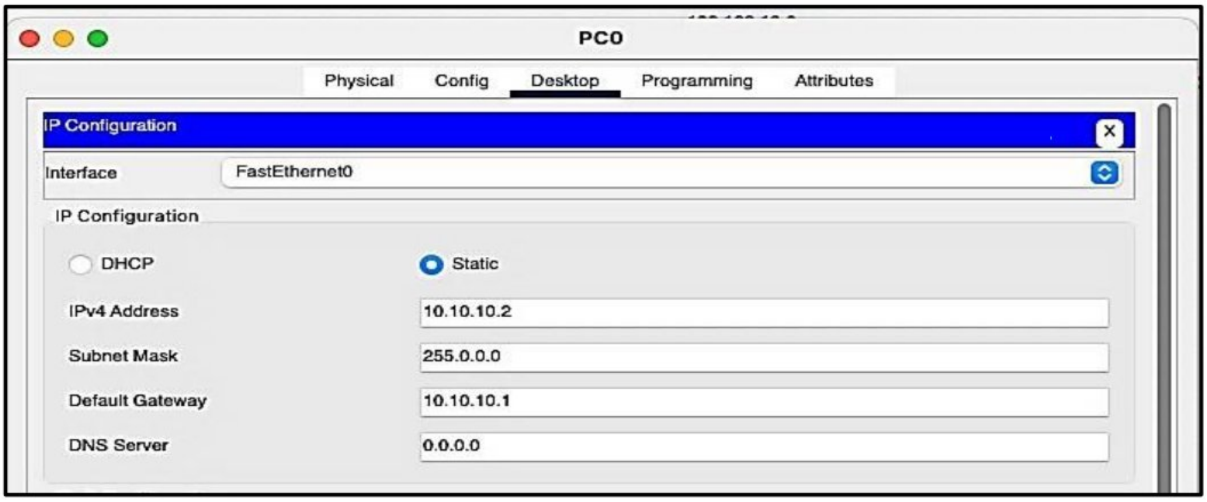
1. **IP Address Conservation:** NAT allows multiple devices within a private network (e.g., a home or office network) to share a single public IP address for communication with external networks, such as the internet. This conserves the limited pool of available public IPv4 addresses.
2. **Private Addressing:** Private IP addresses, as defined in RFC 1918, are reserved for use within private networks. NAT enables devices with private IP addresses to communicate with public networks without exposing their internal addresses.
3. **Security:** NAT acts as a barrier between the public internet and a private network. It hides the internal network structure and IP addresses from external sources, providing a level of security by obscurity. Incoming traffic is typically blocked by default, unless specific rules (port forwarding or DMZ configuration) are applied.
4. **Load Balancing:** Some advanced NAT implementations, such as Network Address Port Translation (NAPT), allow for load balancing. Multiple internal devices can use the same public IP address, and NAT keeps track of which device initiated each connection, forwarding responses to the correct device based on port numbers.
5. **Port Address Translation:** Port Address Translation (PAT), a form of NAT, uses a single public IP address and modifies the source port number in the packet header to distinguish between multiple internal devices. This enables many devices to share a single public IP address simultaneously.
6. **IPv6 Transition:** NAT is often used as an interim solution to extend the life of IPv4 networks while transitioning to IPv6. It allows IPv6-capable devices to communicate with IPv4 devices by performing translation between the two address formats.

NAT is commonly implemented in home routers, corporate firewalls, and other gateway devices. There are different types of NAT, including Static NAT (1:1 mapping of internal and external addresses), Dynamic NAT (maps internal addresses to available external addresses from a pool), and PAT (maps multiple internal addresses to a single external address using port numbers).

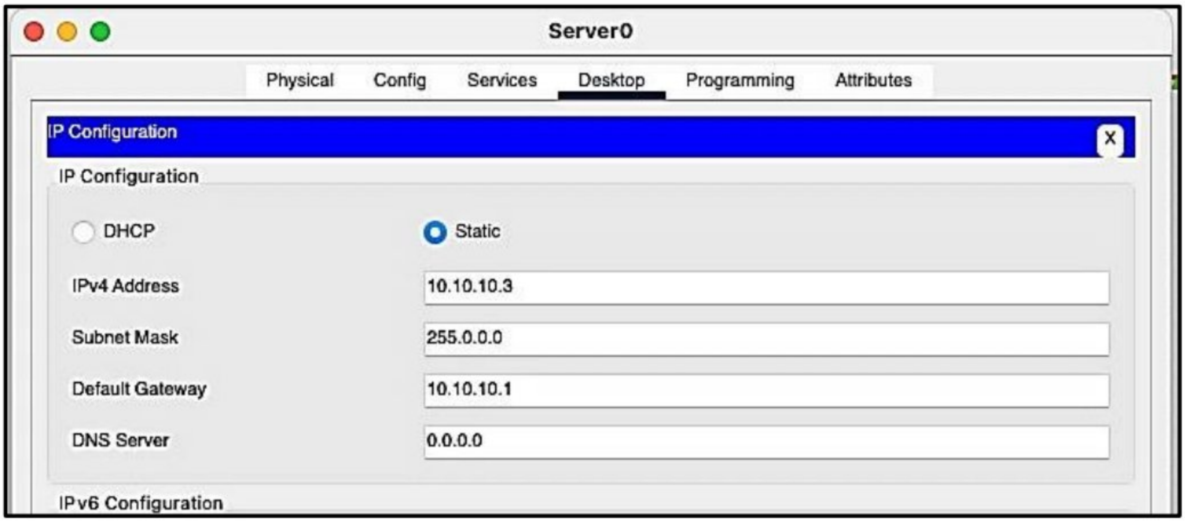
While NAT is effective for its intended purposes, it does introduce certain limitations, such as making it more challenging to host services on internal devices that need to be accessible from the internet. This limitation can be addressed through techniques like port forwarding or using technologies like IPv6, which provides a larger pool of globally routable addresses and reduces the need for NAT. Network Topology



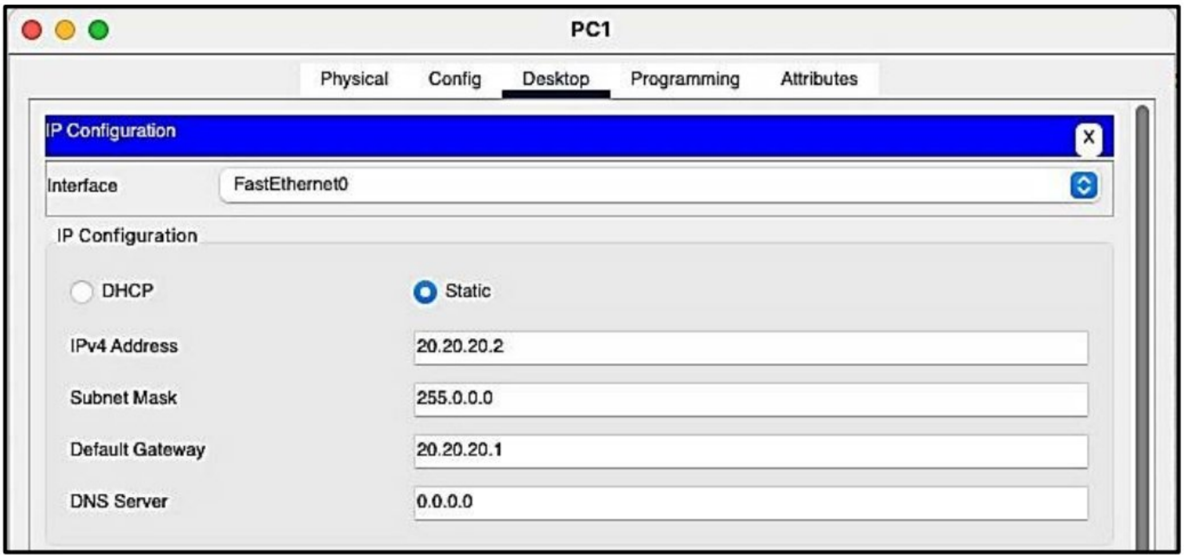

PC0 IP Configuration



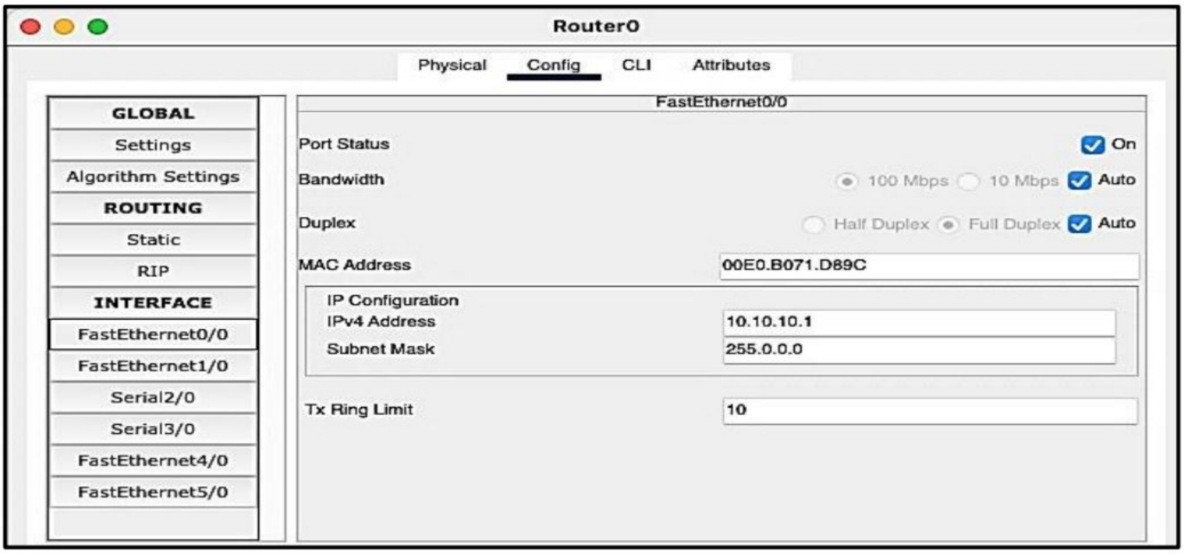

Server0 IP Configuration

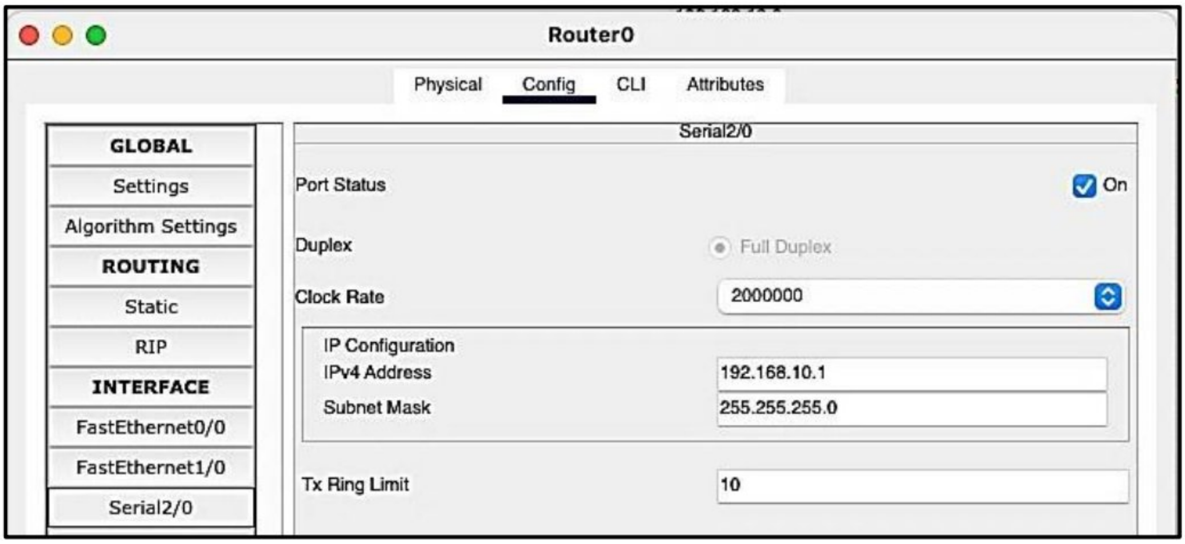



PC1 IP Configuration

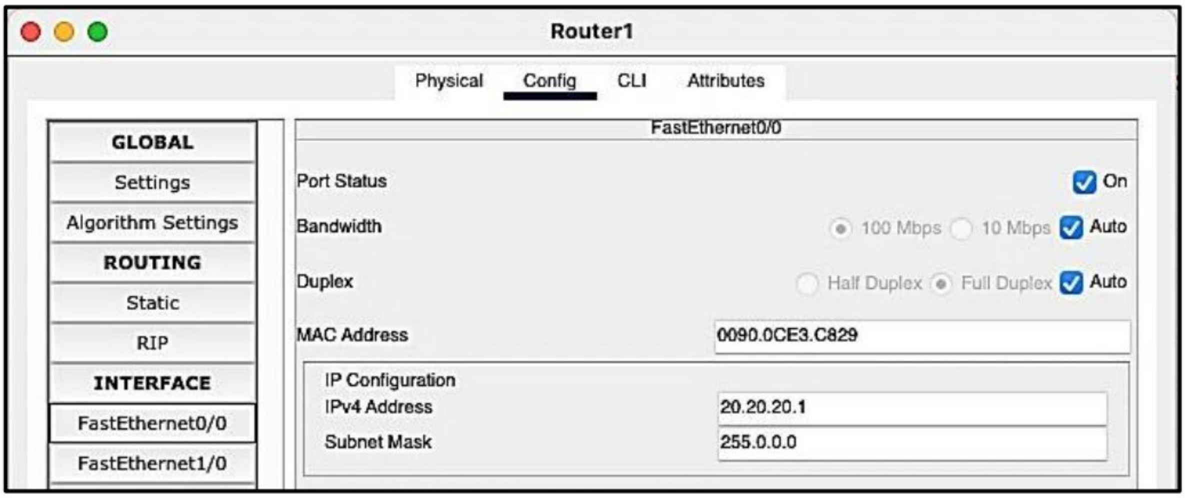


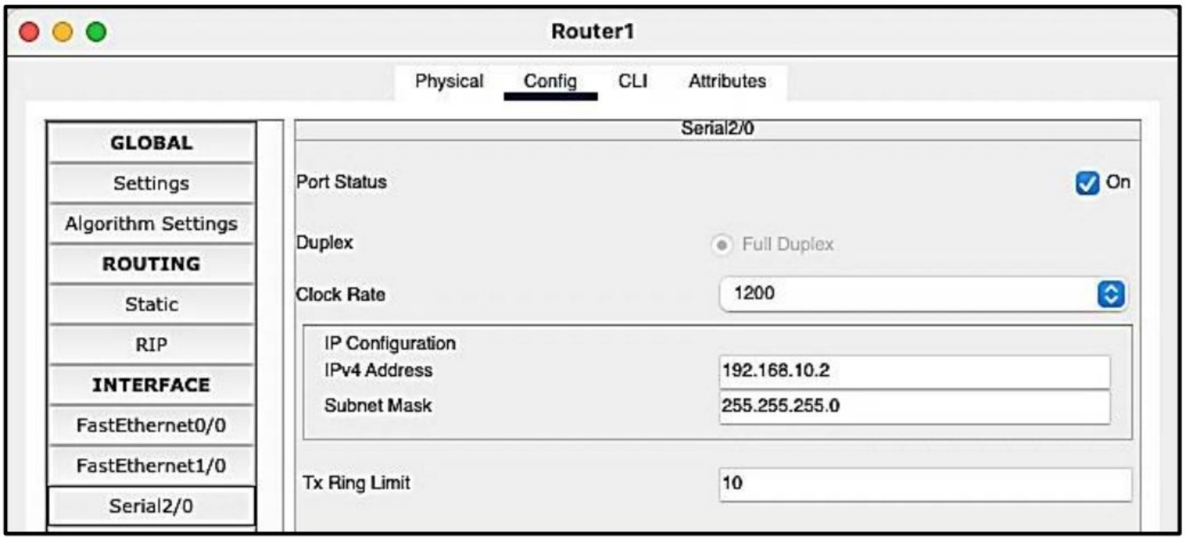

Router0 Configuration




Router1 IP Configuration






Static Route and Nat Configuration Router1

Router(config)#ip nat inside source static 20.20.20.2 60.60.60.2

Router(config)#exit

Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface FastEthernet0/0

Router(config-if)#ip nat inside

Router(config-if)#exit

Router(config)#interface Serial2/0

Router(config-if)#ip nat outside

Router(config-if)#exit

Router(config)#ip route 50.0.0.0 255.0.0.0 192.168.10.1

Router(config)#exit

Router#

%SYS-5-CONFIG_I: Configured from console by console

Router#

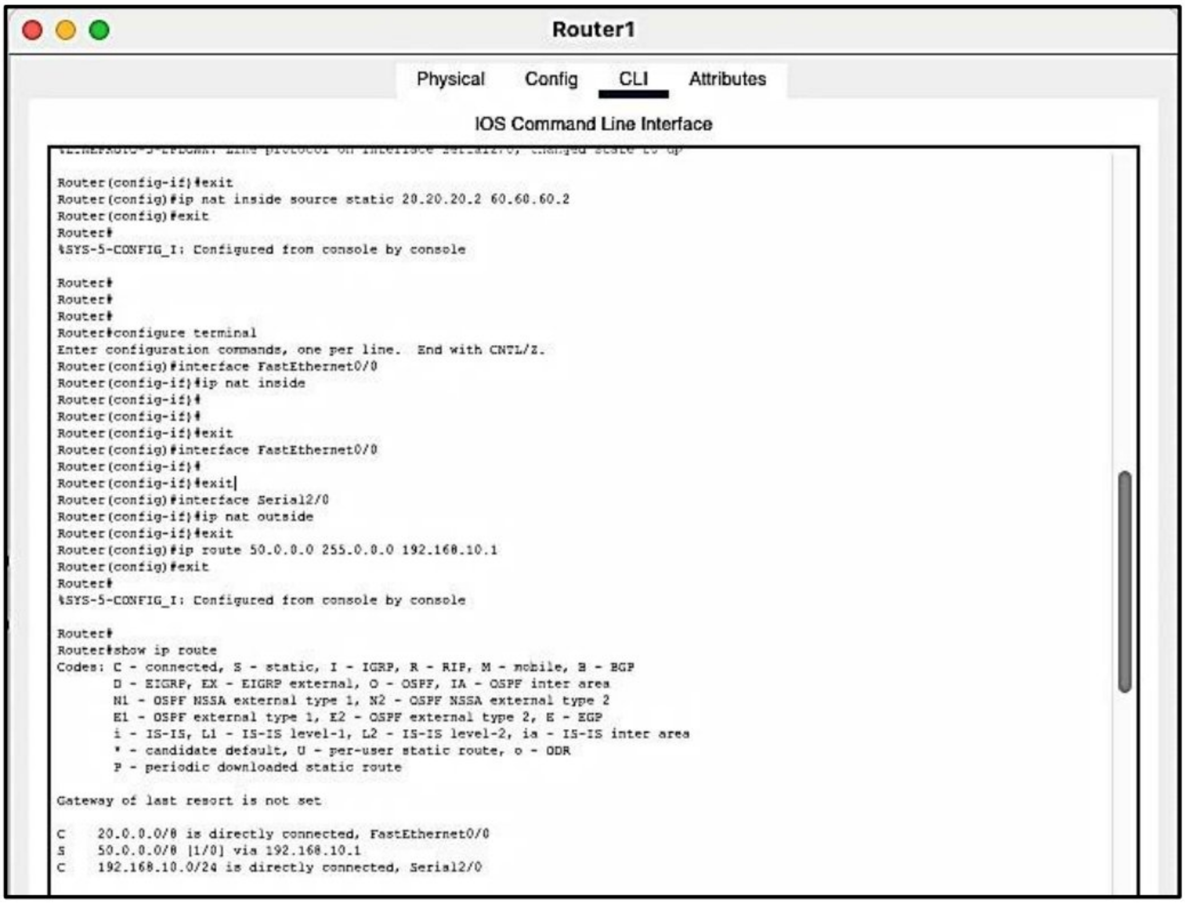
Router#show ip route

Gateway of last resort is not set

C 20.0.0.0/8 is directly connected, FastEthernet0/0

S 50.0.0.0/8 [1/0] via 192.168.10.1

C 192.168.10.0/24 is directly connected, Serial2/0




Static Route and Nat Configuration Router0

Router(config)#ip nat inside source static 10.10.10.2 50.50.50.2

Router(config)#ip nat inside source static 10.10.10.3 50.50.50.3

Router(config)#interface FastEthernet0/0

Router(config-if)#ip nat inside

Router(config-if)#exit

Router(config)#interface FastEthernet0/0

Router(config-if)#exit

Router(config)#interface Serial2/0

Router(config-if)#ip nat outside

Router(config-if)#exit

Router(config)#interface Serial2/0

Router(config-if)#exit

Router(config)#ip route 60.0.0.0 255.0.0.0 192.168.10.2

Router(config)#exit

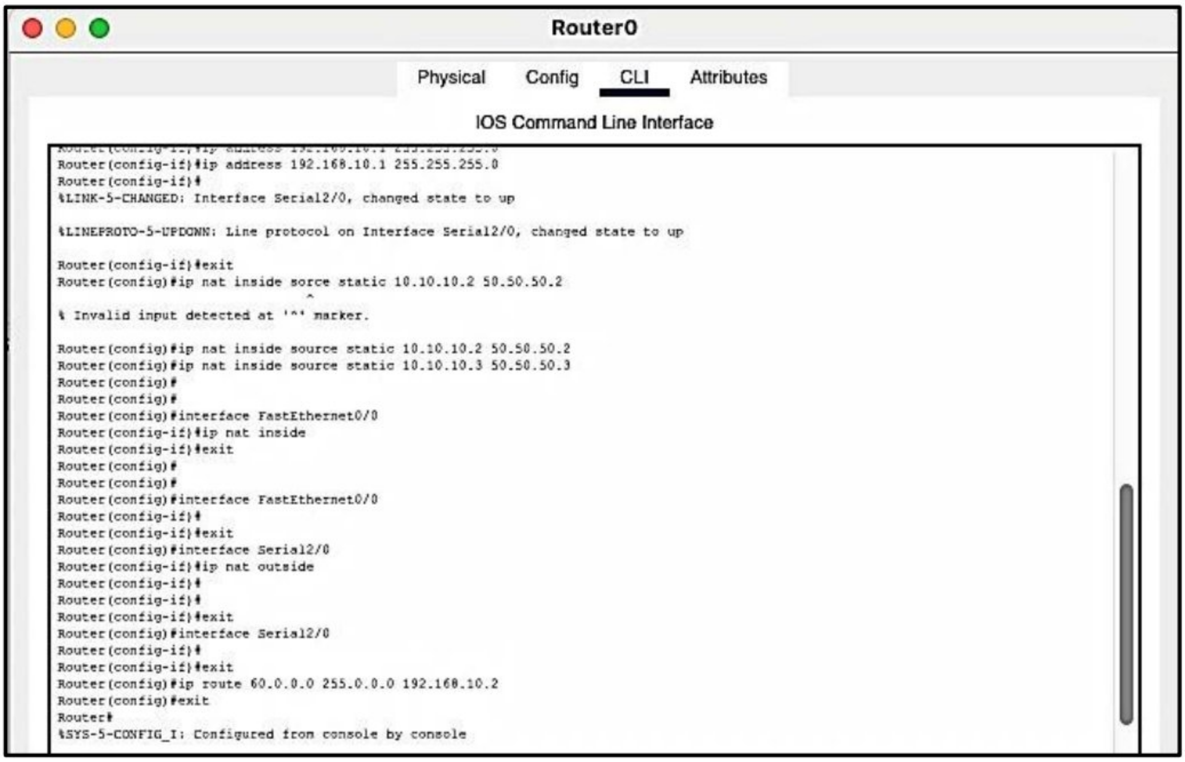
Router# show ip route

Gateway of last resort is not set

C 10.0.0.0/8 is directly connected, FastEthernet0/0

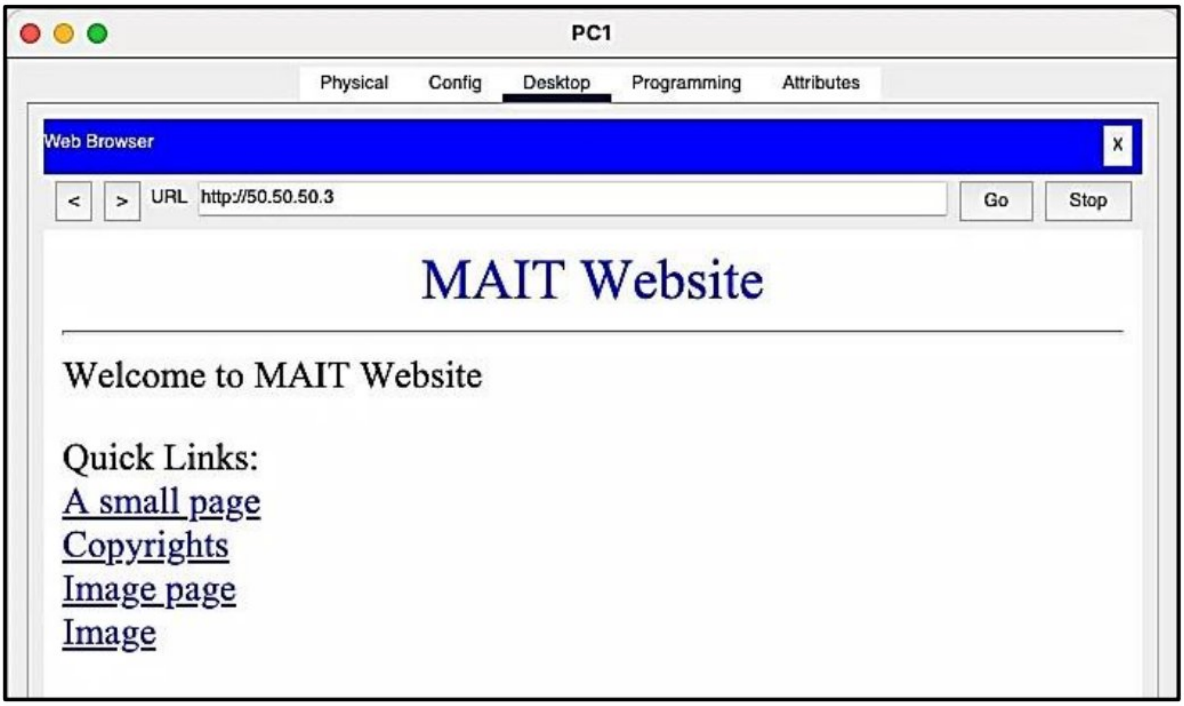
S 60.0.0.0/8 [1/0] via 192.168.10.2

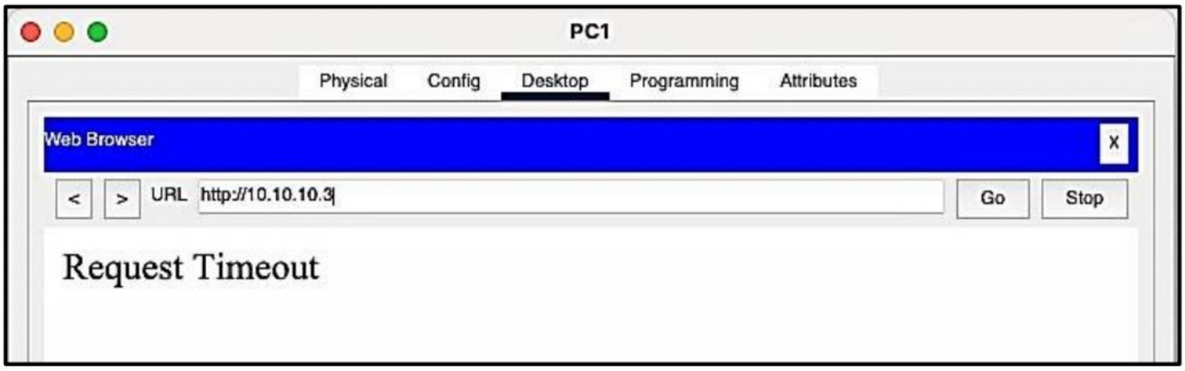
C 192.168.10.0/24 is directly connected, Serial2/0




NAT Output

We should be able to visit the website hosted on Webser0 using its public IP only




Hence The private IP of the website is hidden, and no one can access the website from outside the network without the public IP of the website.

# Lab Exercise-10.

## Aim :- Conducting A Network Capture And Monitoring With WireShark Tool .

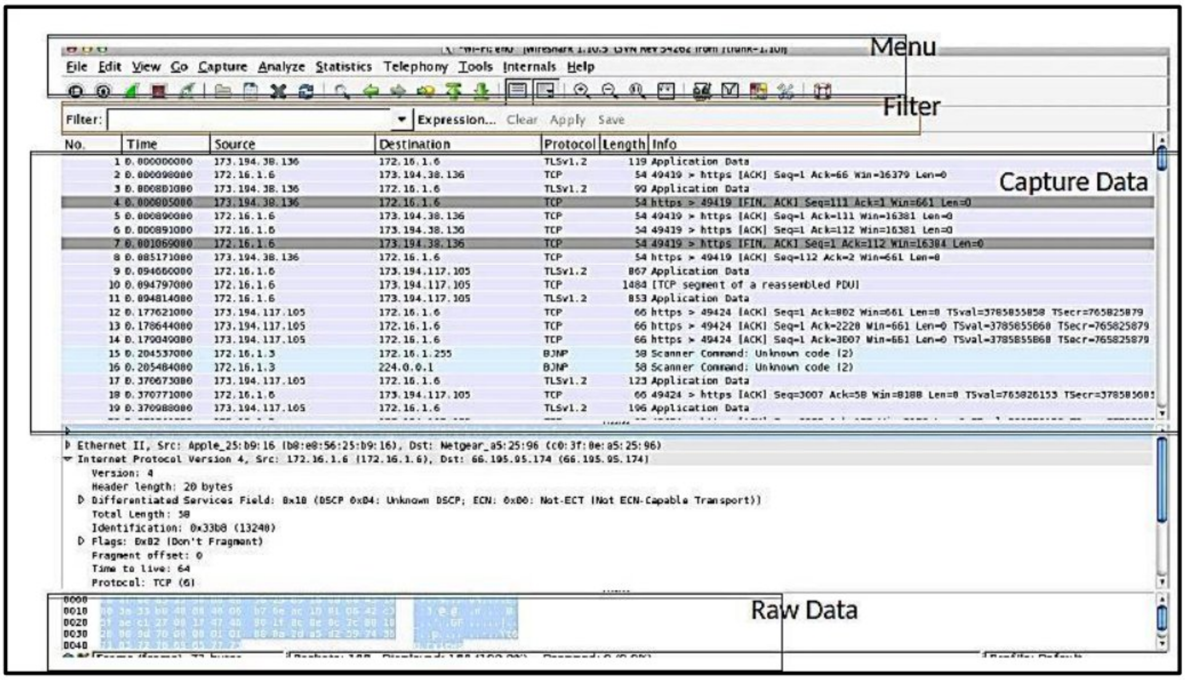
### Wireshark

Wireshark is a widely used open-source network protocol analyzer. It allows users to capture and analyze the data traffic flowing through a computer network. Here are the key aspects of Wireshark:

1. **Packet Capture:** Wireshark enables the real-time capture of data packets as they traverse a network. It can capture packets from various network interfaces, including wired (Ethernet) and wireless (Wi-Fi).
2. **Packet Analysis:** Once captured, Wireshark provides a detailed and interactive view of network packets. Users can examine individual packets to inspect their headers, payloads, and other relevant information.
3. **Protocol Analysis:** Wireshark is capable of dissecting and interpreting a wide range of network protocols, including common ones like TCP/IP, UDP, HTTP, DNS, and many more. It can decode these protocols and present the data in a human-readable format, making it easier to troubleshoot network issues.
4. **Filtering and Searching:** Users can apply filters and search for specific packets or conditions within the captured data. This functionality is valuable for isolating and focusing on specific network events or issues.
5. **Packet Playback:** Wireshark can replay captured packets to simulate network traffic. This feature is useful for testing and troubleshooting network configurations and security measures.
6. **Statistics and Graphs:** Wireshark provides statistical analysis and visualization tools to help users understand network behavior. It can generate graphs, charts, and tables summarizing various aspects of network traffic.

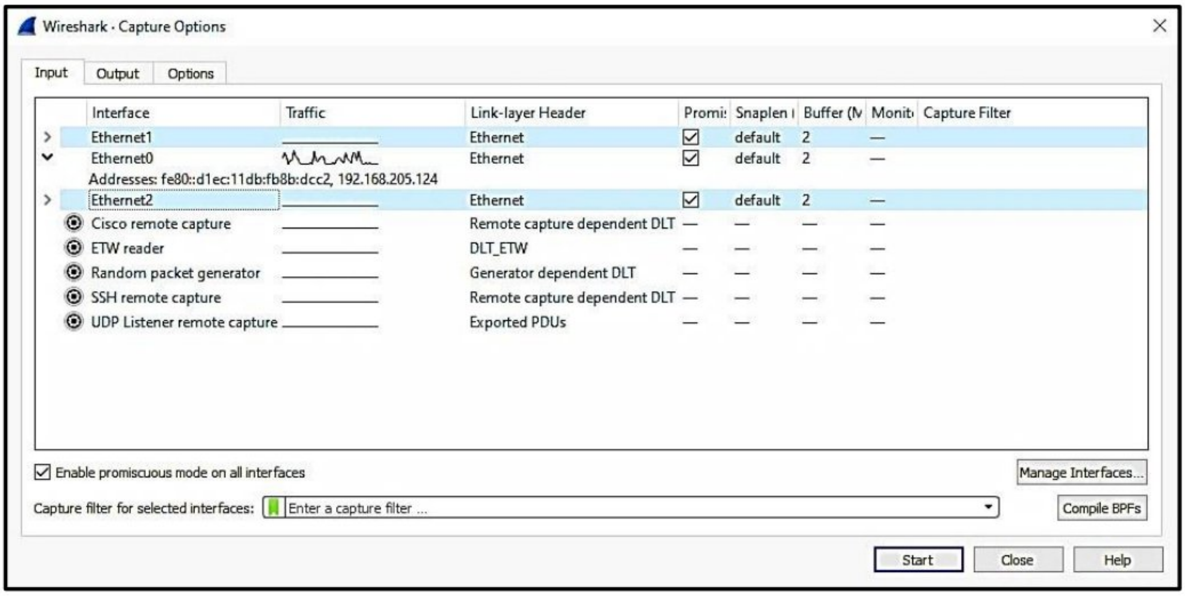
7. **Exporting Data:** Users can export captured data or analysis results to various file formats, making it easy to share findings with colleagues or integrate the data into other tools and reports.
8. **Security:** Wireshark is a versatile tool, but it can also be used maliciously. Therefore, it's essential to use it responsibly and ethically, respecting privacy and legal regulations. Additionally, encrypted network traffic (e.g., HTTPS) is challenging to analyze with Wireshark due to encryption.
9. **Platform Availability:** Wireshark is available for multiple platforms, including Windows, macOS, and various Linux distributions. This cross-platform compatibility makes it accessible to a wide range of users.

Wireshark is a valuable tool for network administrators, security professionals, developers, and anyone tasked with diagnosing or monitoring network performance and behavior. It can help identify network issues, security threats, and performance bottlenecks, making it an indispensable asset in network troubleshooting and analysis.




Exercises to be performed on Wireshark

1. Packet Capture over any connection, Wired or Wireless




2. F ilter packets using filter command

1. TCP
2. HTTP
3. UDP
4. ICMP

3. Run Tracert Command on Terminal and Capture and Analyze the packets